

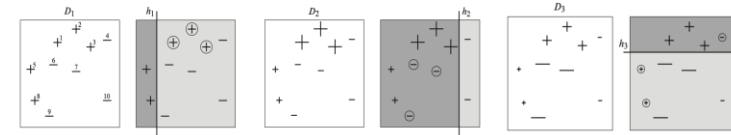
Machine Learning

Rudolf Mayer
December 18th, 2025

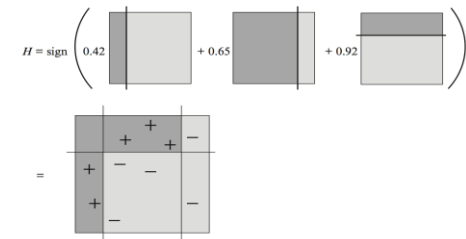
Recap: Ensemble Learning



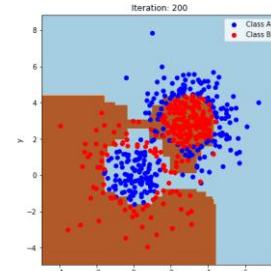
- Combining multiple classifiers
 - Often “weaker” models
 - Goal: improve overall effectiveness
 - Similar, but different from Model/Algorithm Selection: use all (most) classifiers
 - Homogenous vs. heterogenous
 - Bagging vs. Boosting
 - Bagging: need classifiers with some variance



- AdaBoost: higher weights to wrongly predicted samples → more importance for next model

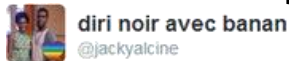


- GradientBoosting: learn a model that corrects previous mistakes directly
 - By using pseudo residuals



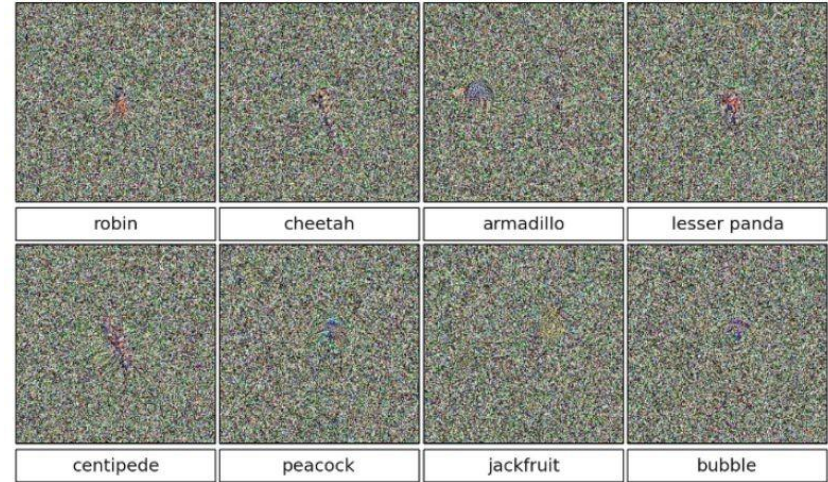
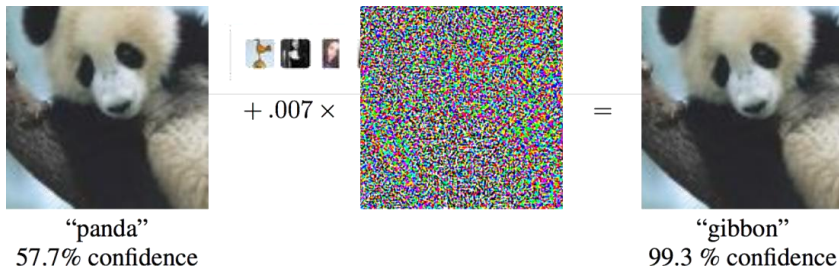
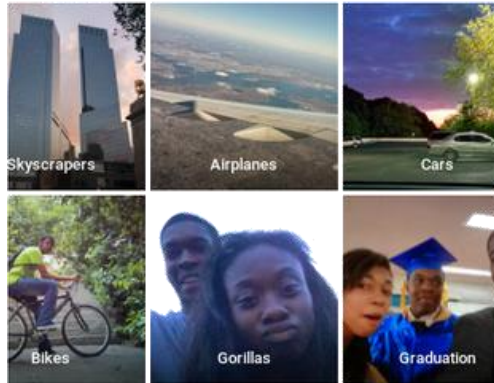
Recap: Robustness/Adversarial ML

- ML Models make ridiculous mistakes & can be easily fooled



Google Photos, y'all fucked up. My friend's not a gorilla.

Voir la traduction



- Attacks are “easy”, defending is hard!

Recap: Explainable AI

- Complex ML models are powerful, but **black-box**

- Cannot understand decisions
- Trust decision-support systems?

→ Explainable AI

- Intrinsic, ex-ante, post-hoc
- Explain locally / globally
- Single or multiple samples
- Model-specific or agnostic



This face is Angry

because it is similar to these examples

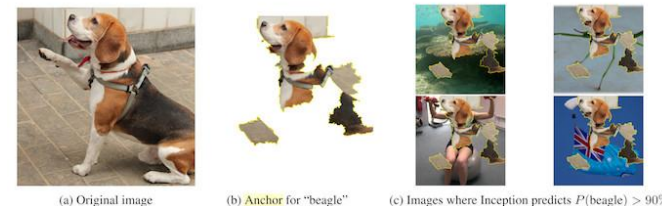


and dissimilar to these



This bird is a **Crested Auklet** because this is a black bird with a small orange beak and it is not a **Red Faced Cormorant** because it does not have a long flat bill.

This bird is a **Parakeet Auklet** because this is a black bird with a white belly and small feet and it is not a **Horned Grebe** because it does not have red eyes.

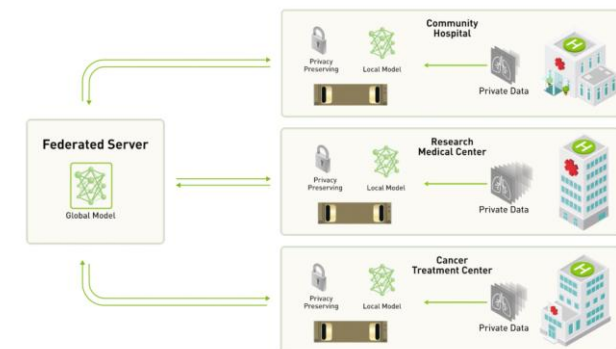
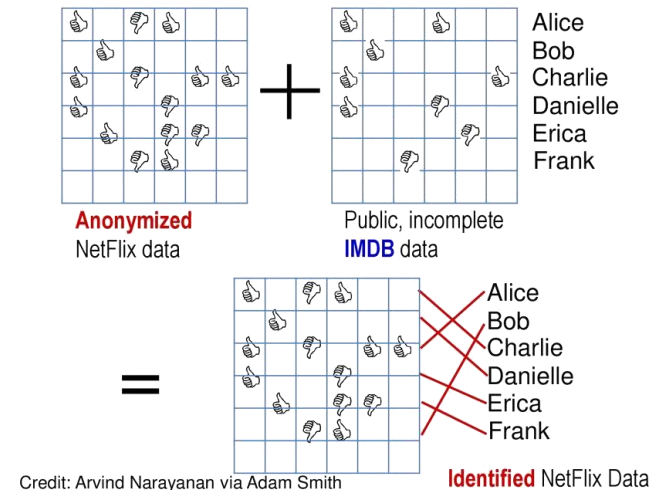


Recap: Privacy

- ML very useful e.g. in health-care
 - But requires large amounts of highly sensitive data
 - Data protection regulations restrict sharing & usage

→ Privacy-preserving ML

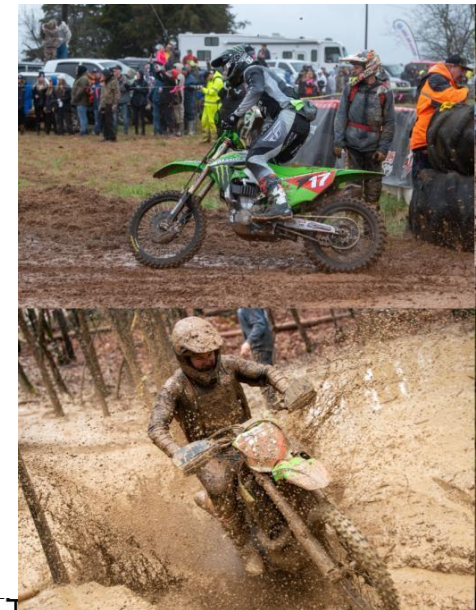
- Privacy-preserving data publishing
 - k-anonymity, ...
 - Trade-off data utility vs. privacy
- Privacy-preserving computation
 - Federated learning
 - Cryptographic methods (SMPC, HE)



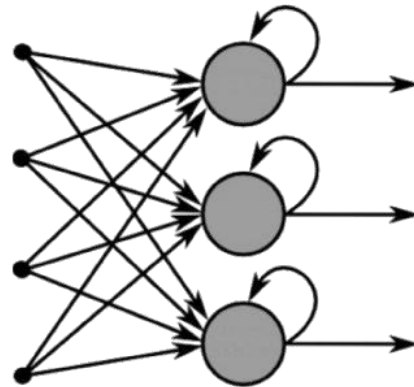
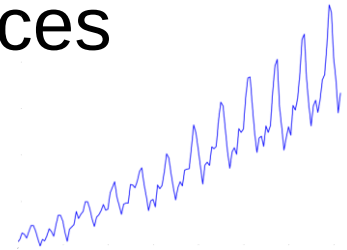
- Recurrent Neural Networks
- Support Vector Machines
- Statistical Significance Testing
- Feature Selection
- Challenge: Reproducibility
- Summary
- Advertisements: ML lectures, internships, thesis,

...

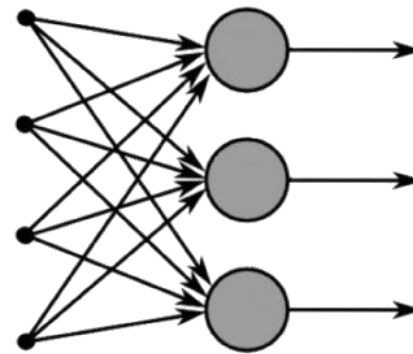
- Feed forward networks accept **fixed-sized** vector as input and produce **fixed-sized** vector as output
 - E.g. train a CNN for ImageNet
 - 224x224x3 input dimensionality
 - 1000 dimensional output: classes
- These are fixed, can't easily adapt to different sizes
 - If **some** images are larger / smaller:
 - Down-/upscale images
 - If **generally** larger / smaller images:
 - Train / fine-tune model with fitting input layer
 - Different output dimensionality?
 - I.e. different number of target classes
 - Train/fine-tune with new output layers (and potentially further FC layers)



- Feed forward networks: **fixed-sized** input & output
 - Fixed amount of computational steps
- **Recurrent** networks operate over sequences
(vectors, matrices, generally: tensors)
 - Useful for sequential / time series data
 - Text, sound, video, ...



Recurrent Neural Network

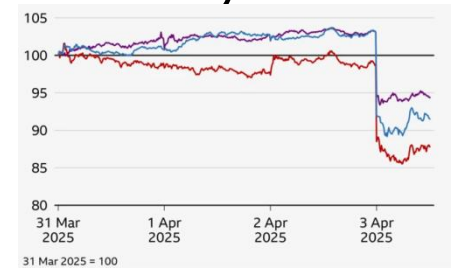


Feed-Forward Neural Network

- How to feed stock-exchange data to MLP?

- Just consider the current day

- Lose all context / history / memory of previous days – but they likely influence the prediction!
- Or maybe not... ?

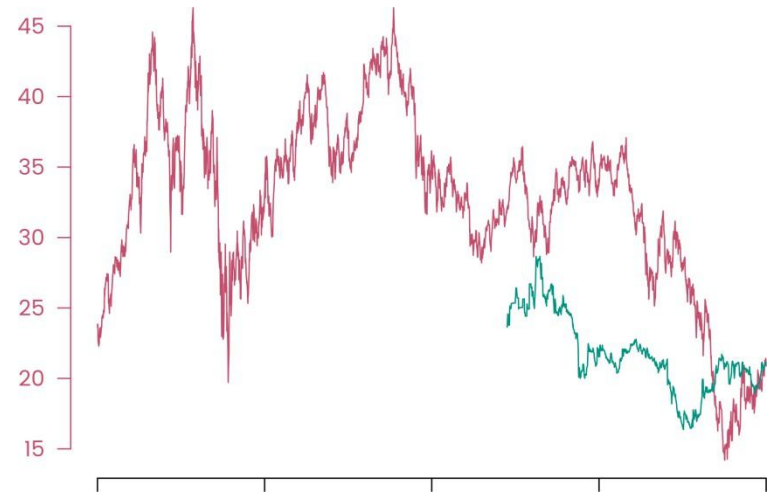


- Consider a fixed-length window, e.g. past 5 days
 - Very limited; difficult to chose window length
- Compute statistics over all previous days
 - Mean, variance, or higher-order moments,

- How to feed stock-exchange data to MLP?

Input is a sequence, variable length

- E.g. companies started trading at different points in time

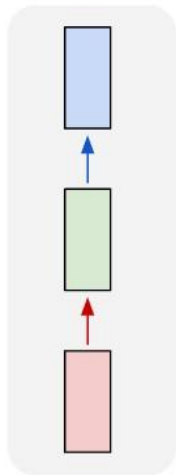


- Use models that can handle sequences per design – enter Recurrent Neural Networks!

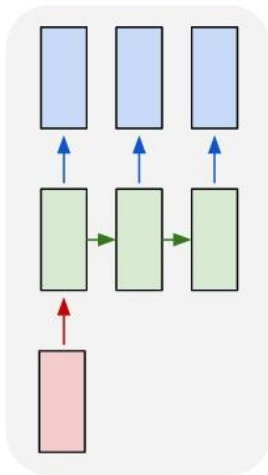
Different Sequential In/output Settings

- Sequential information (as **input** and/or **output**)

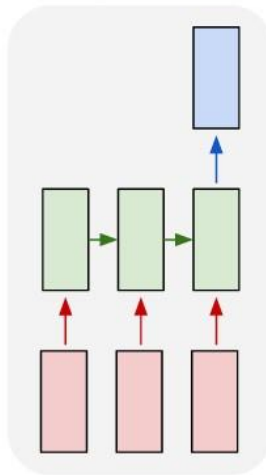
one to one



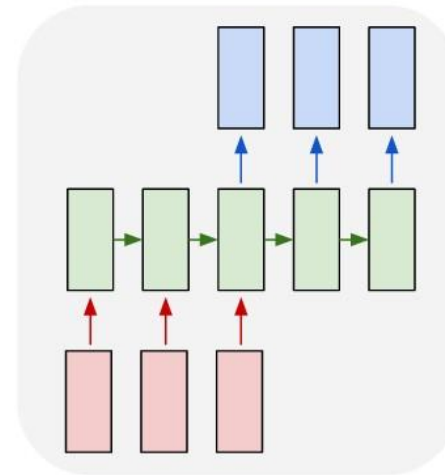
one to many



many to one



many to many



many to many

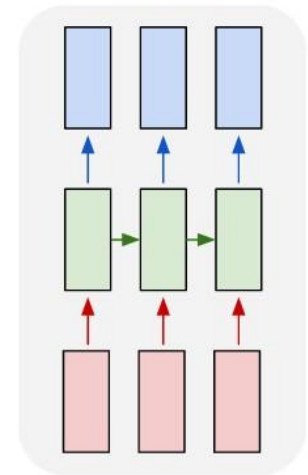


Image
classification

(one image → one label)
E.g. via CNN

Sentiment analysis
(many words → one label)

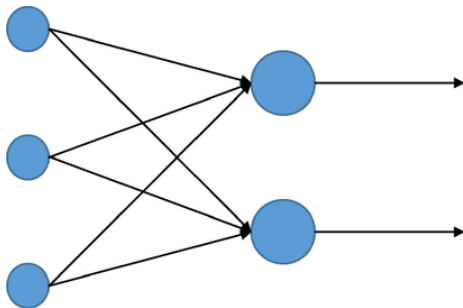
Image captioning
(one image → many words)

Machine translation
(many words → many words)

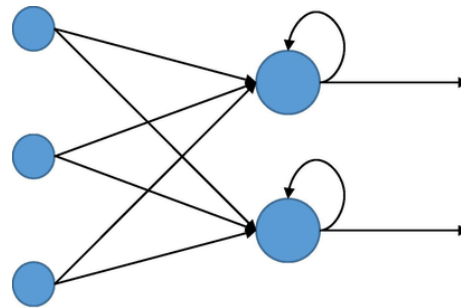
Synced sequence
(video classification, each frame)

- Convolutional Networks (Feed-Forward Networks)
 - Neurons organised in layers
 - Weights, bias & activation functions
 - Input: fixed-sized vector/tensor (e.g. an image of 224x224x3)
 - Output: fixed-sized vector/tensor (e.g. probabilities of classes)
- Recurrent Neural Networks:
 - Neurons organised in layers
 - Weights, bias & activation functions
 - **PLUS: feedback**
 - Thus: can operate over **sequences** of vectors/tensors
 - Input & output **size** is **variable** (“infinite”)
 - Have a **state** (“memory”)

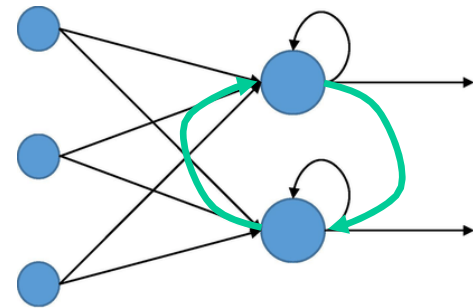
- Recurrent Neural Networks:
 - Weights, bias & activation functions
 - **Feedback via connections**
 - **Have a state (“memory”)**



Feed forward
neural network



Simple
recurrent
neural
network



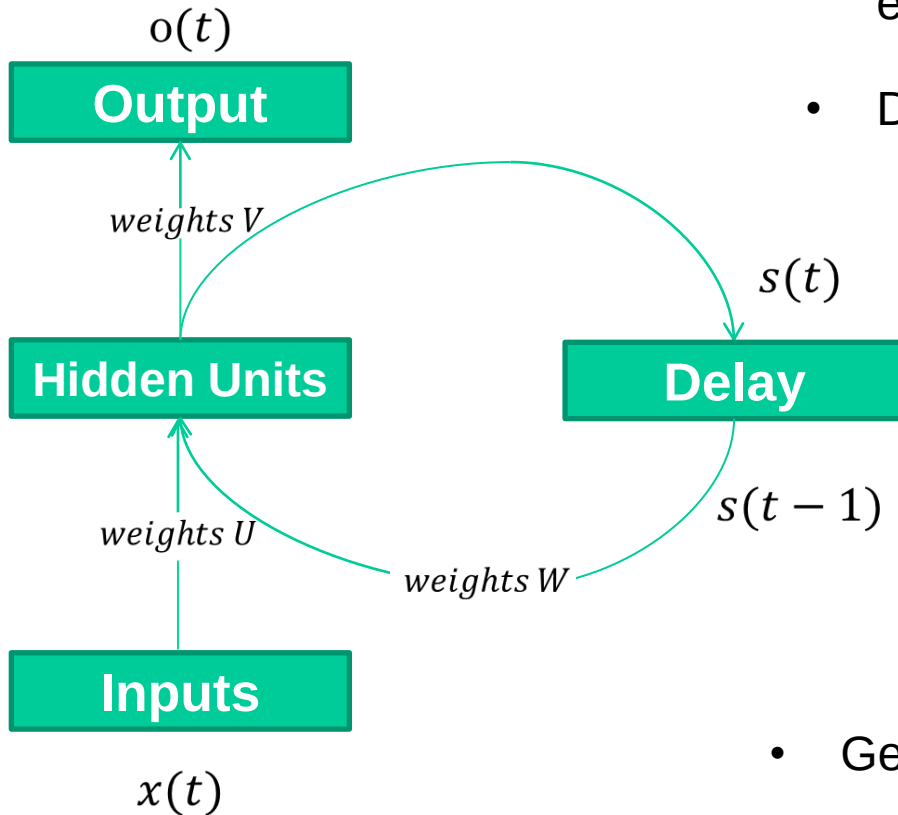
Fully connected
recurrent neural
network

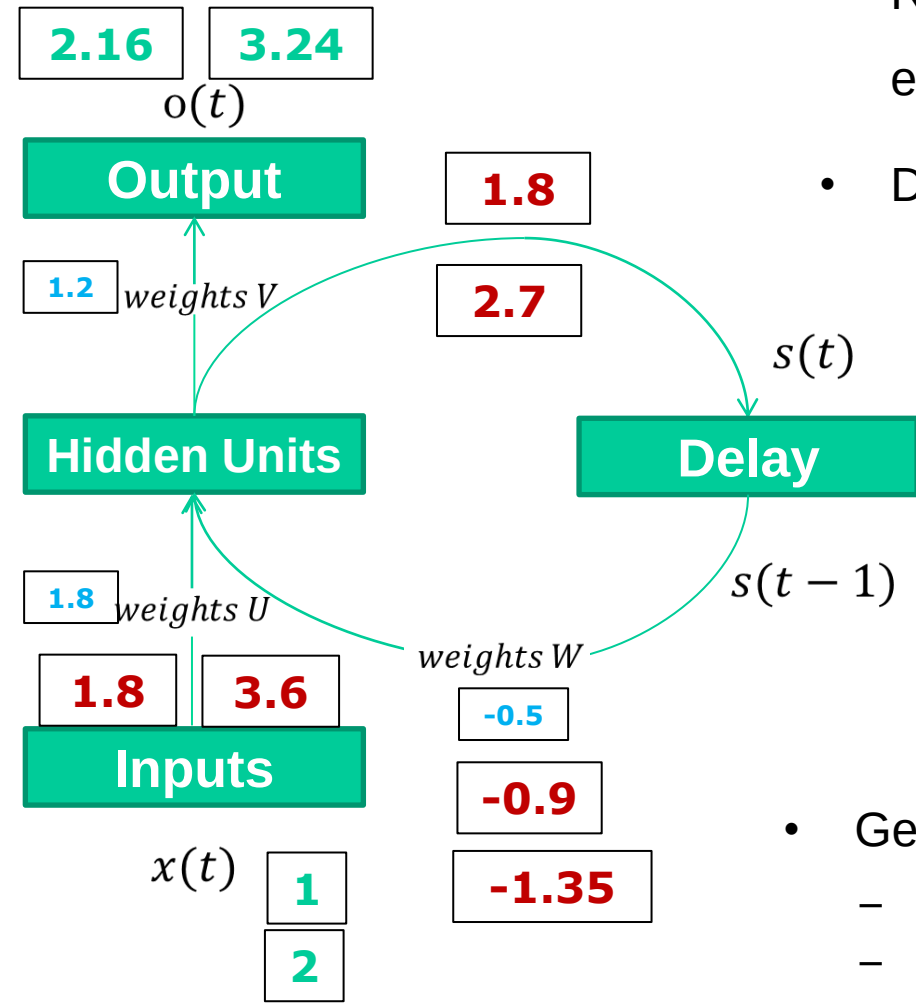
.....

- Recurrent, as they perform the same task for every element of the sequence

- Delay unit: holds **activation** until next step
 - Recurrent units: 2 sets of weights & inputs
 - **Weights: U (input) & W (memory)**
 - **Inputs $x(t)$** (t-th element of the sequence) and $s(t-1)$ (memory)
 - Weights V : computes output at t

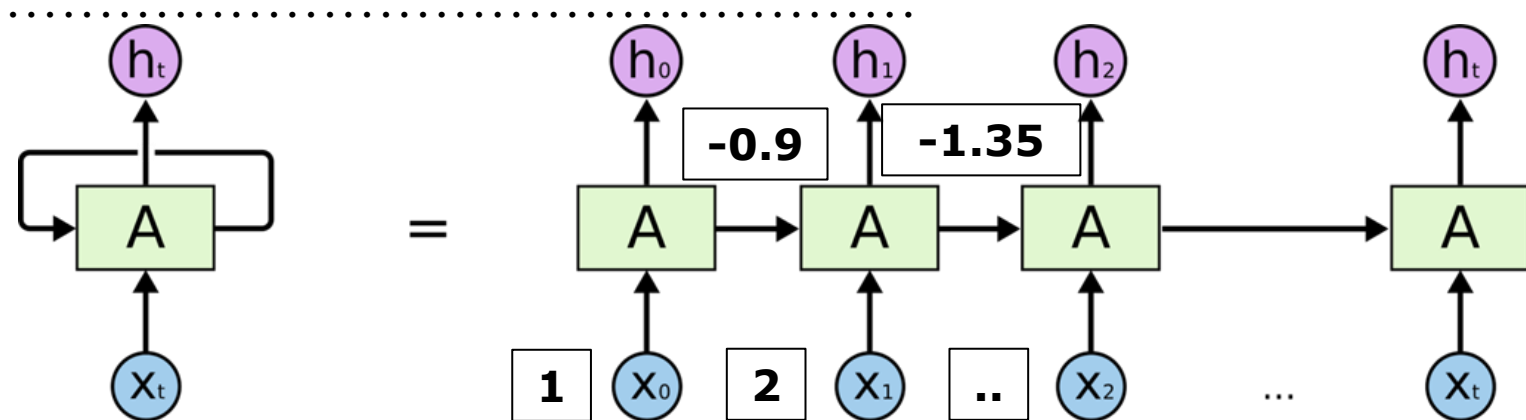
- Generally: only one set of W
 - Weights are reused! **Consequence?**
 - Number of parameters is independent of length of input
- Potentially also U and V are the same as W





- Recurrent, as they perform the same task for every element of the sequence
- Delay unit: holds **activation** until next step
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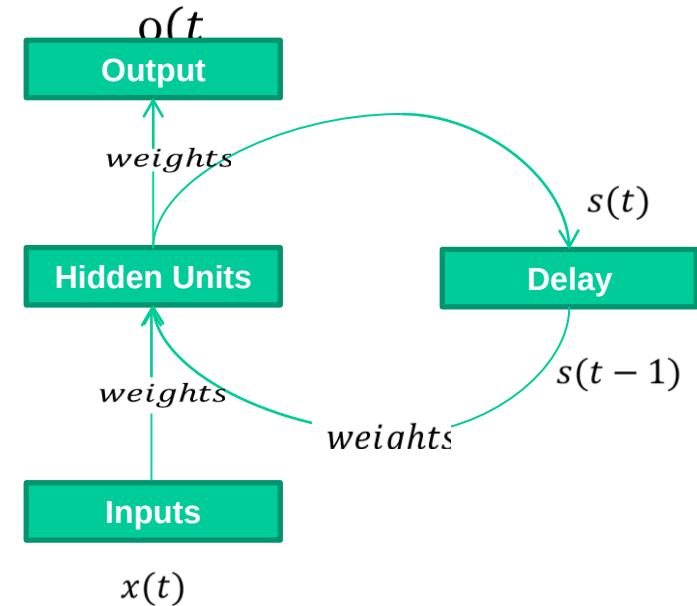
Recurrent Neural Networks (RNNs)



- RNNs process **sequences** of same-sized elements
- **Recurrent layers** process an input element together with their own output from the previous element
- Can be seen as a deep network with shared weights, unrolled in time
 - depth = length of sequence

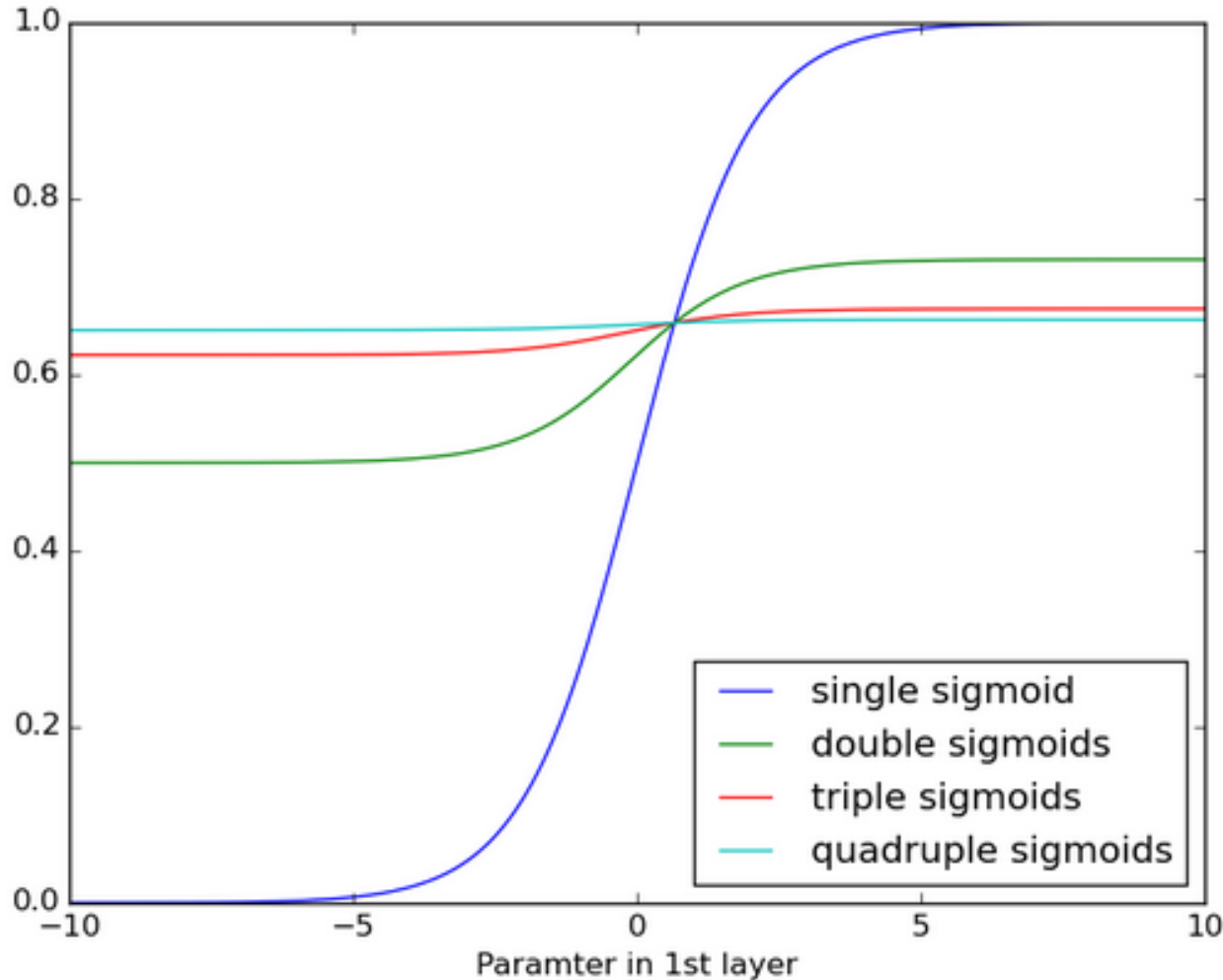
- Very similar to feed-forward networks (e.g. MLP)
- Forward pass: compute error/loss
 - On “unfolded” version of network
 - Intermediate steps & final output h_t
- Back-propagation through (in) time (BPTT)
 - Similar to MLP: derivatives, parameter updates
 - But on the unrolled version of the network
 - ➔ Cyclic graph transformed into an acyclic one
 - ➔ Can apply standard back-propagation

- Network input at time t :
 - $a_h(t) = Ux(t) + Ws(t-1)$
- Activation of input at time t :
 - $s(t) = f_h(a_h(t))$
- Output unit at time t :
 - $A_o(t) = V_s(t)$
- Output of network at time t :
 - $O(t) = f_o(a_o(t))$



- Long sequences: back-propagation consists of many steps (like a very deep network)
- Consequences?
 - Exploding / vanishing gradients
 - Each sequence element: multiplied by $w \rightarrow w^{|sequence|}$
 - E.g. 50 elements: $w > 1 \rightarrow ?$ $w < 1 ?$
 - Very large (w^{50}) resp. close to 0
 - Exploding: “simple” solution of gradient clipping (e.g. max 1)
 - Exploding & Vanishing:
 - Specialised “cells” proposed
 - In architectures such as LSTM (Long Short-Term Memory) and GRU (Gated Recurrent Unit)

RNN: Vanishing Gradients



- Effect: does not learn long-term dependencies

- LSTM: Long Short-Term Memory
 - Do not use same feedback loop for early & recent inputs
 - E.g. not the same for the stock market price at the beginning of the sequence, and the one just yesterday
 - Instead: different memories for long & short-term
 - More powerful **unit** than before

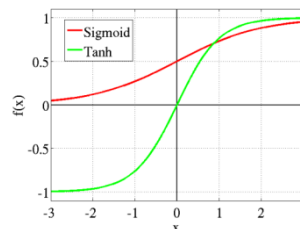


**SHORT-TERM
MEMORY**

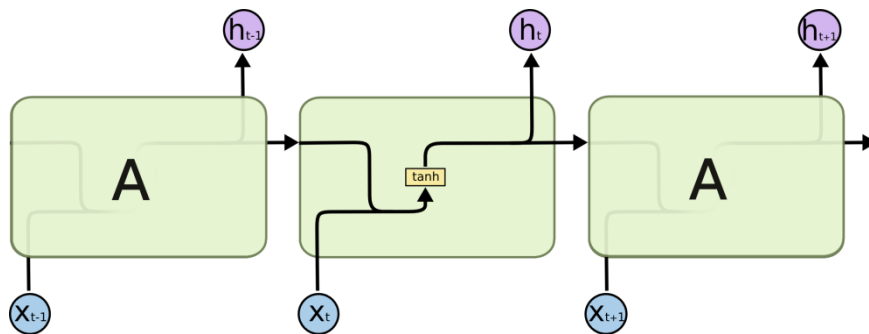


**LONG-TERM
MEMORY**

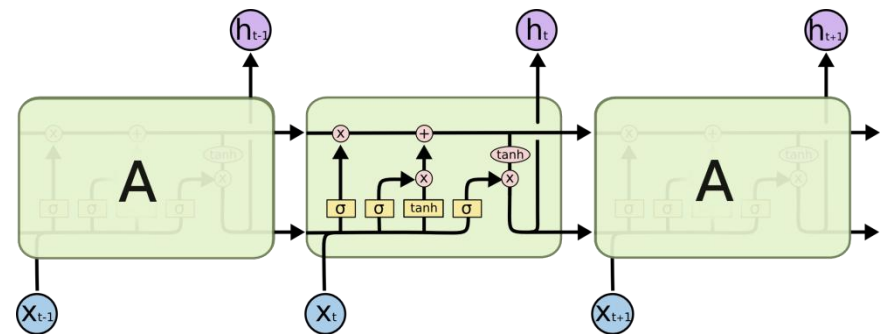
- LSTM uses tanh & sigmoid activation functions



- LSTM: Long Short-Term Memory
 - Long-term memory (LTM): scalar, not multiplied by weight
 - No issues with vanishing / exploding gradients
 - Short-term memory (STM): multiplied by weight
 - Units: Instead of neurons → memory blocks
 - Can store longer term memory
 - At each step: (i) keep old or (ii) update to new value



Standard RNN



LSTM

contains 4 interacting layers

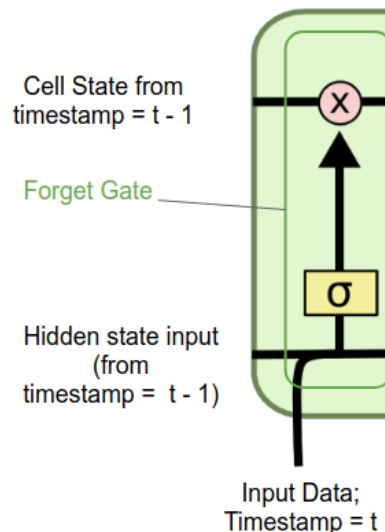
Details: <http://colah.github.io/posts/2015-08-Understanding-LSTMs/>

LSTM: Hochreiter and Schmidhuber 1997

FACULTY OF **INFORMATICS**

1. *Forget* gate computes how much of LTM to keep
 - *Forget*, though that gate computes how much to remember
2. Input gate (though it decides on how to update the memory...)
 1. Computes new potential LTM
 2. Multiplies by percentage of potential to keep (control)
3. Output gate decides on new STM & output

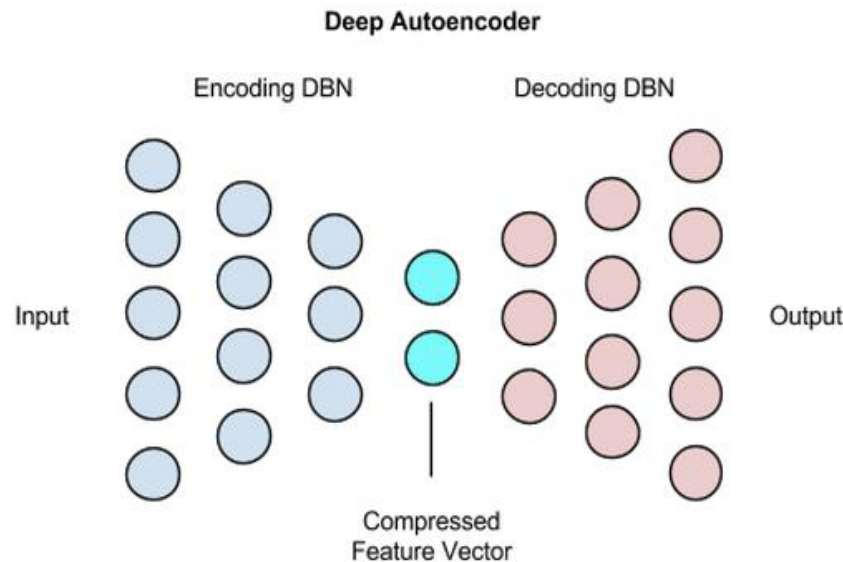
Different weights
for each gate



- LSTM: Long Short-Term Memory
- GRU: **Gated Recurrent Units**
 - Gating mechanism for RNNs (2014)
 - Similar performance as LSTM
 - But fewer parameters
- RNNs and CNNs can also be combined
 - e.g. an RNN processing sequences of CNN outputs

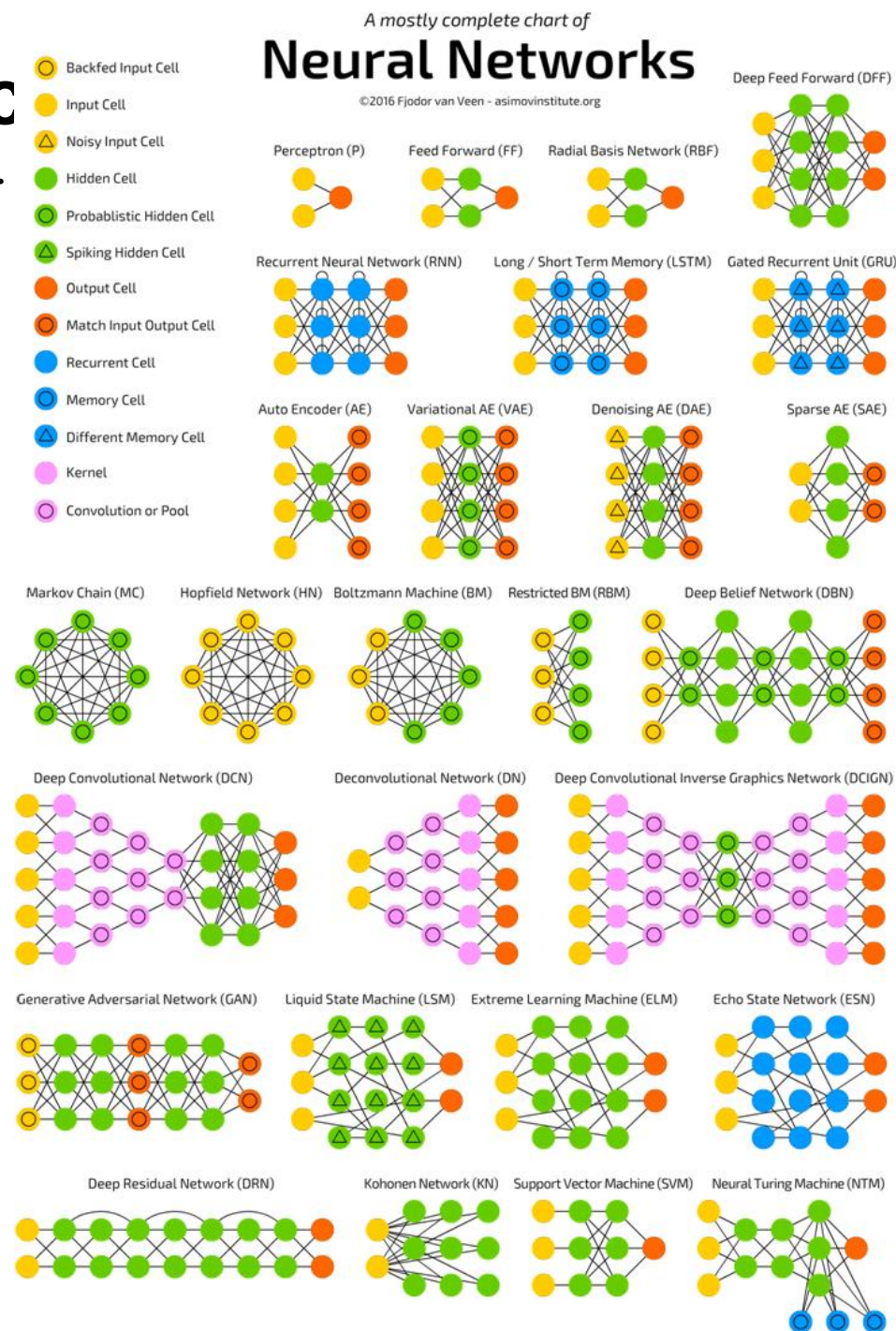
Other Architectures: Auto-Encoder

- Neural Network where input and output are (nearly) the same
- How is that useful? How is it different of just using the identity function?
- Purpose: typically dimensionality reduction / compression
- Learn a representation (encoding) for a data set
 - (Somewhat) related: Variational Auto Encoder (VAE), learning probabilities in compressed space

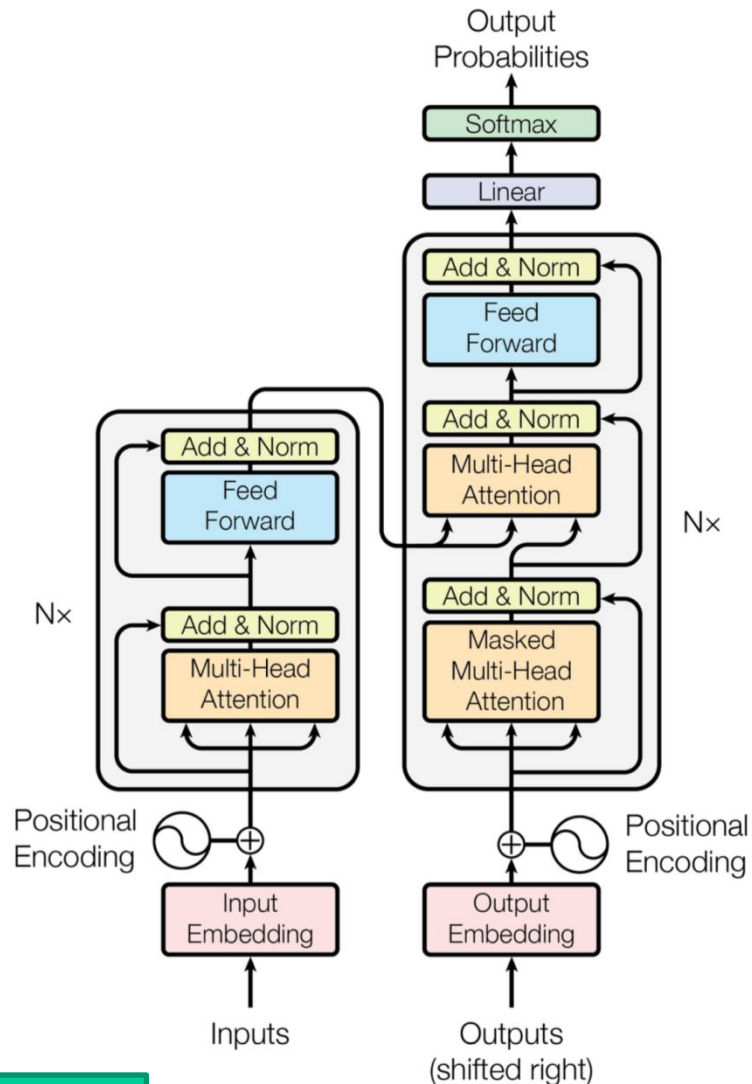


- Plenty of different architectures
 - For specific tasks
 - Supervised & unsupervised
 - Very active research field
 - Chart does e.g. not include Transformers (BERT; GPT, ...)

- <https://towardsdatascience.com/the-mostly-complete-chart-of-neural-networks-explained-3fb6f2367464>

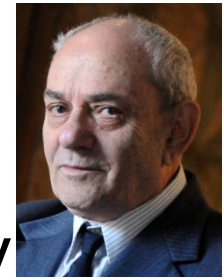


- Transformers, GPT, ...



- Recurrent Neural Networks
- Support Vector Machines: Intro
- Statistical Significance Testing
- Feature Selection
- Challenge: Reproducibility
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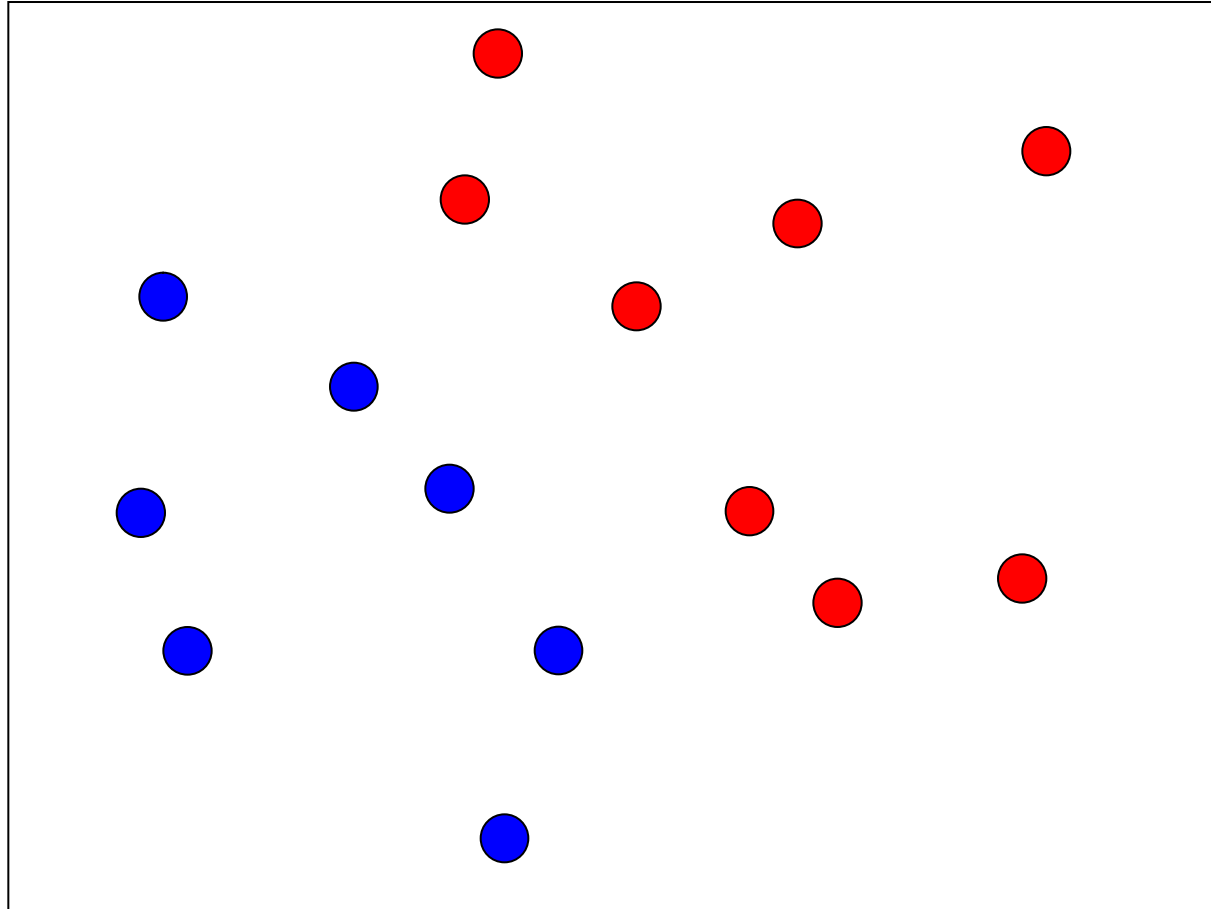
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- Introduced by Vladimir Vapnik & Alexey Chervonenkis in early 1960s (!)
 - Kernel-trick added in 1992
- Also known as **maximum margin** classifiers
- Heavily used & researched in last decade(s)
- Rather sophisticated mathematical model
- Basic concepts
 - Linear separation
 - Optimisation of hyperplane
 - Soft margin & kernel function
 - When linear separation is not possible

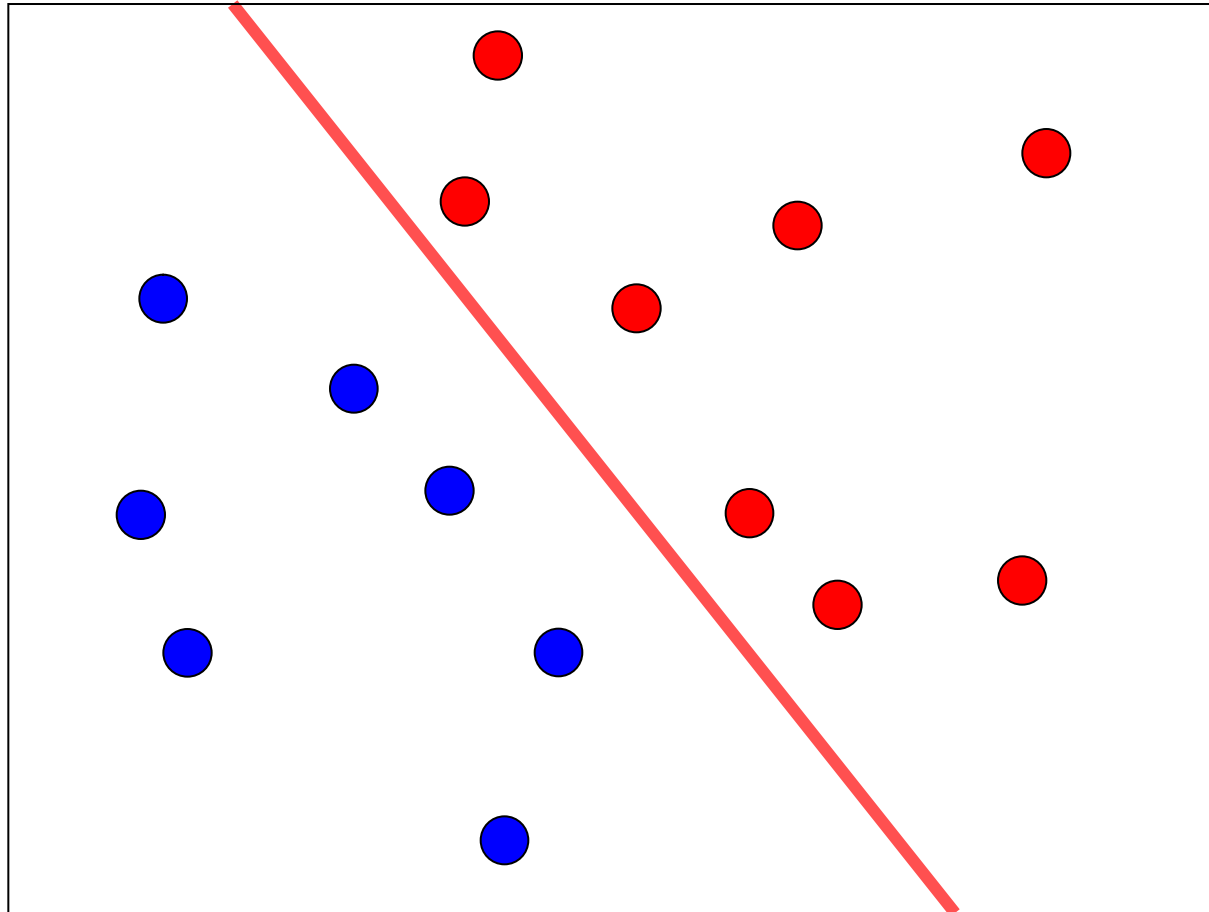
Linear separation

.....



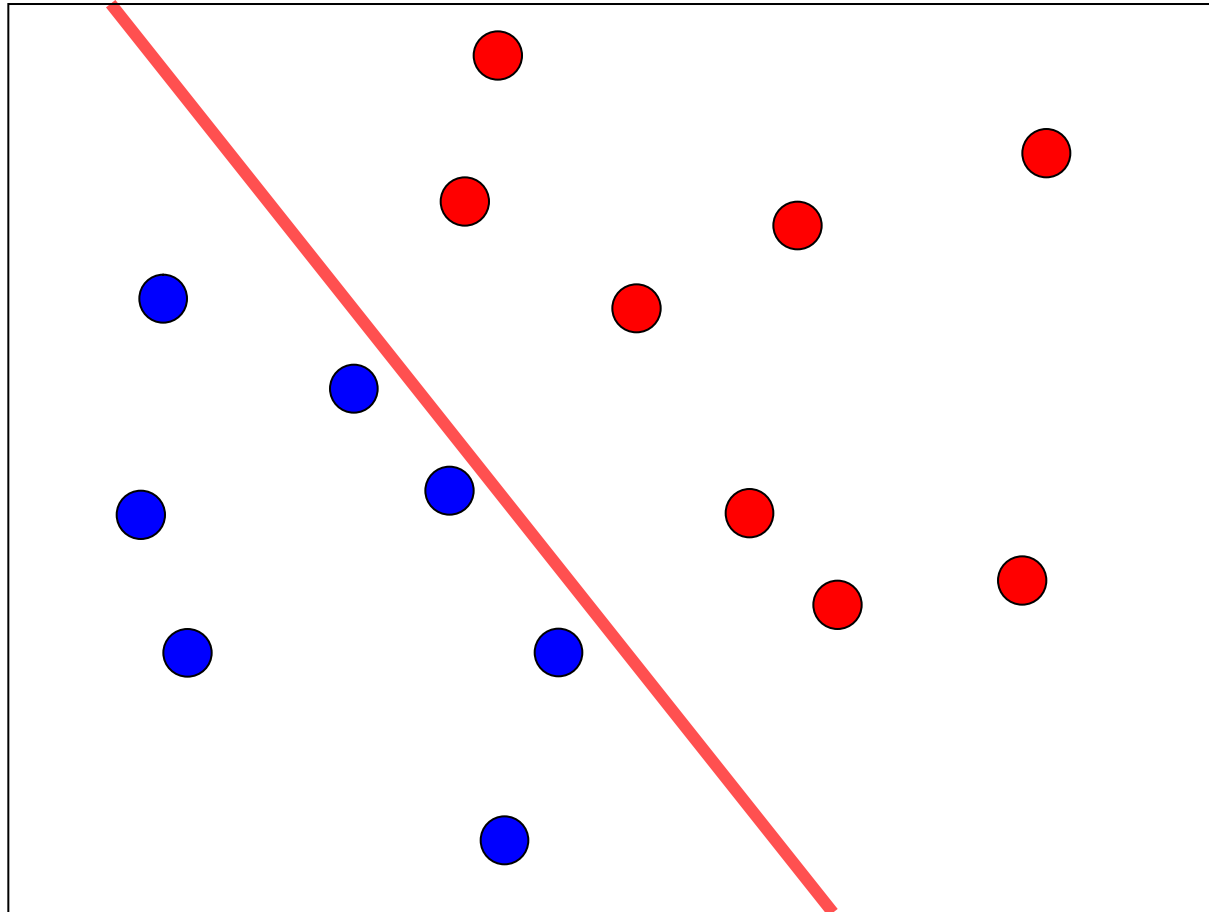
Linear separation

- Separating line #1



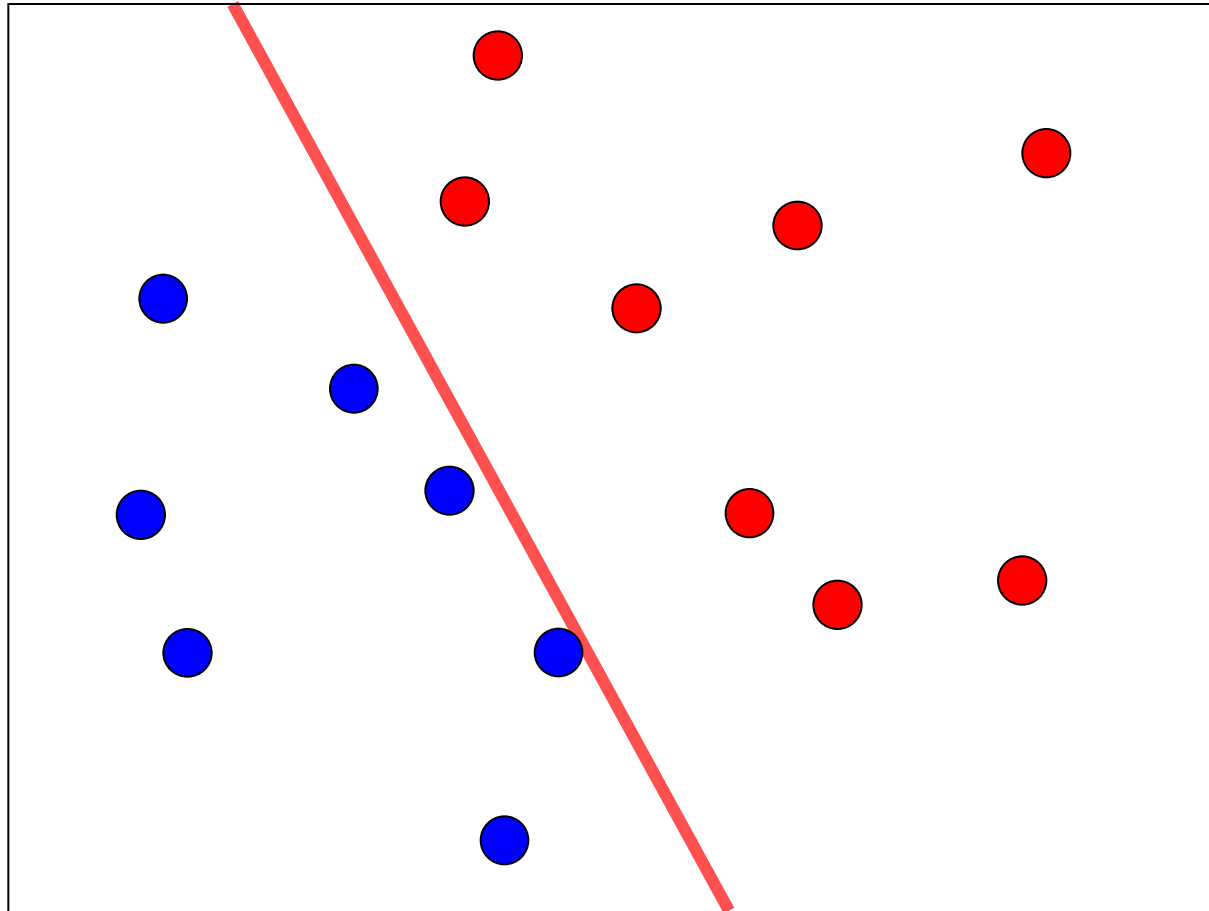
Linear separation

- Separating line #2



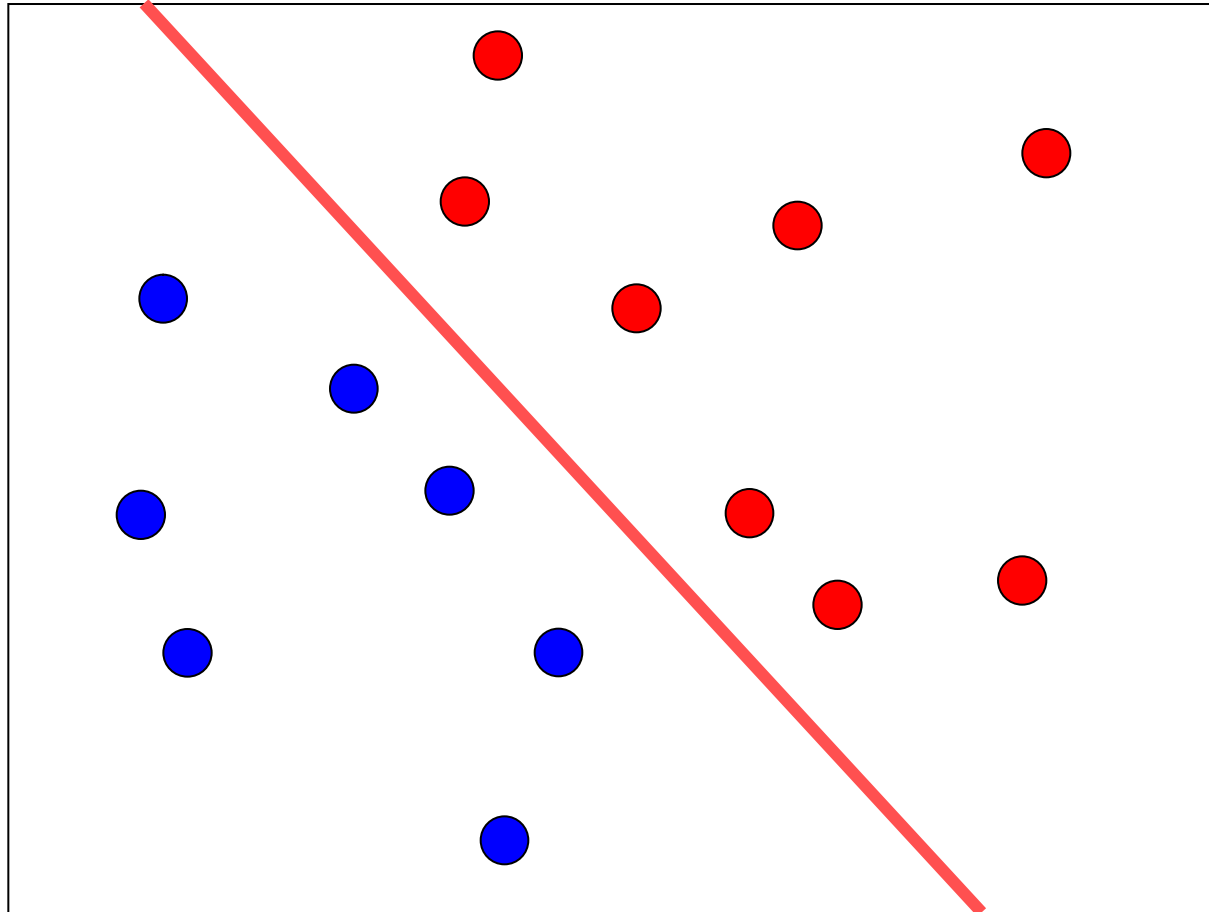
Linear separation

- Separating line #3



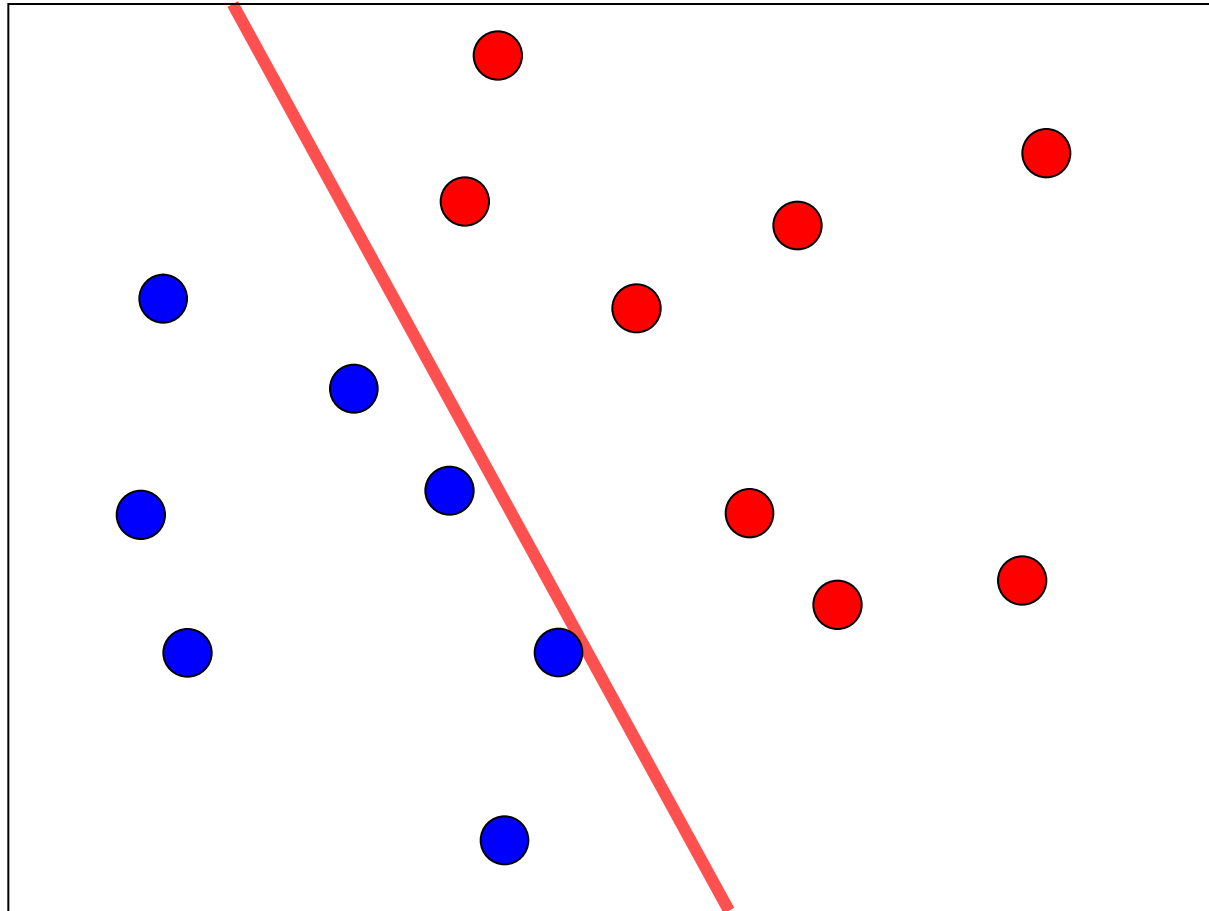
Linear separation

- Separating line #4



Linear separation

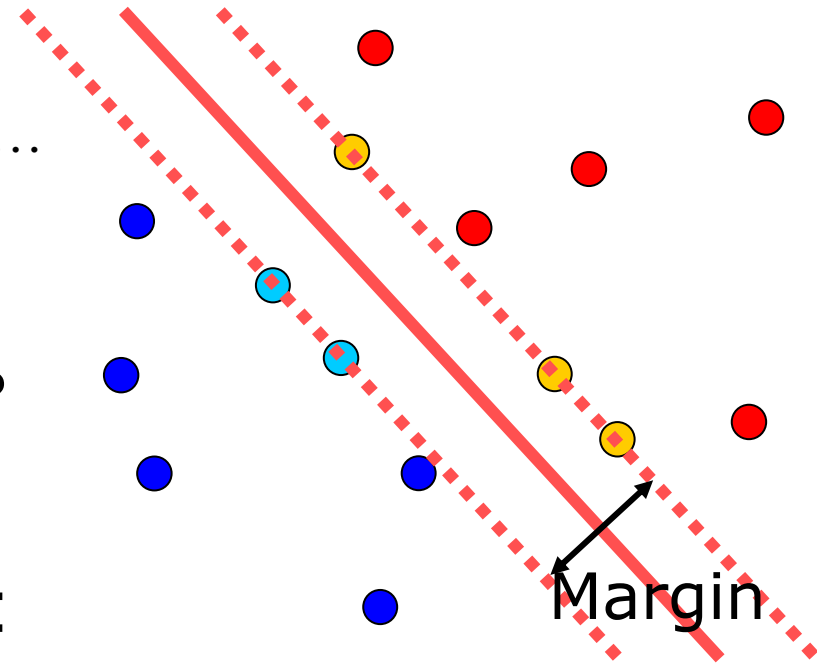
- Separating line #5



- *Which other classifier(s) use linear separation?*
- *What's the difference to SVMs?*
- Optimization
 - maximisation of margin separating items
- *Later: soft margin, kernels, ...*

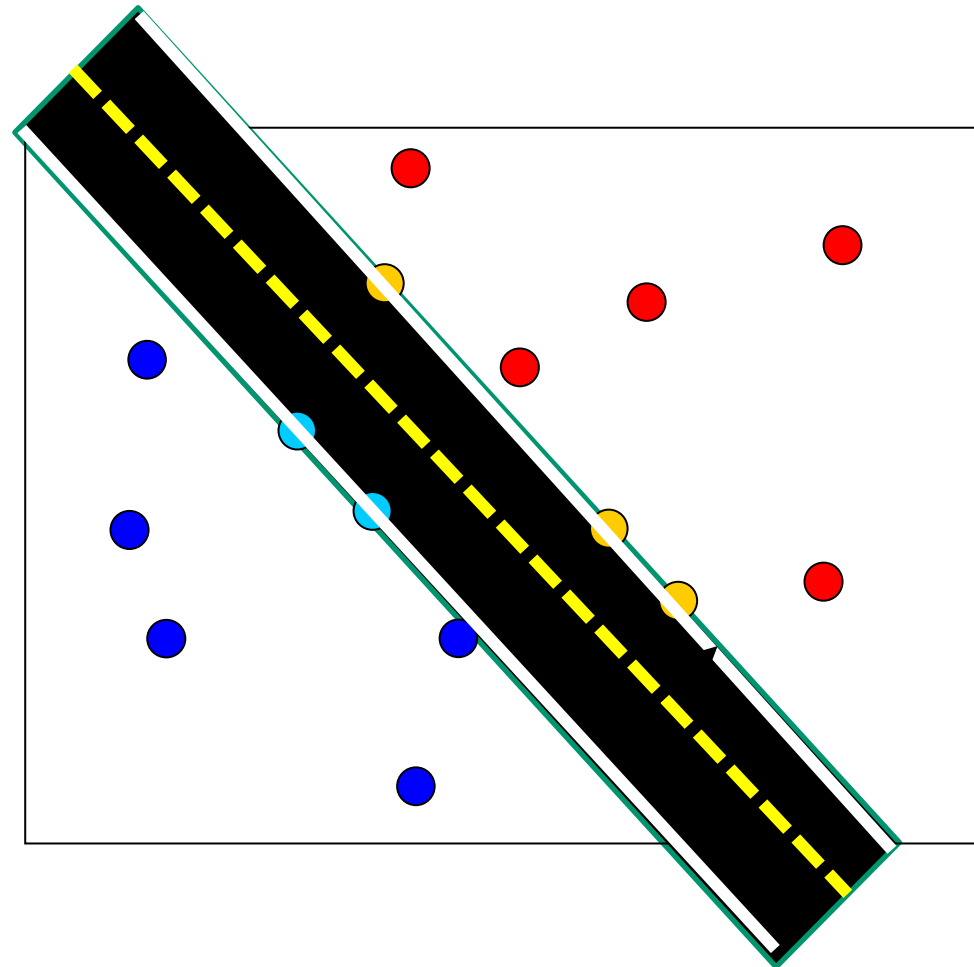
Largest Margin

- All separations are valid
 - Which separation is the best?
- **Margin** of a linear classifier:
width that boundary could be increased to
 - before hitting any data-point
- **Support Vectors** are those data-points that the margin pushes up against
- *What's the minimum number of support vectors?*



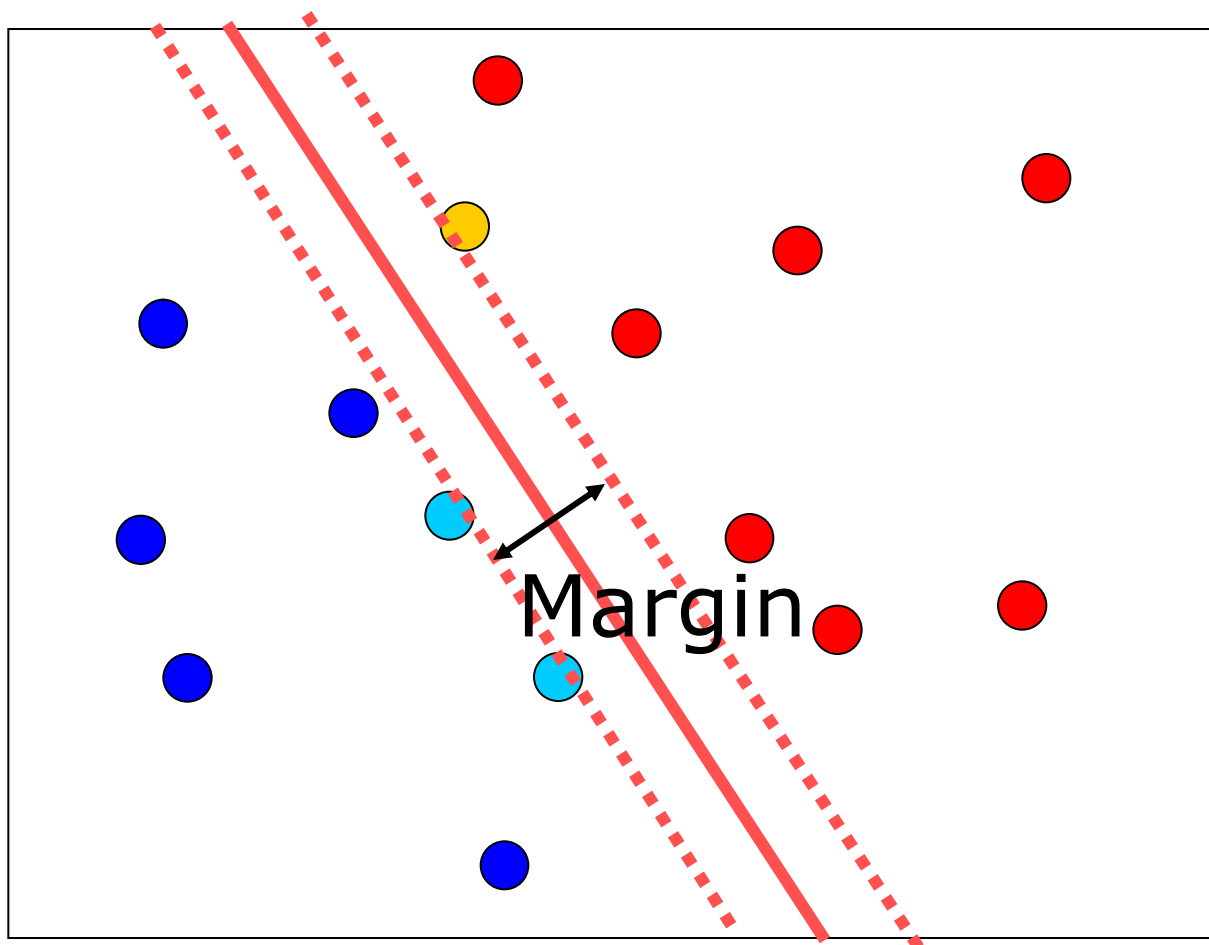
Margin – a simple analogy

- Margin is a road
 - Decision boundary is the *median separator*
 - Support vectors are *reflector posts*
 - Margin = width of the road

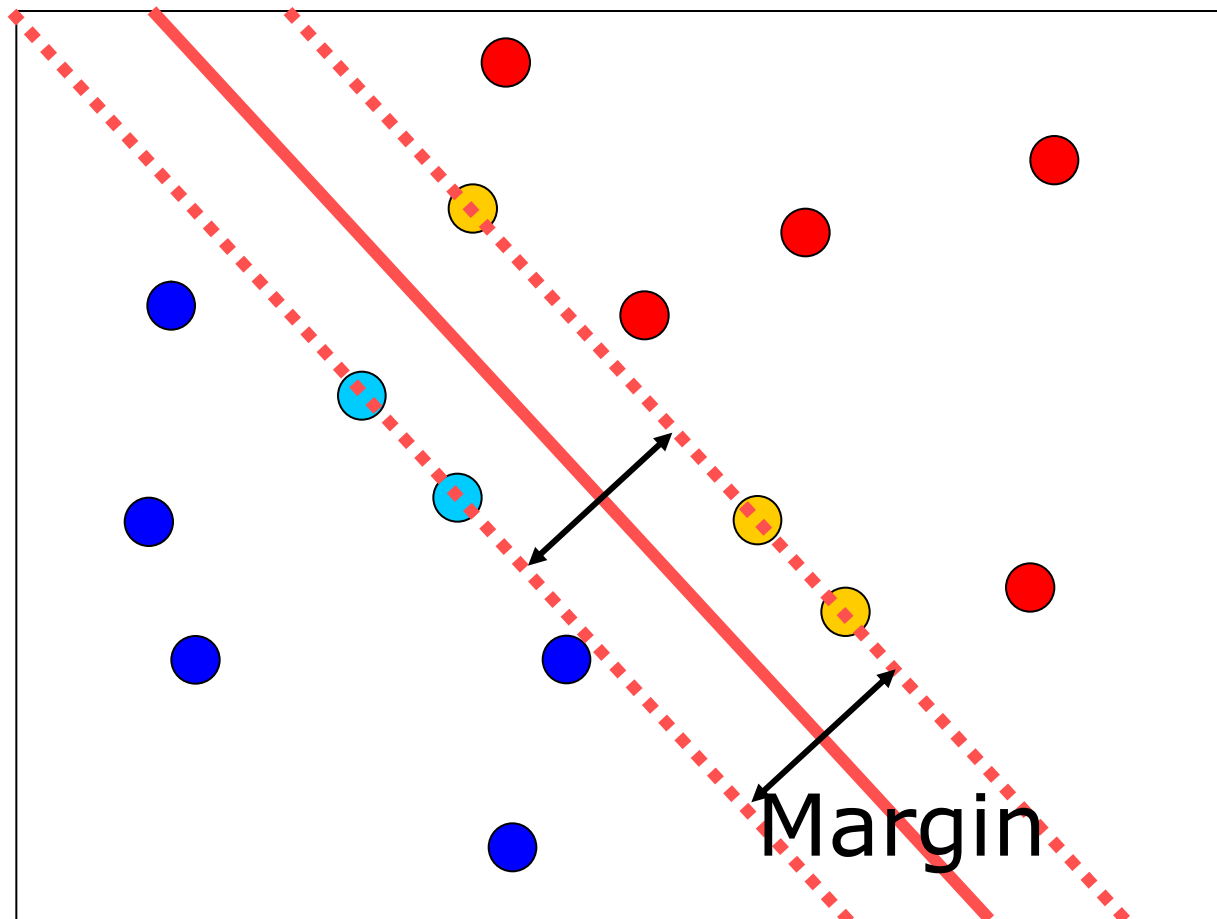


Largest Margin

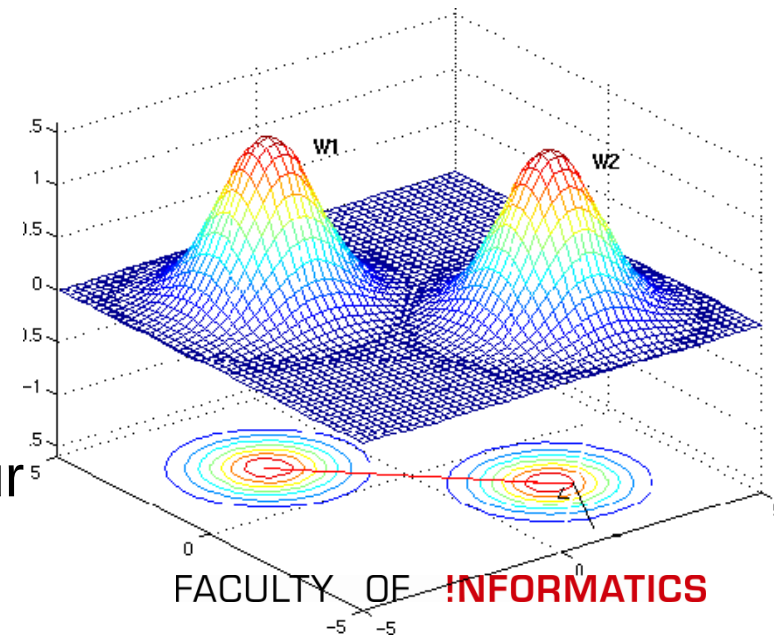
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Largest Margin

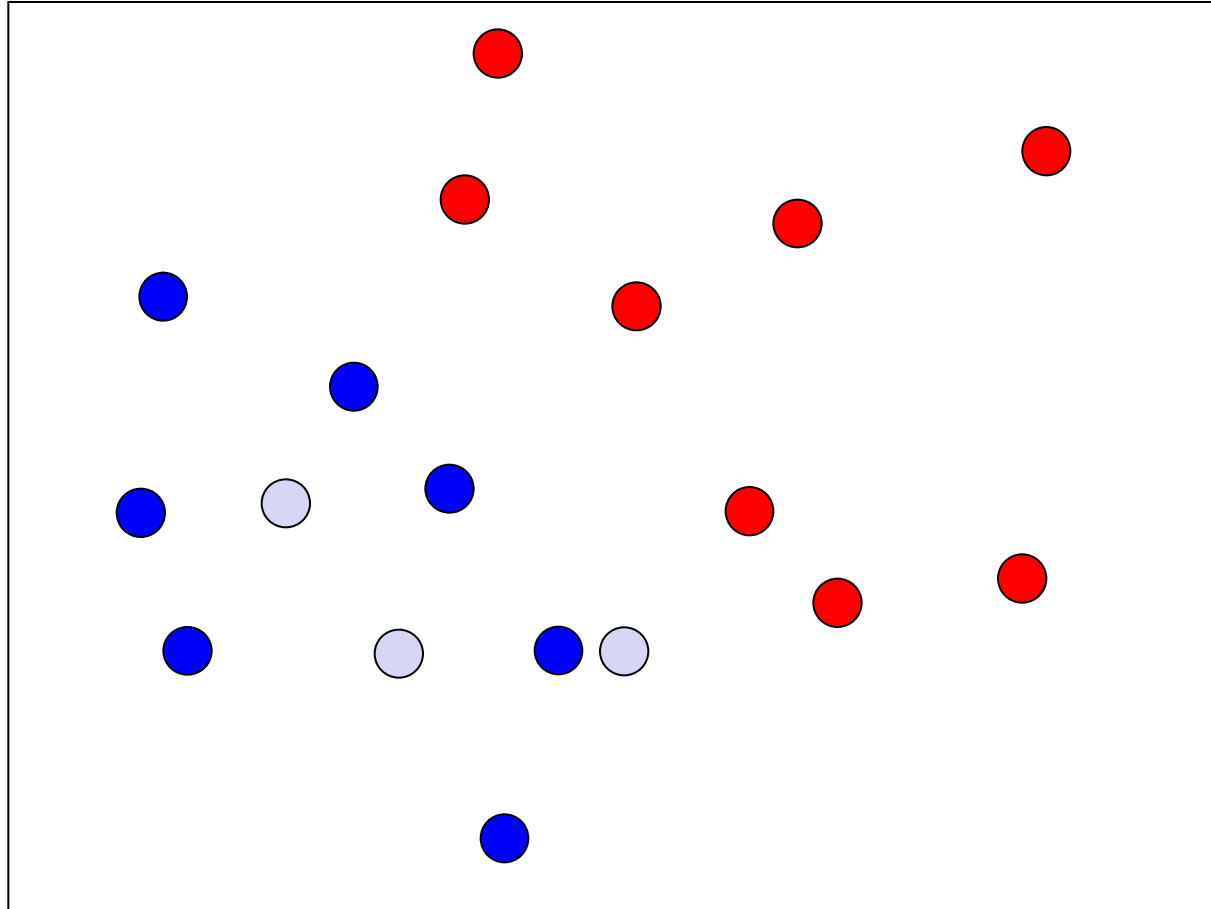


- *Which separation/margin is the best?*
- Claim: **bigger margin is better**
- Intuitive illustration example
 - Assumption in previous dataset: samples are drawn from probability distribution
 - E.g. two Gaussians with different means (& variances)
 - Now, draw more samples from these distributions to increase our data set (training/testing)



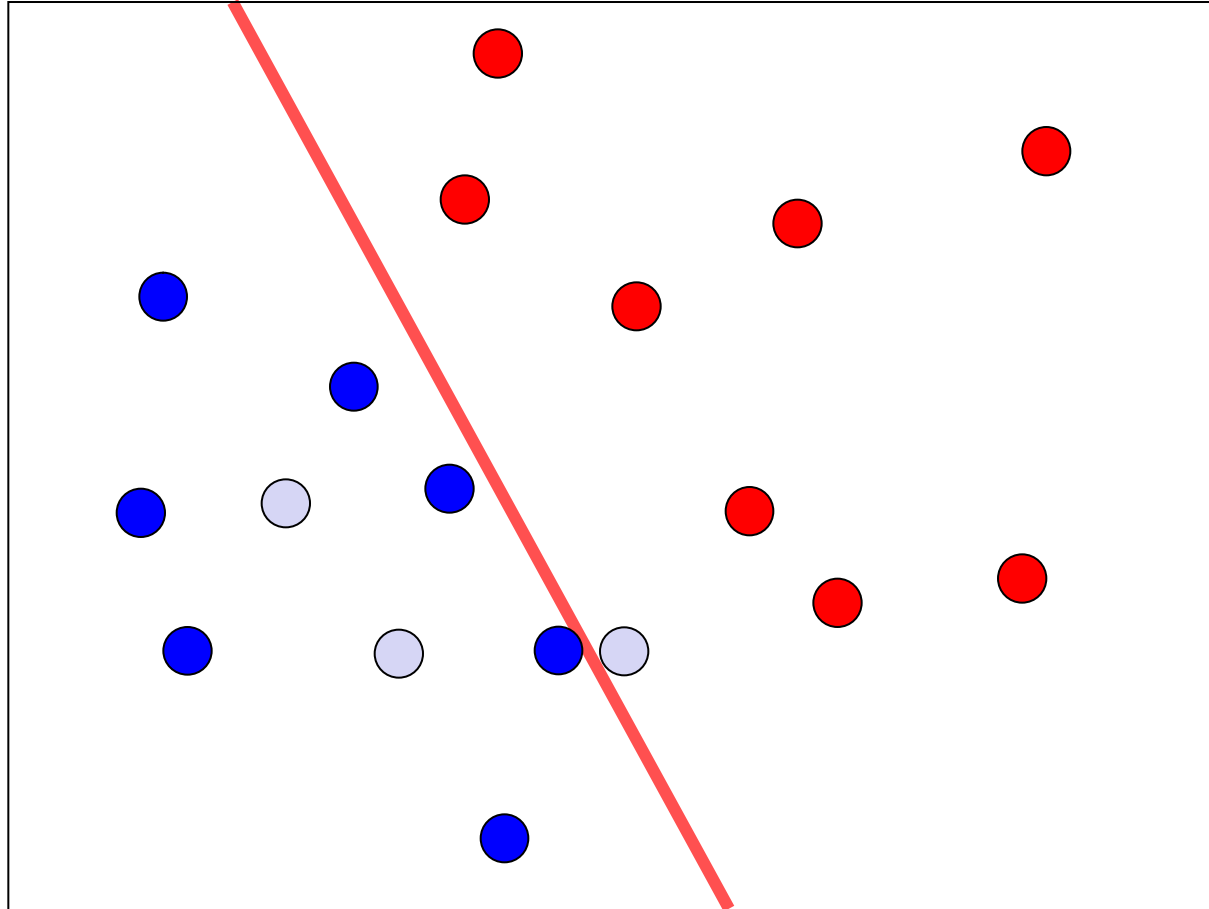
Linear separation

- Draw more samples from the distribution



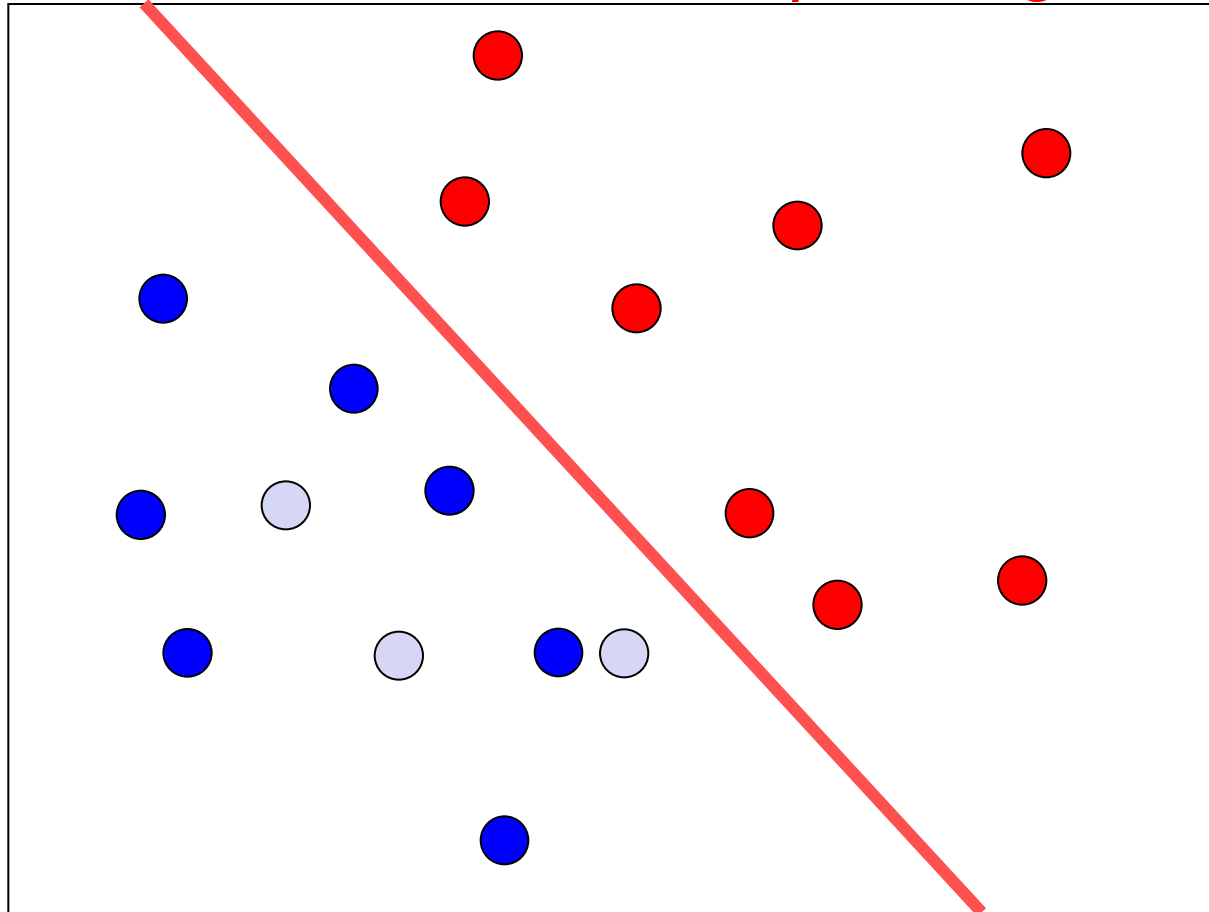
Linear separation

- Draw more samples from the distribution
 → *Line #3 not separating anymore*



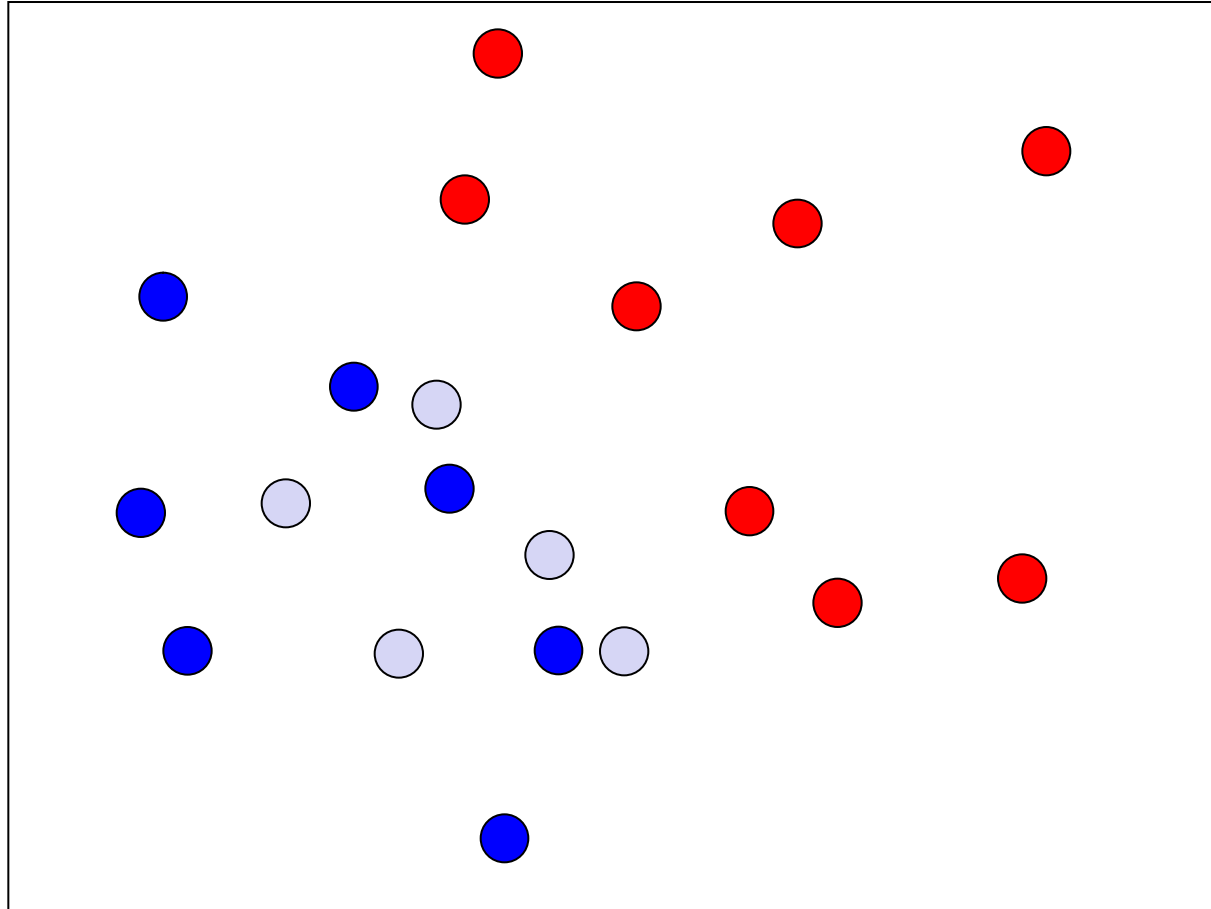
Linear separation

- Draw more samples from the distribution
 → *Line #4 still separating*



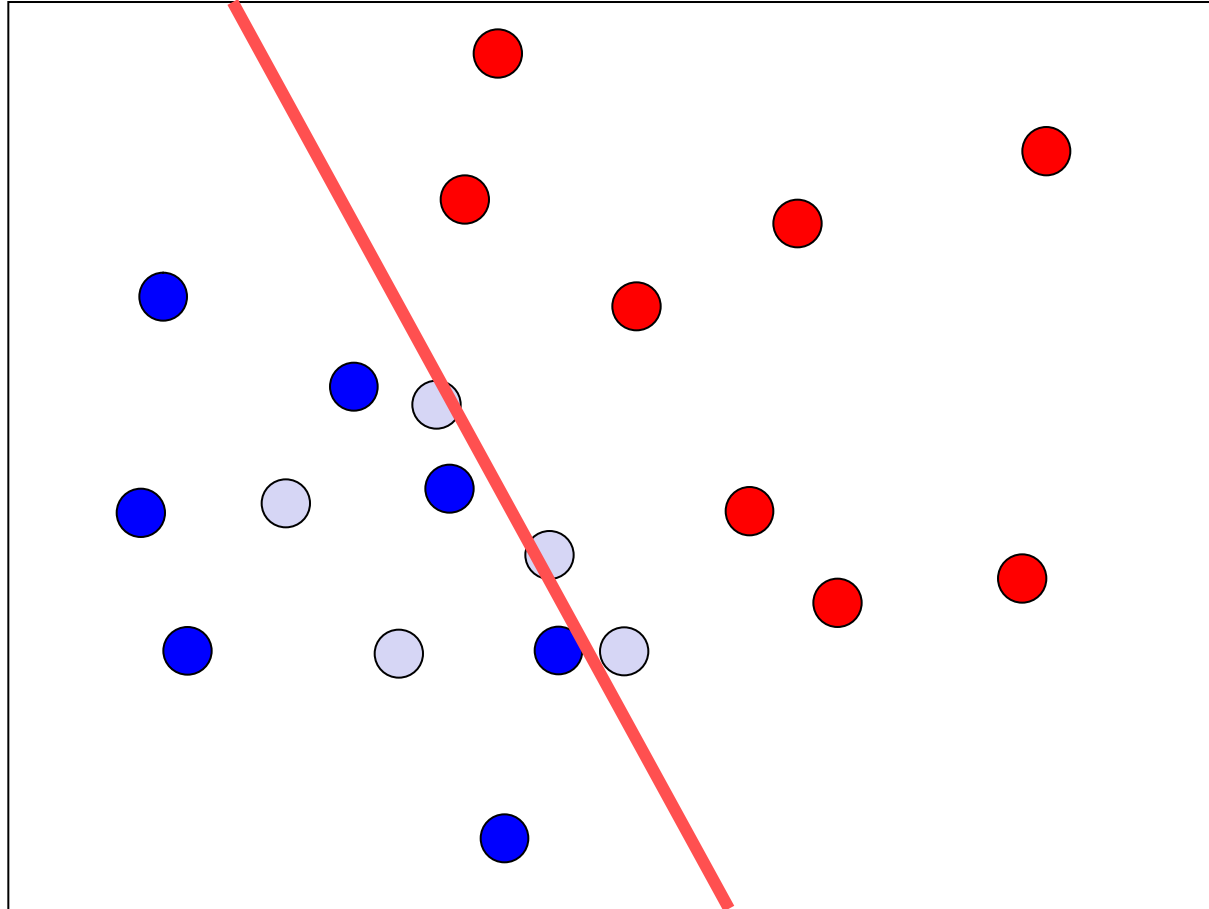
Linear separation

- Draw even more samples from the distribution



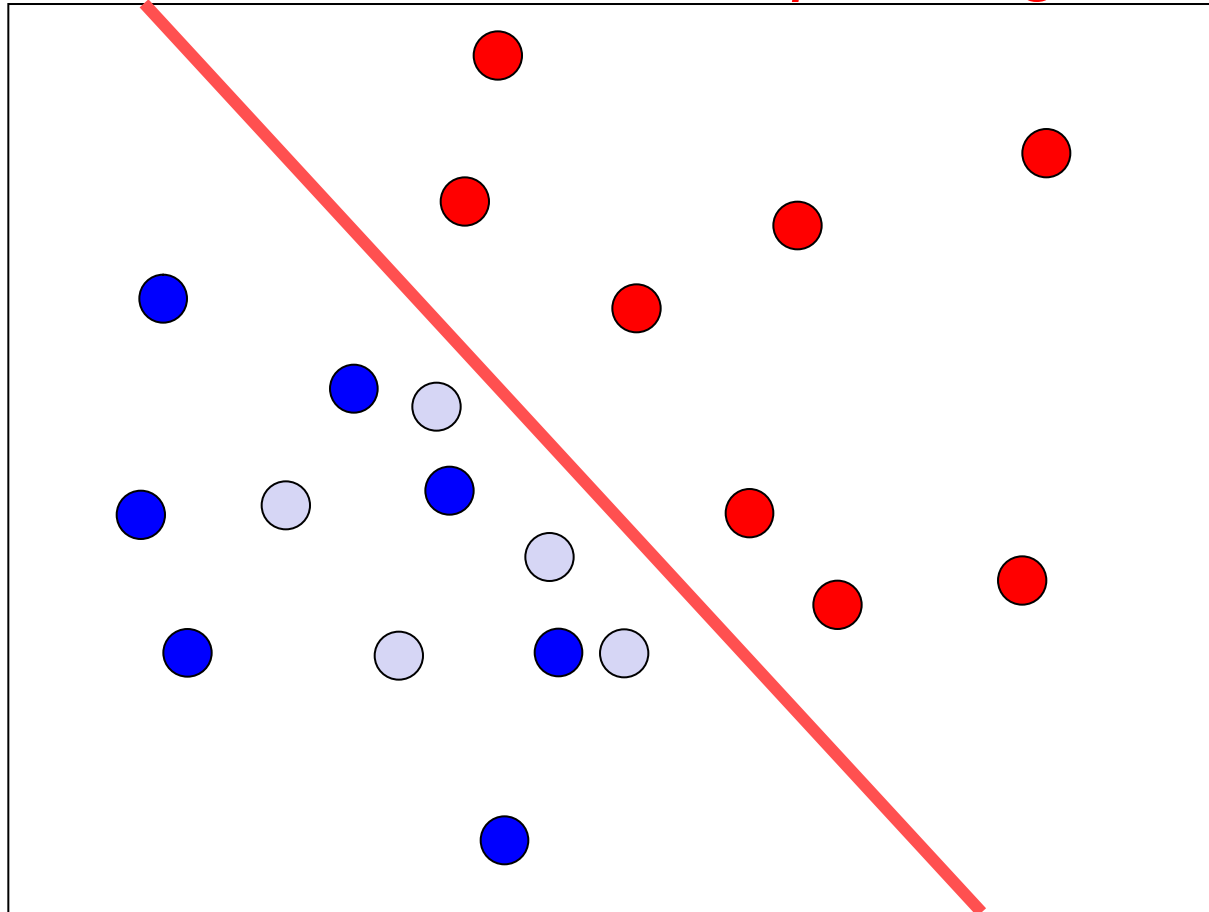
Linear separation

- Draw even more samples from the distribution
 → *Line #3 separates even worse*

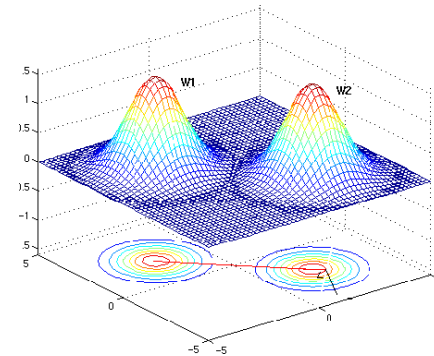


Linear separation

- Draw even more samples from the distribution
 → *Line #4 still separating*



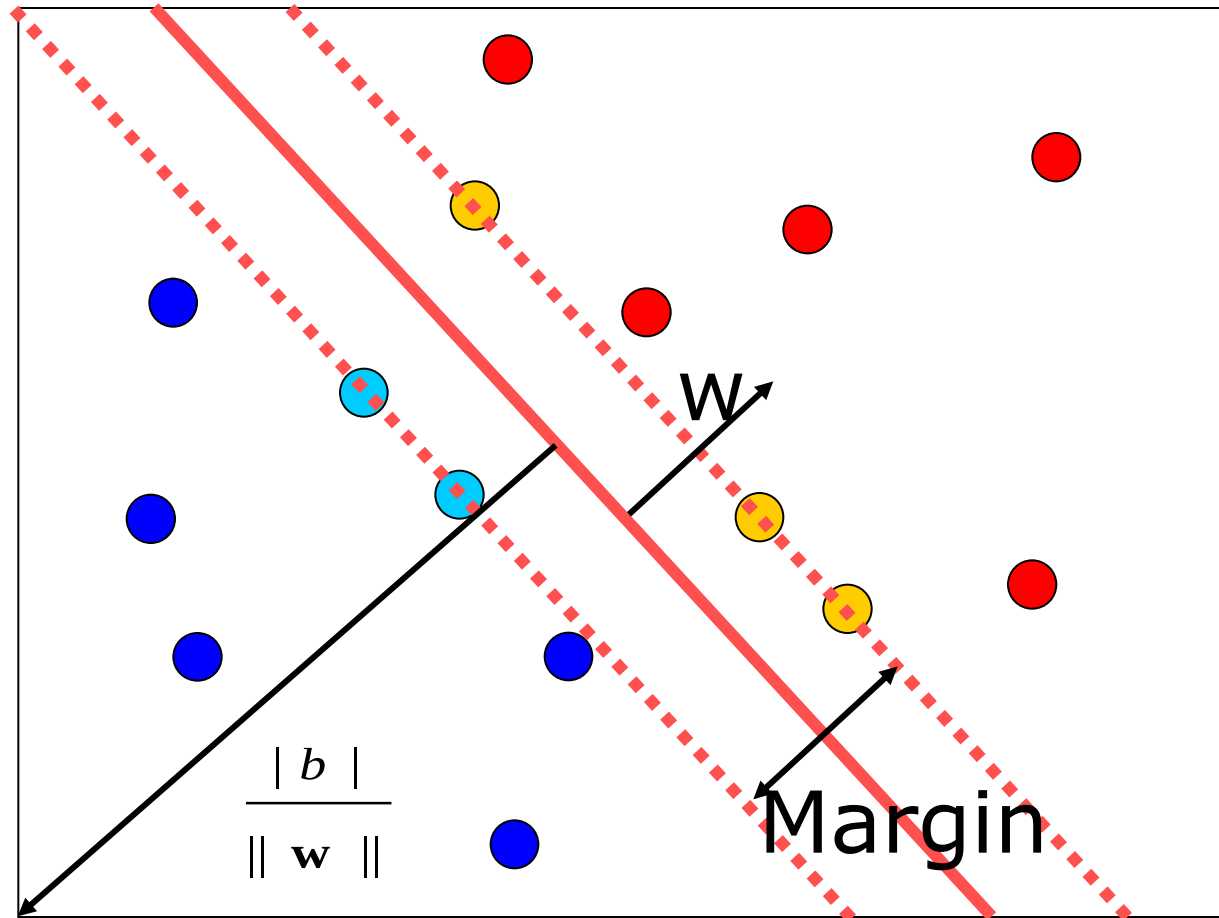
- *Which separation/margin is the best?*
- Claim: **bigger margin is better**
- Intuitive demonstration
 - The bigger the margin $\rightarrow$ the better is the separating plane fitting to slightly different data
 - I.e. less “overfitting”, more generalization
- *Later: how to optimise the margin*
 - Using Lagrange multipliers, quadratic programming, ...



Finding largest margin

- Given set of points $\mathbf{x}_i$ (n-dimensional)
 - belonging to binary class: $y_i = 0$ or $y_i = 1$
- Hyperplane dividing points
 - Satisfying: $\mathbf{w} \cdot \mathbf{x}_i + b = 0$
 - $\mathbf{w}$ is normal to the hyperplane (weight vector)
 - b determines offset from origin (intercept/bias)
 - $\frac{|b|}{\|\mathbf{w}\|}$ perpendicular (normal) distance from origin
 - In general: *infinite number of possibilities*

Finding largest margin



Origin

Excursus: Finding largest margin

- Recap *Perceptron*: Activation $a = \sum_{i=1}^n w_i x_i$
 Classification $f(x) = \begin{cases} 1 & \text{if } a \geq 0 \\ 0 & \text{if } a < 0 \end{cases}$
- SVM: require **known** points to be $\geq +1$ / < -1
- Classification function: $f(\vec{x}) = \text{sign}(\vec{w}^T \vec{x} + b)$
 – Alternative formulation: $f(x) = \begin{cases} 1 & \text{if } a \geq 1 \\ 0 & \text{if } a < -1 \end{cases}$
- Hyperplane dividing points in -1 & +1
 – Satisfying: $\vec{w} \cdot \vec{x}_i + b = 0$
 with $\vec{a} \cdot \vec{b} = \vec{a}^T \vec{b}$

$$\Rightarrow \vec{w}^T \vec{x}_i = -b$$

Excursus: Finding largest margin

- Task: find hyperplane with *largest margin*
 - **minimise** $\| \mathbf{w} \|$ (check additional material why)
- Given the *constraints*

$$\mathbf{w} \cdot \mathbf{x}_i + b \geq +1 \quad \text{for } \mathbf{y}_i = +1 \quad (1)$$

$$\mathbf{w} \cdot \mathbf{x}_i + b \leq -1 \quad \text{for } \mathbf{y}_i = -1 \quad (2)$$
- Constraints (1) and (2) combined:

$$y_i (\mathbf{w} \cdot \mathbf{x}_i + b) - 1 \geq 0 \text{ for all } i$$
 - *if* $y_i = +1$: $\mathbf{w} \cdot \mathbf{x}_i + b - 1 \geq 0 \rightarrow \mathbf{w} \cdot \mathbf{x}_i + b \geq 1$
 - *if* $y_i = -1$: $(-1) (\mathbf{w} \cdot \mathbf{x}_i + b) - 1 \geq 0 \rightarrow (-1) (\mathbf{w} \cdot \mathbf{x}_i + b) \geq 1$
 $\rightarrow \mathbf{w} \cdot \mathbf{x}_i + b \leq -1$

- *How to find the largest margin?*
- Analytical solution, e.g. via quadratic programming and Lagrange multipliers
 - SMO (Sequential Minimal Optimisation) algorithm
 - 1998 by John Platt (Microsoft Research)
 - Breaks the optimisation in a series of sub-problems
 - Used e.g. in WEKA, LibSVM, ...
 - *More on that later*
- Heuristic Optimisation
 - Sub-gradient descent, coordinate descent
 - Sub-gradient when there are many training examples
 - Coordinate descent when the dimensionality is high
- *Difference in the two approaches?*

- Recurrent Neural Networks
- Support Vector Machines: Soft Margin
- Statistical Significance Testing
- Feature Selection
- Challenge: Reproducibility
- Summary
- Advertisements: ML lectures, internships, thesis,

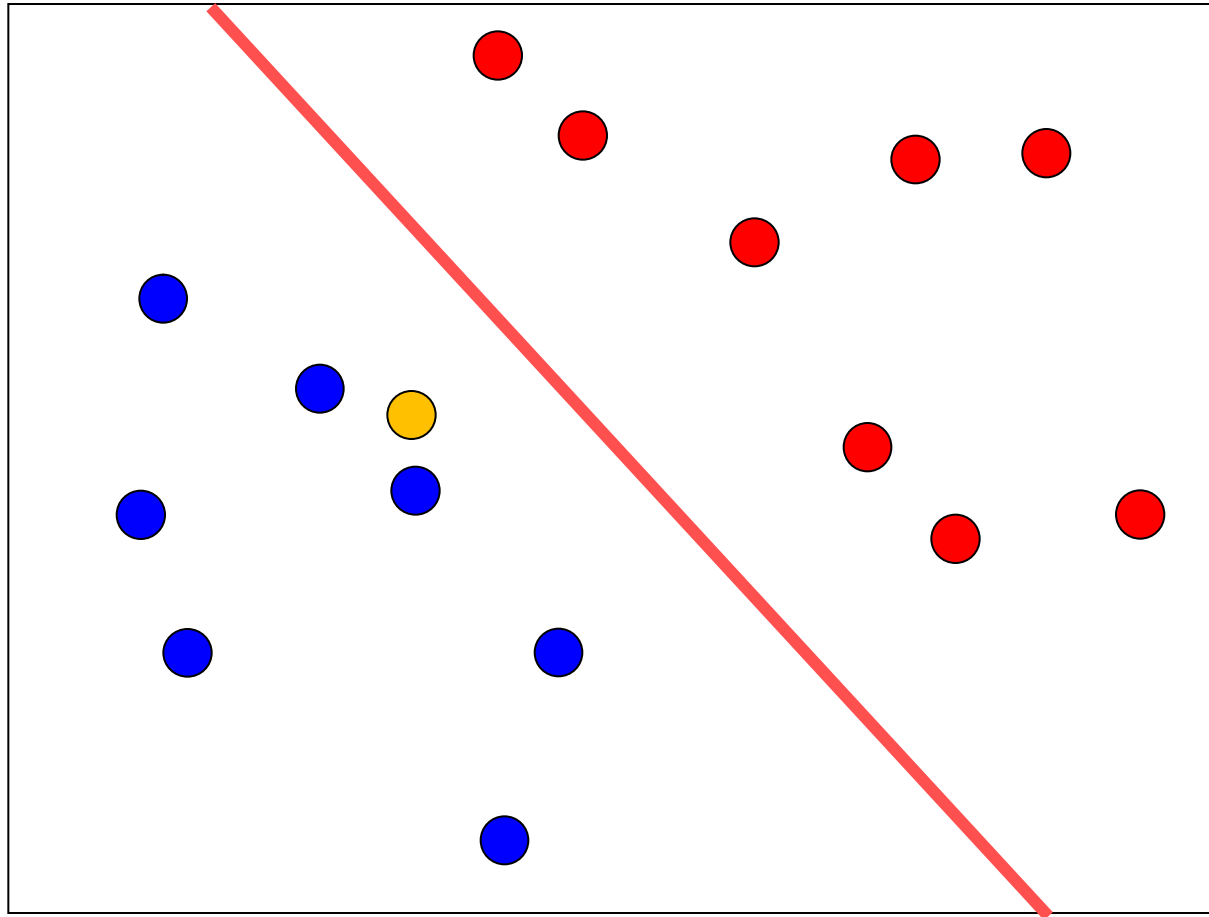
...

- SVMs optimise the decision boundary ...
 - ... but still rely on ***linear separation*** !
1. Sometimes linear separation not possible
 2. Sometimes, linear separation would lead to a badly generalising model
 - When?



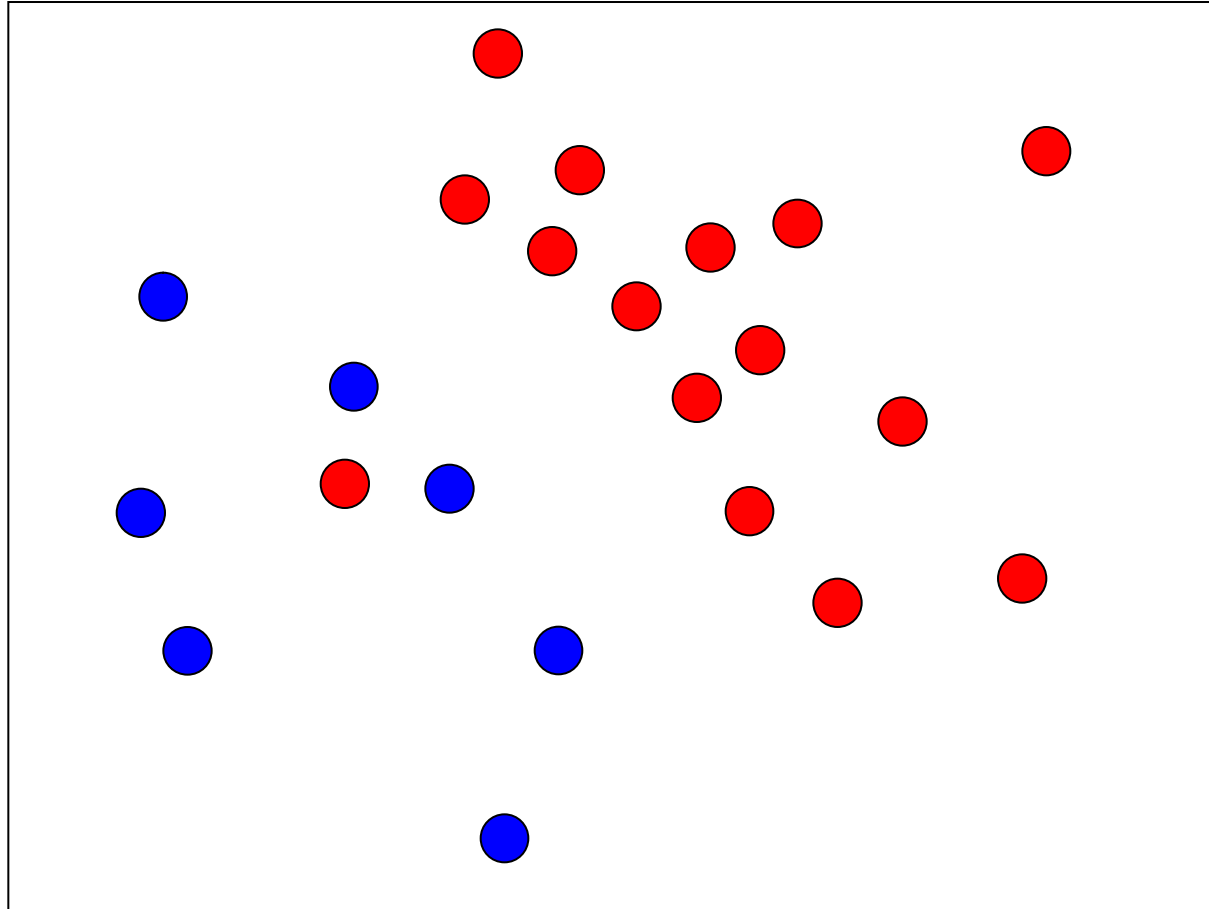
Bad generalisation

- *Wider margin might be better*



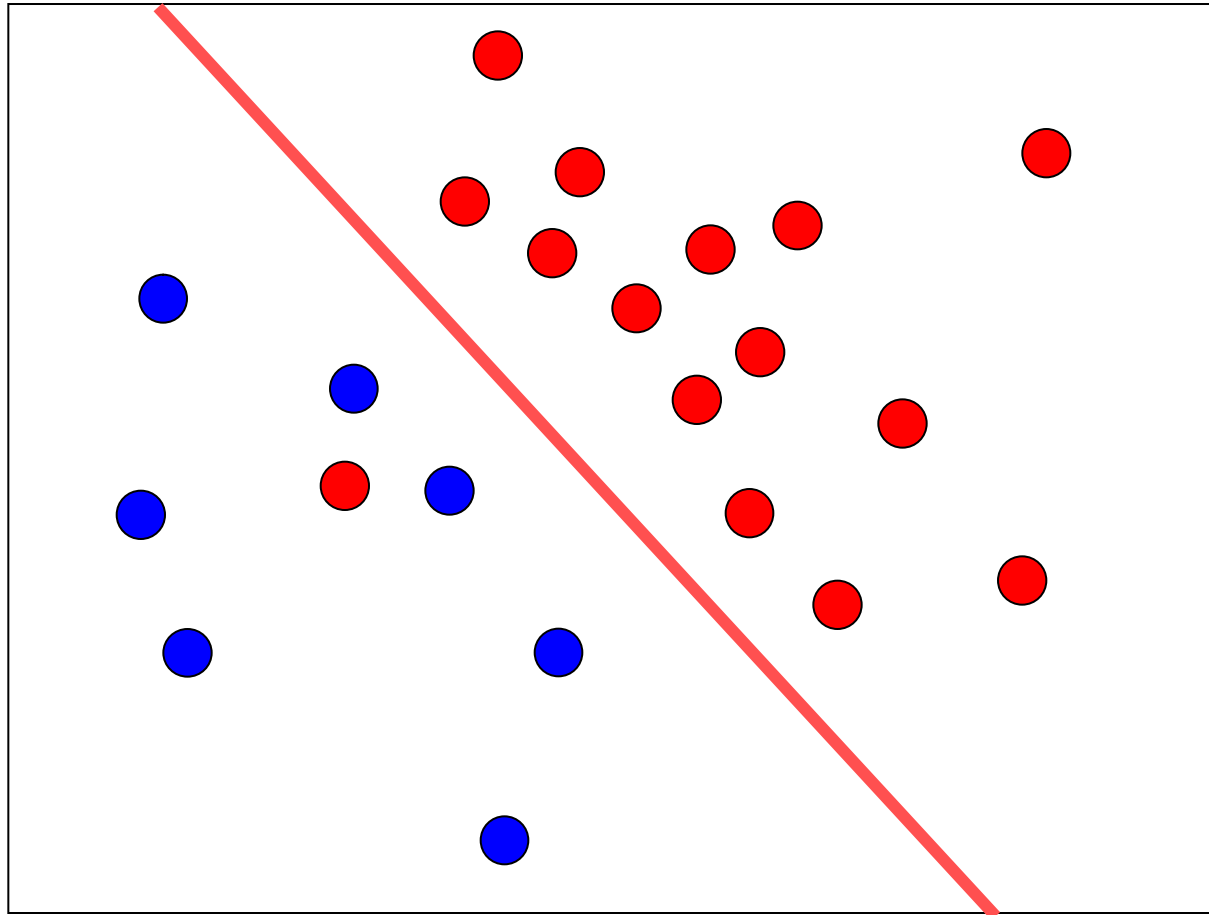
Linear separation not possible

.....



Linear separation not possible

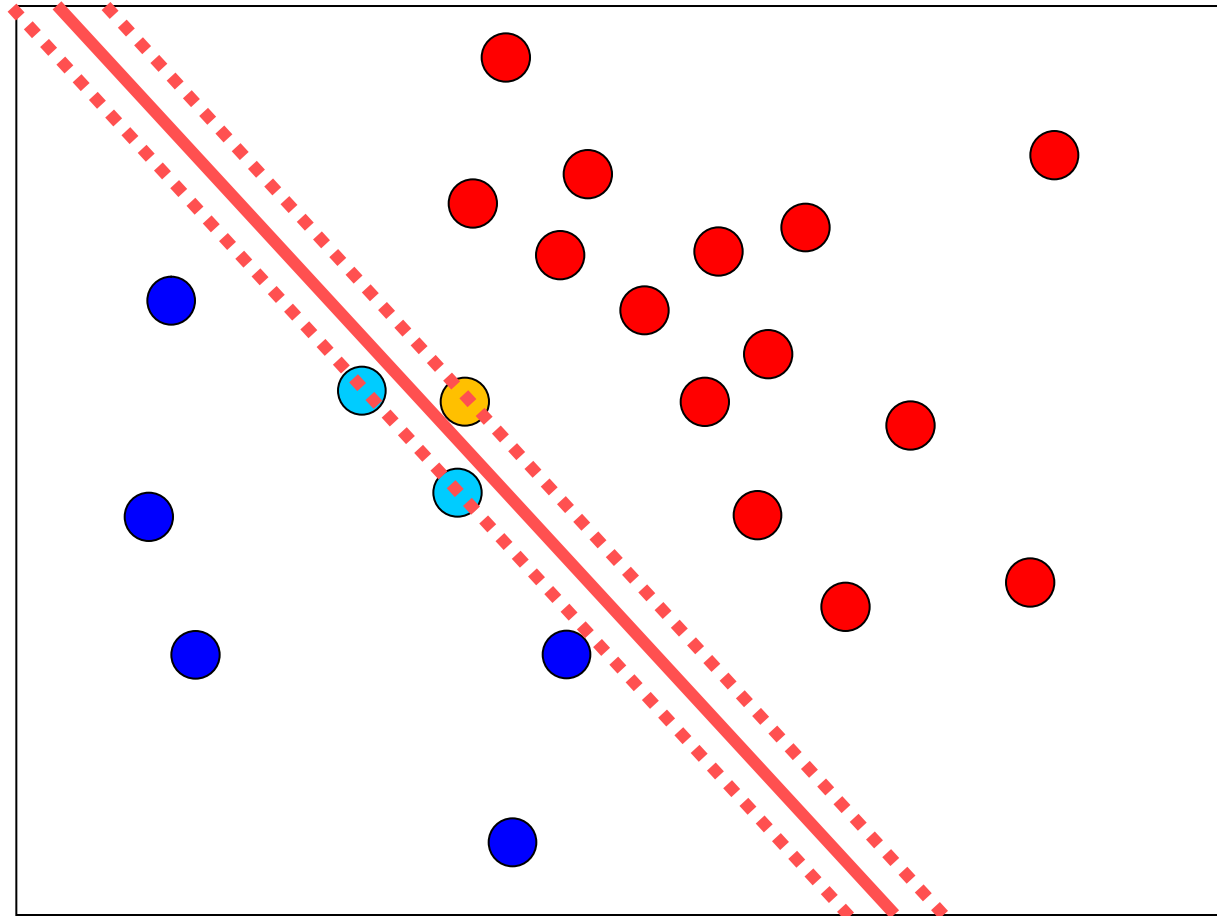
- *“Acceptable” hyperplane could still be found*



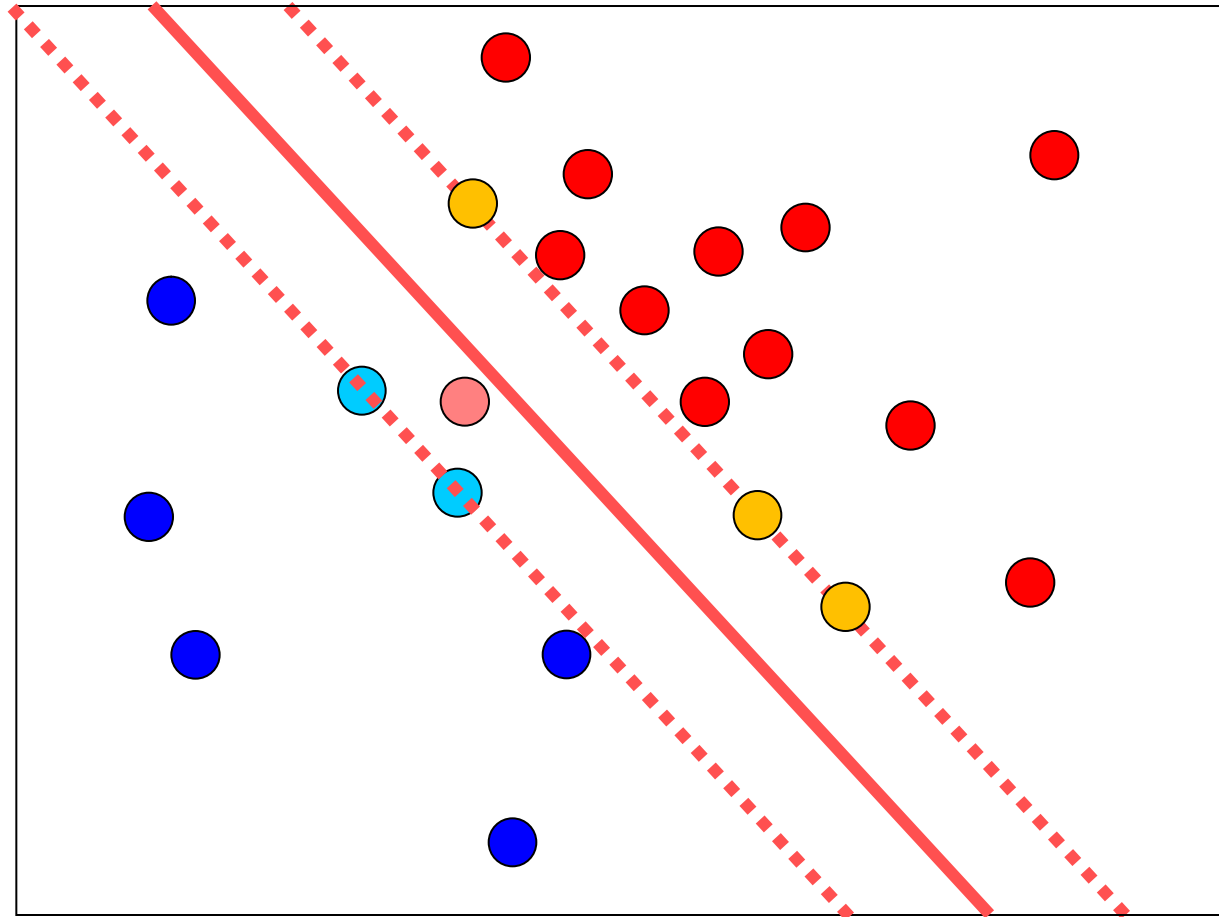
- Sometimes linear separation not possible, or
- Linear separation would lead to a badly generalising model

➔ Soft margin

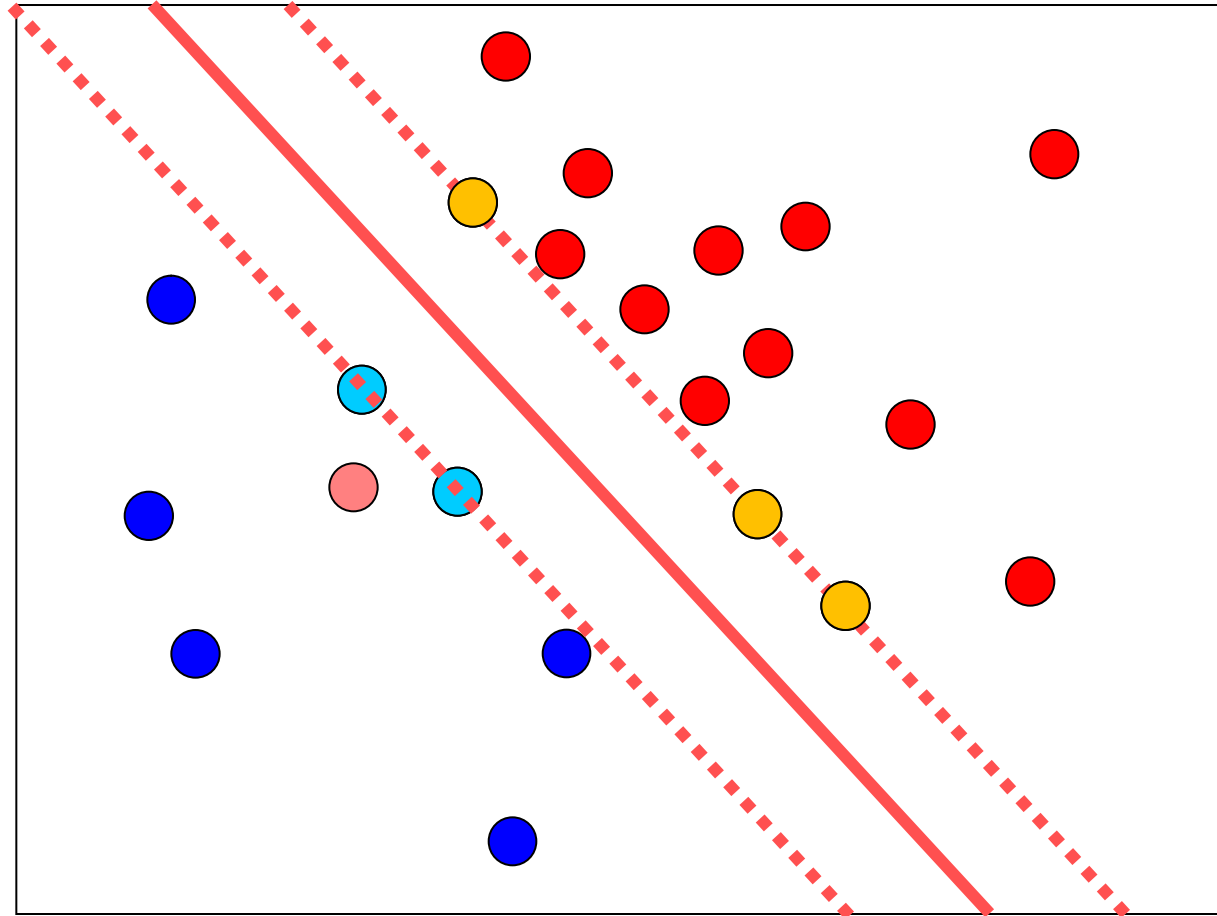
- Hyper plane that splits “as cleanly as possible/desirable”
- While maximising margin



Soft Margin: for larger margin

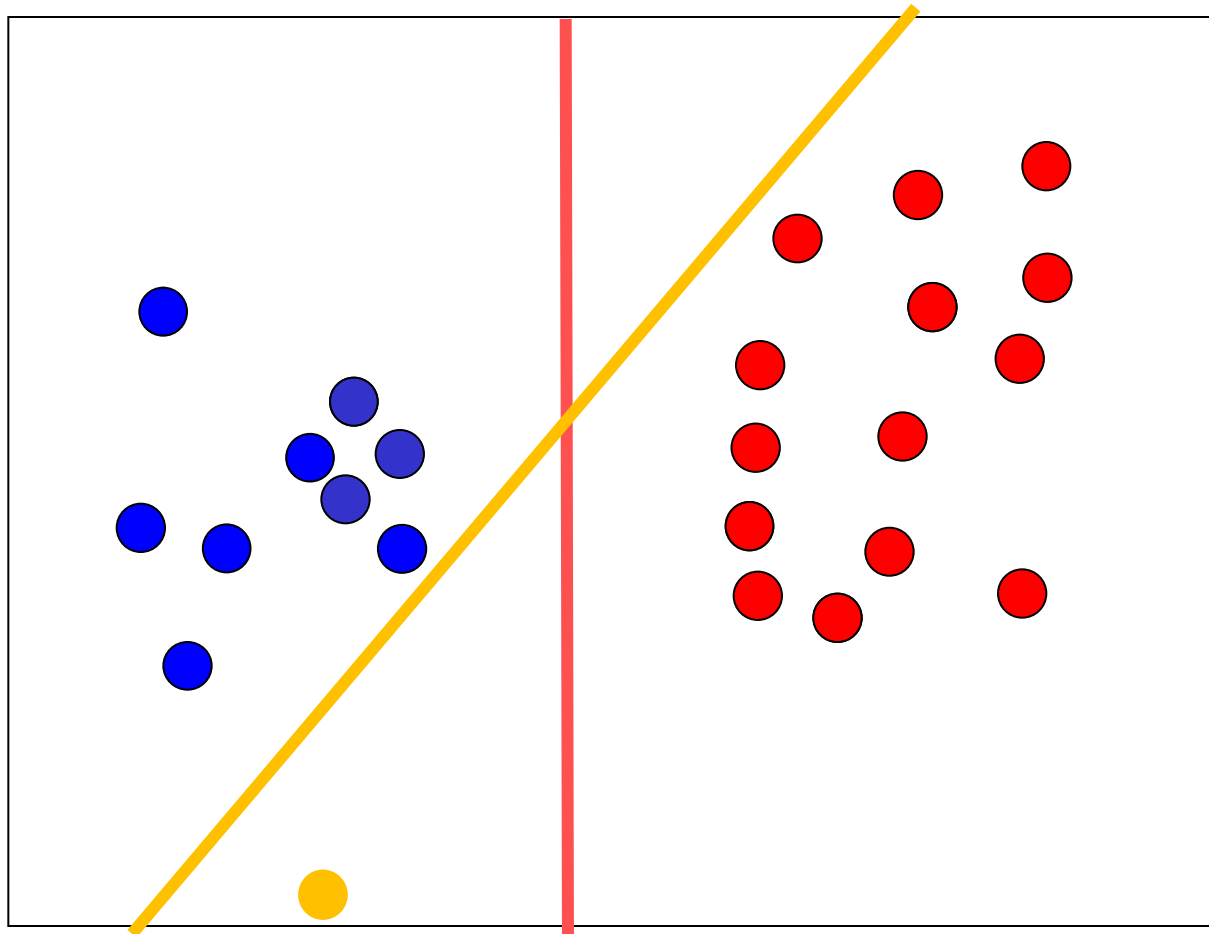


Soft Margin: to find separation



Soft Margin: larger margin with outliers

.....



- Introduction of slack variables
 - penalises misclassification
 - adapt constraint

$$y_i (\mathbf{w} \cdot \mathbf{x}_i - b) \geq 1 - \xi_i \quad \text{for all } i$$

- Penalise non-zero ξ_i

$$\rightarrow \min \quad \|\mathbf{w}\| + C \sum_{i=1}^n \xi_i$$

- How to solve it?
 - *Similar to “hard-margin” case*

- *Implications?*

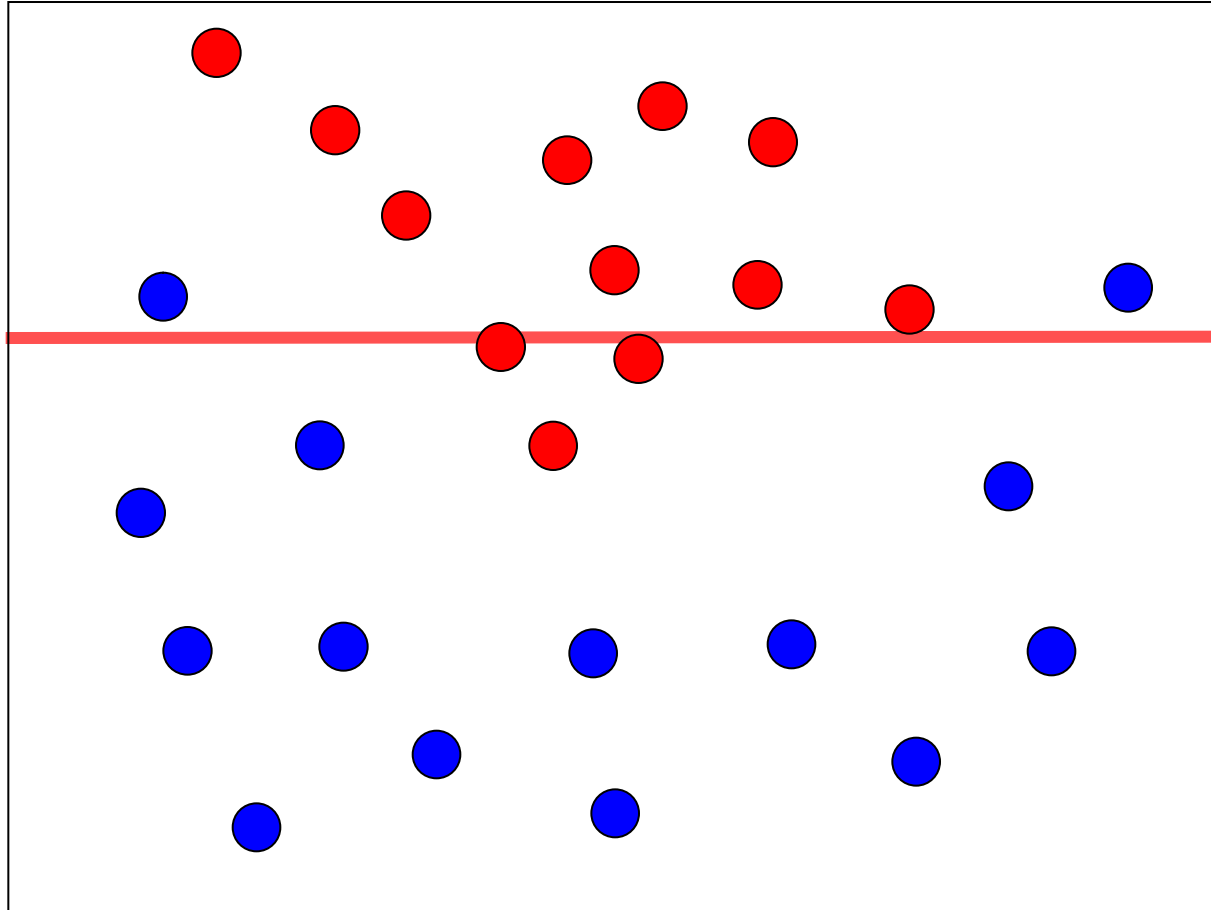
- Introduction of slack variables for optimisation problem
 - penalises misclassification
- *Implications ?*
 - Not necessarily 100% classification accuracy on training set
 - Even if linearly separable
- Optimisation: trade off between large margin and small error penalty (controlled by C)
 - Trade-off between fitting to training data and general model
- C becomes part of optimisation problem
 - Sometimes called the “complexity parameter”
 - Large C : penalising errors more

- Recurrent Neural Networks
- Support Vector Machines: Kernels
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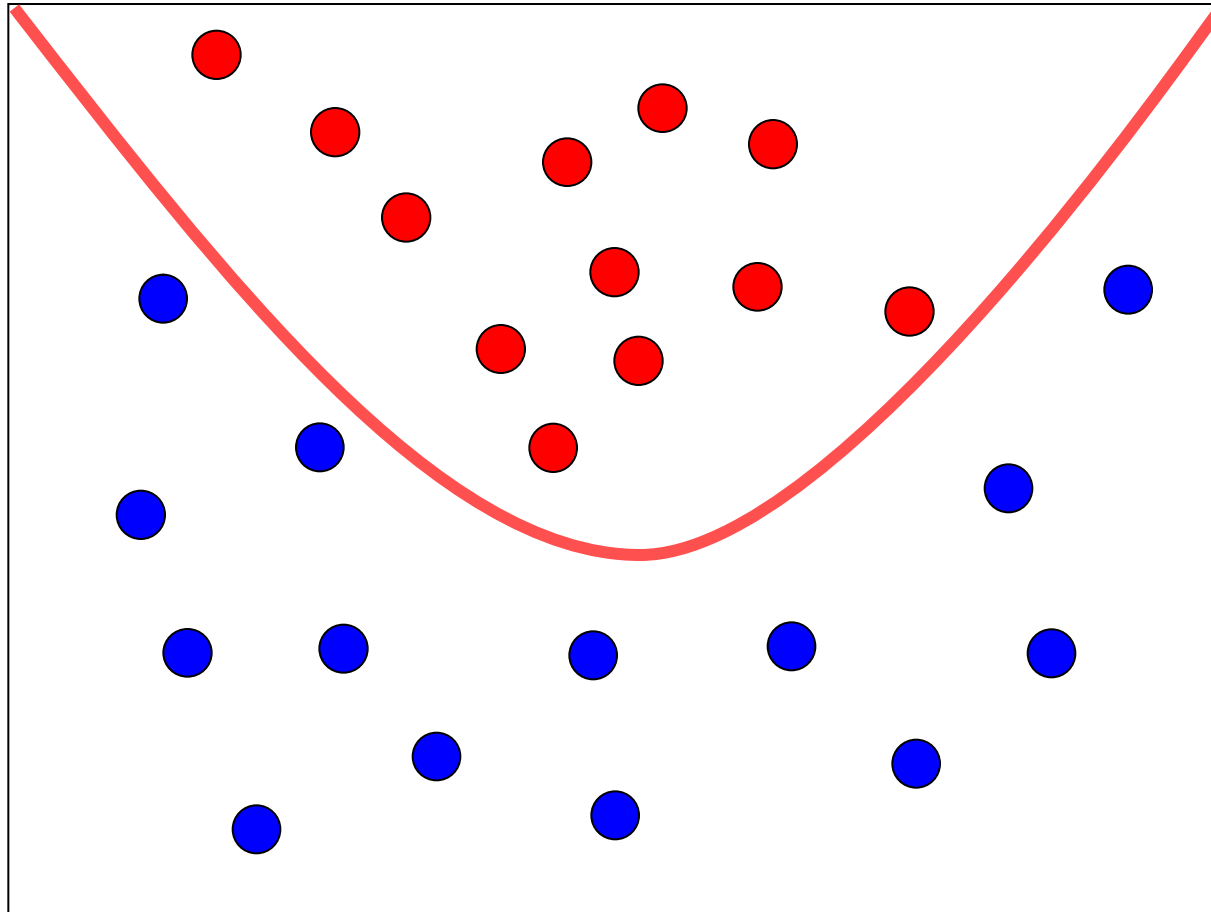
Non-linearly separable data

- “Acceptable” hyperplane can ***not*** be found

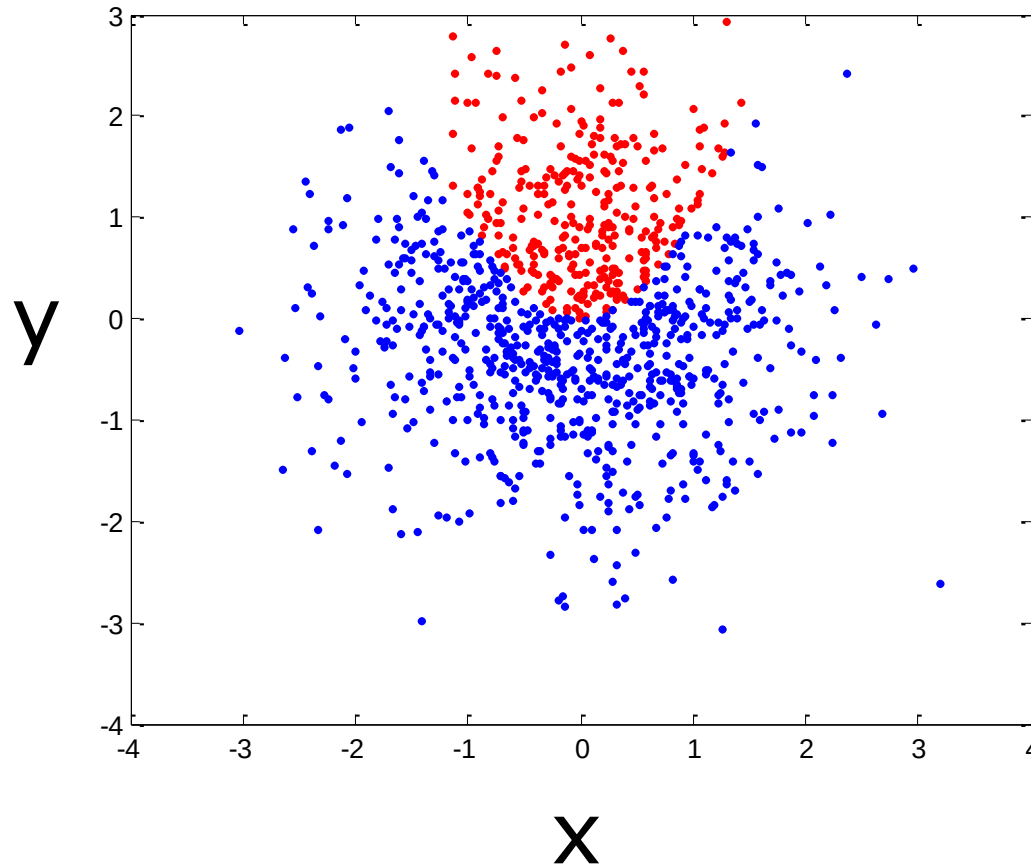


Non-linearly separable data

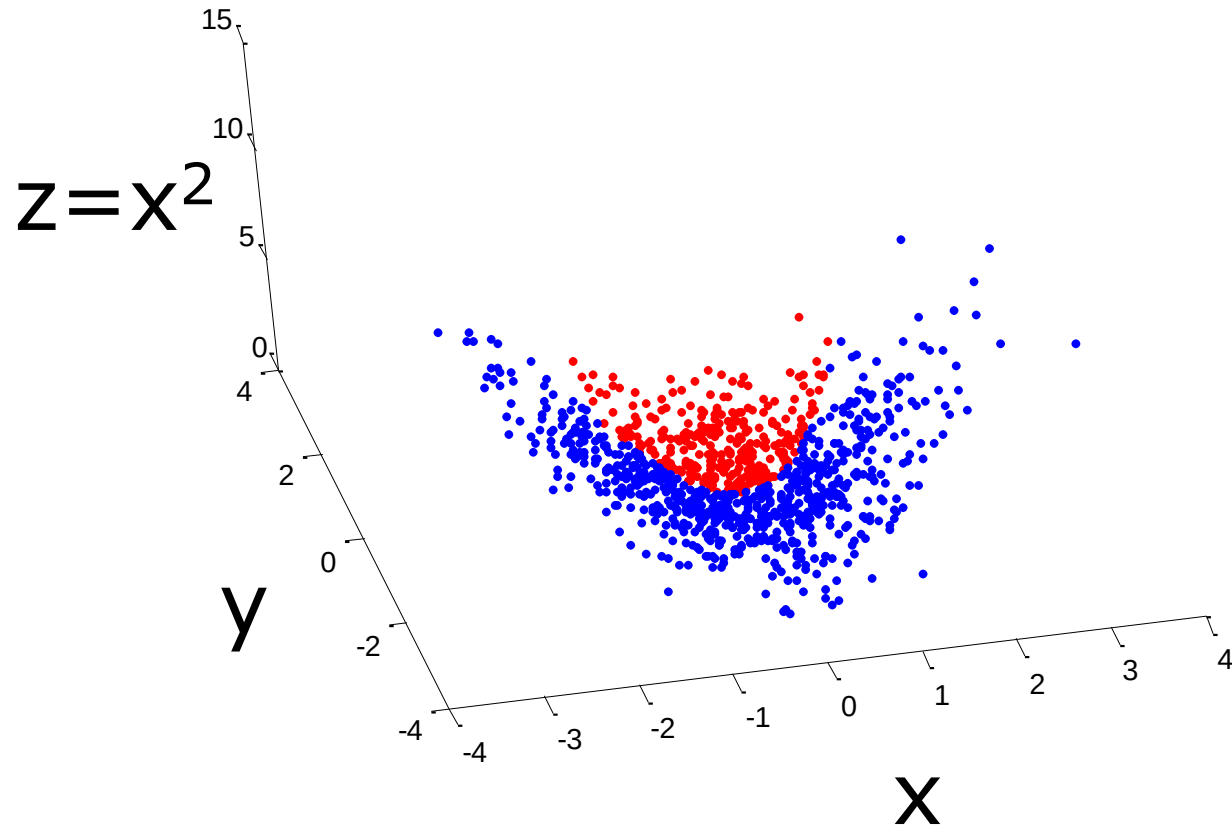
- Data would be easily separable by a polynomial



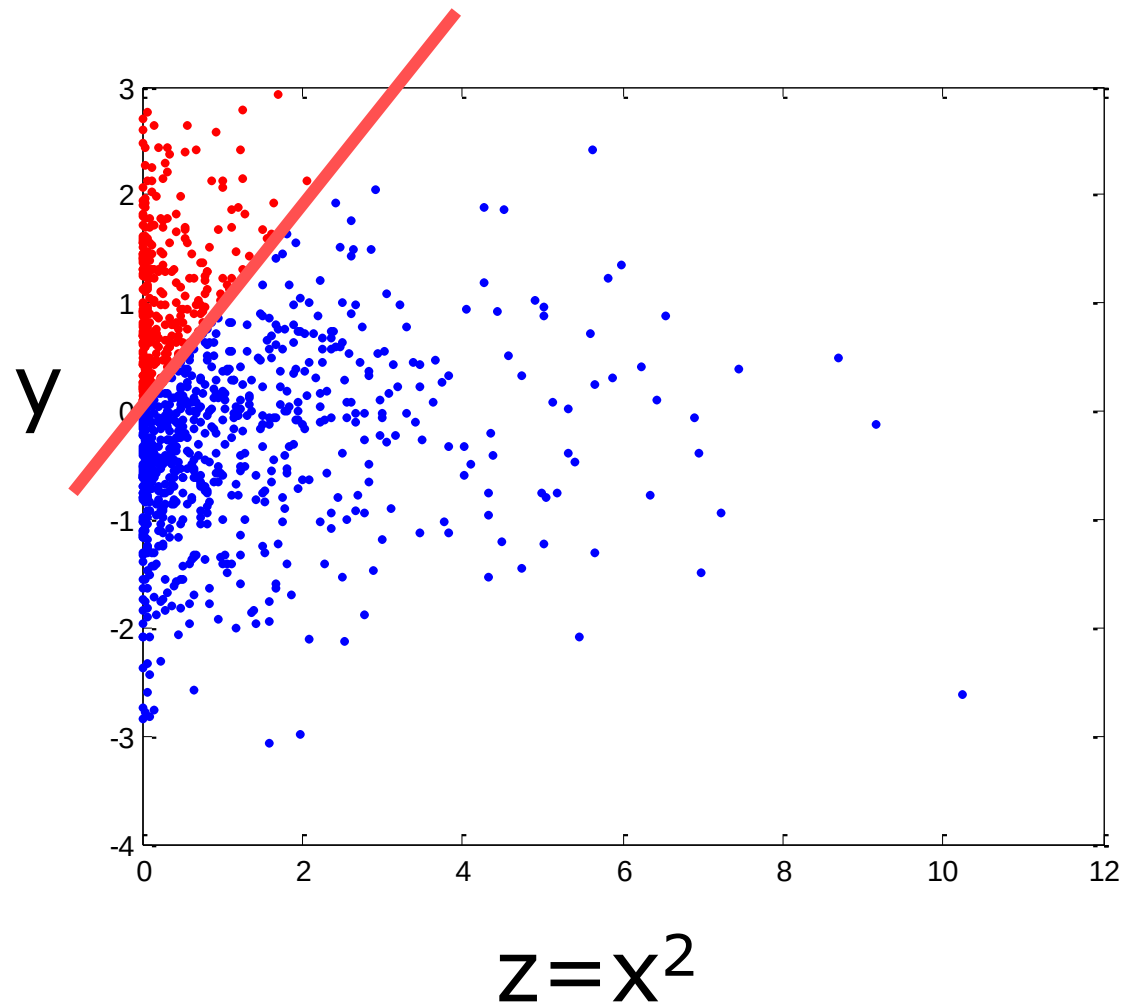
Non-linearly separable data



New coordinate $z=x^2$

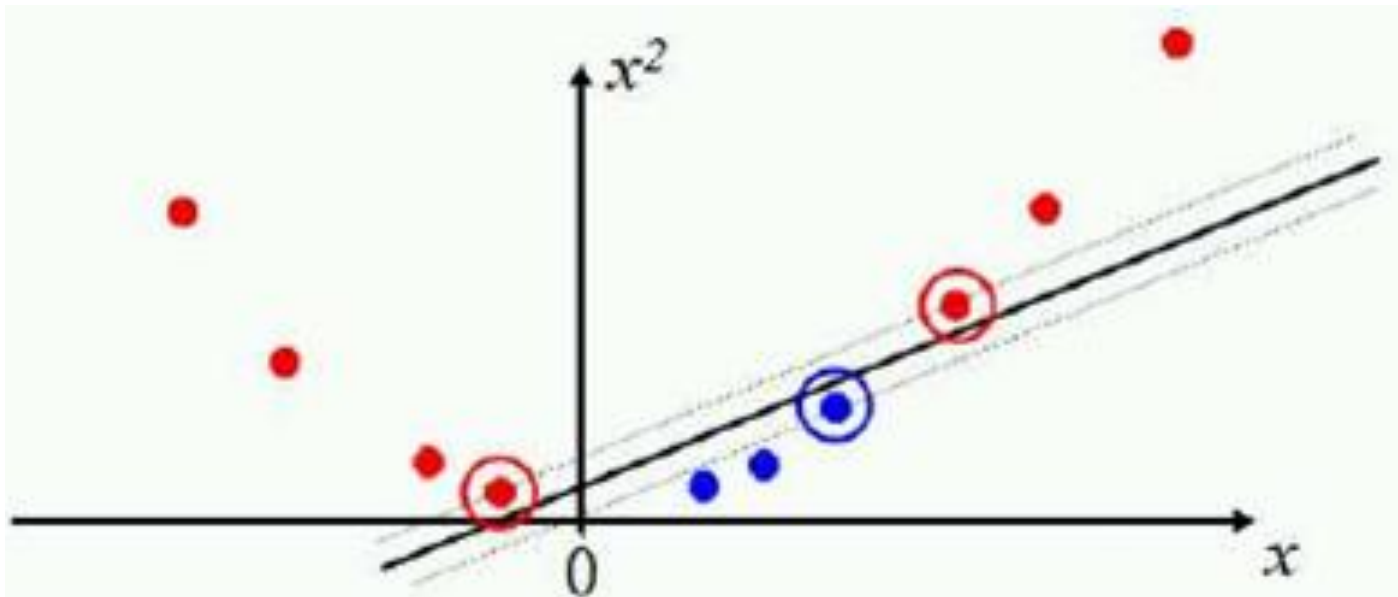


New coordinate $z=x^2$

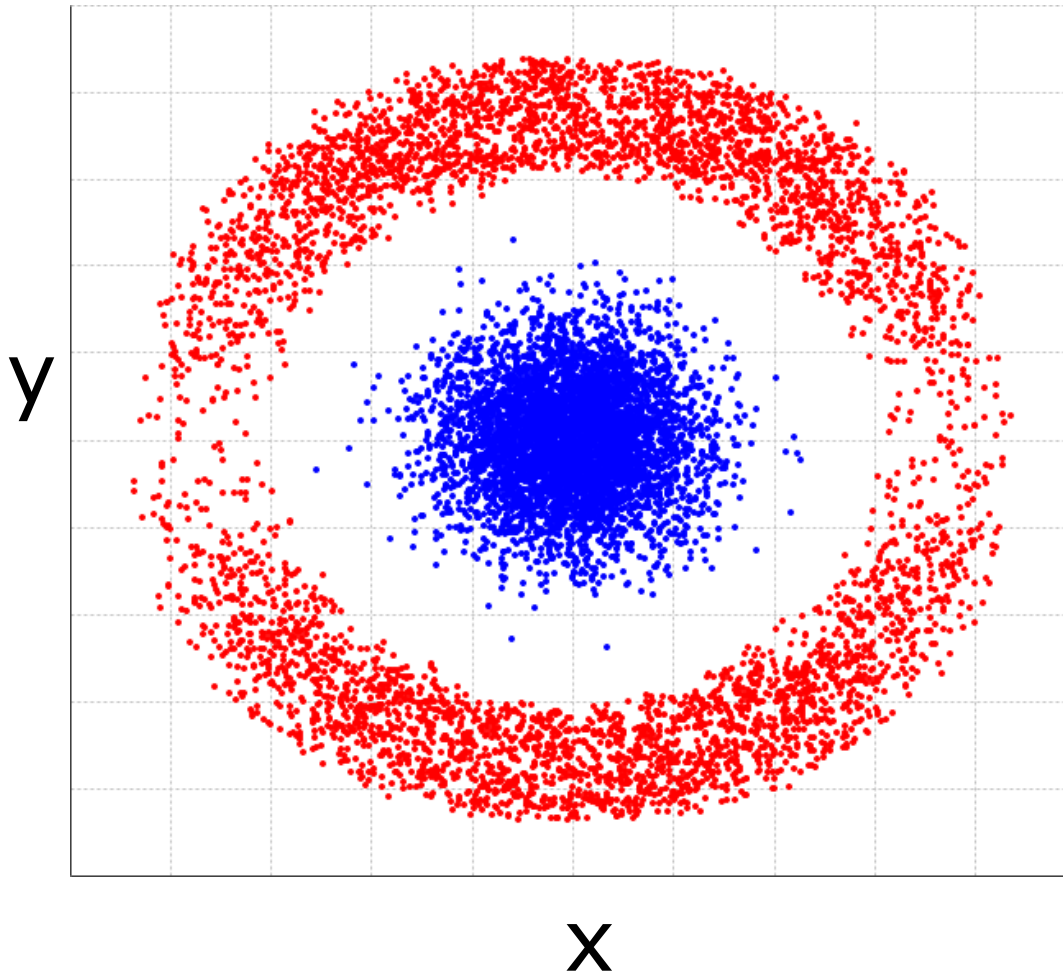


Non-linearly separable data

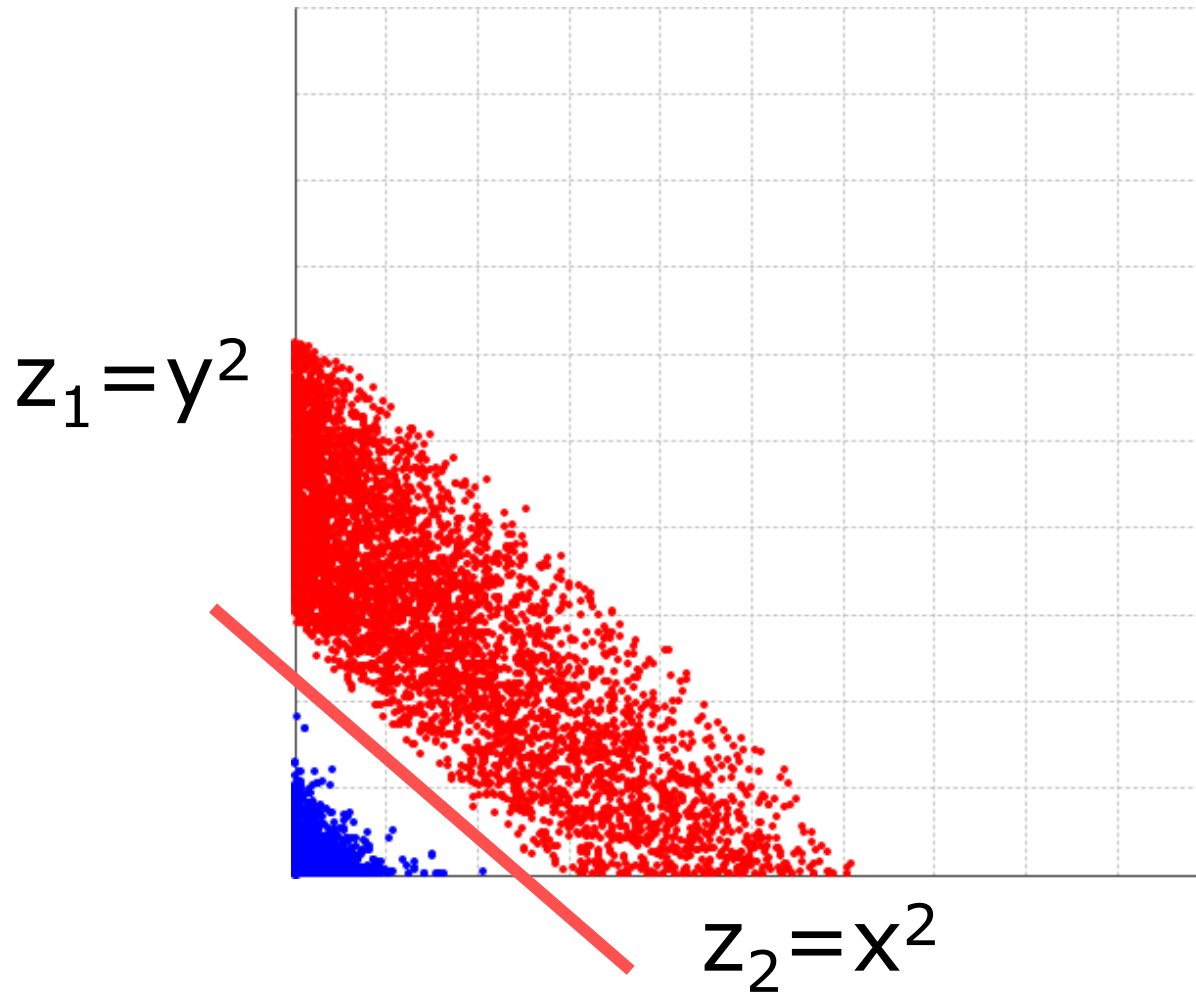




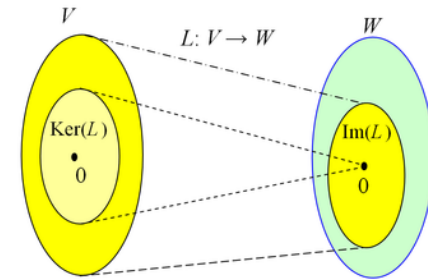
Non-linearly separable data

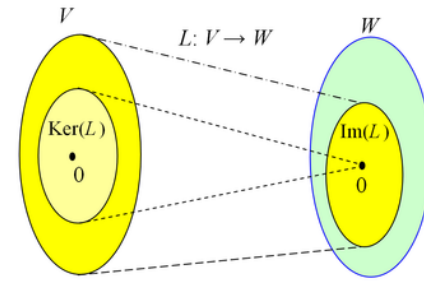


New coordinates $z_1=x^2$ & $z_2=y^2$



- Projection of data into higher dimensional space
 - Data may be separable in this space
- Projection: *multiplication* of vectors with **kernel matrix**
- Often: projected space has dimensionality equal to number of training vectors
- Kernel matrix determines shape of possible separators
 - In previous example: polynomial (quadratic)





- Quadratic
 - e.g. $k(x,y) = (x \cdot y)^2$
- General Polynomial (arbitrary degree)
 - $k(x,y) = (x \cdot y)^d$ (homogenous)
 - $k(x,y) = (x \cdot y + 1)^d$ (inhomogenous)
- Gaussian $k(x, y) = \exp\left(-\frac{\|x - y\|^2}{2\sigma^2}\right)$
- Radial Basis Function $k(x, y) = \exp(-\gamma \|x - y\|^2)$, $\gamma > 0$
- Sigmoid $k(x, y) = \tanh(\kappa x \cdot y + c)$
 for some (not every) $\kappa > 0, c < 0$

- Best kernel depends on the data and its underlying distribution
- **How to chose**
 - **Kernel family?**
 - **Exact parameters?**
 - Model selection:
 - Train several models with different kernels
 - Chose best performing
- Linear kernels work well with sparse data (e.g. Text)

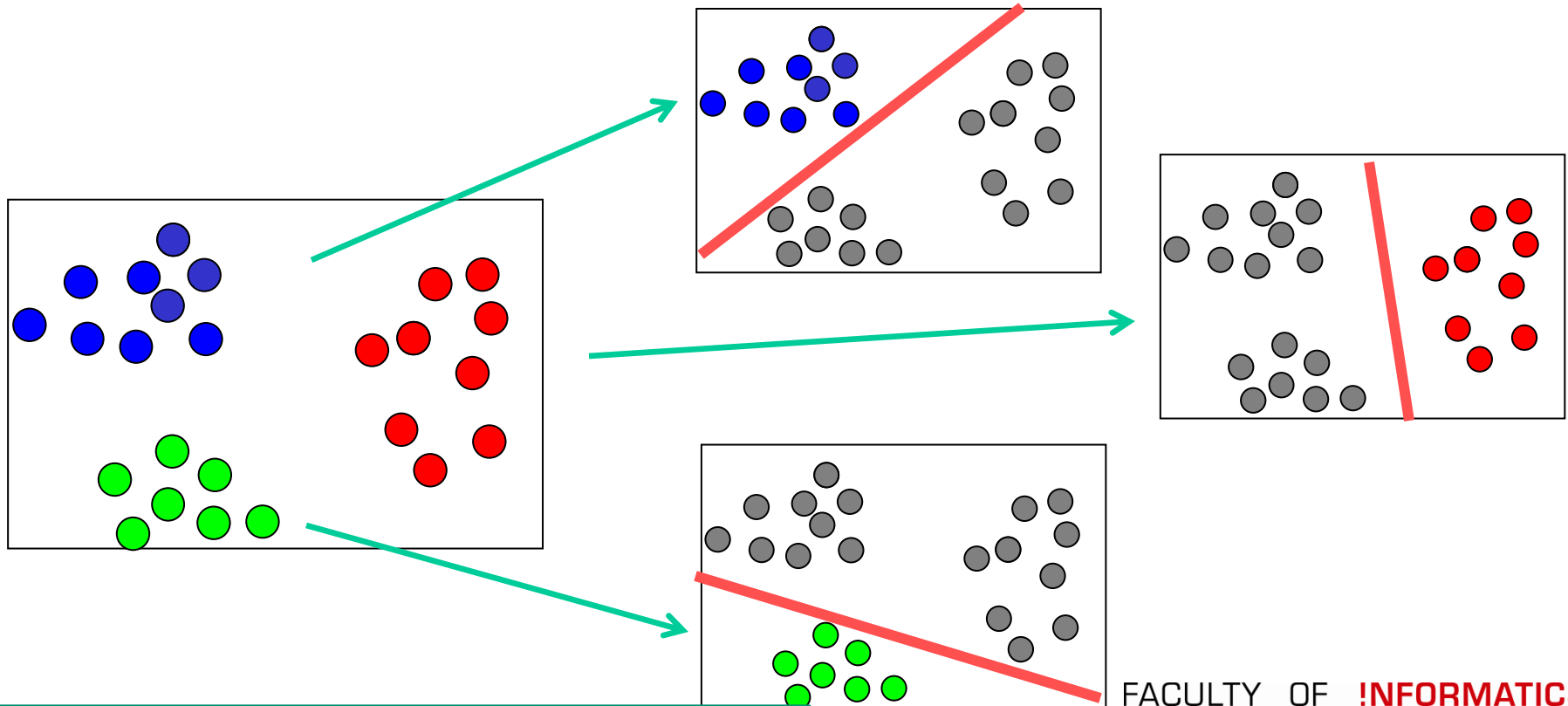
- Other ML algorithms could work with projected (high dimensional) data
 - So why bother with (rather) complicated SVM?
- Working with higher dimensional data is problematic: increased computational complexity

- Working with high dimensional data is problematic (computational complexity)
- *Outlook:* SVMs depend only on *dot-product* between vectors
 - **Kernel Trick:** replace dot-product with kernel function
 - This is computationally inexpensive (relatively)

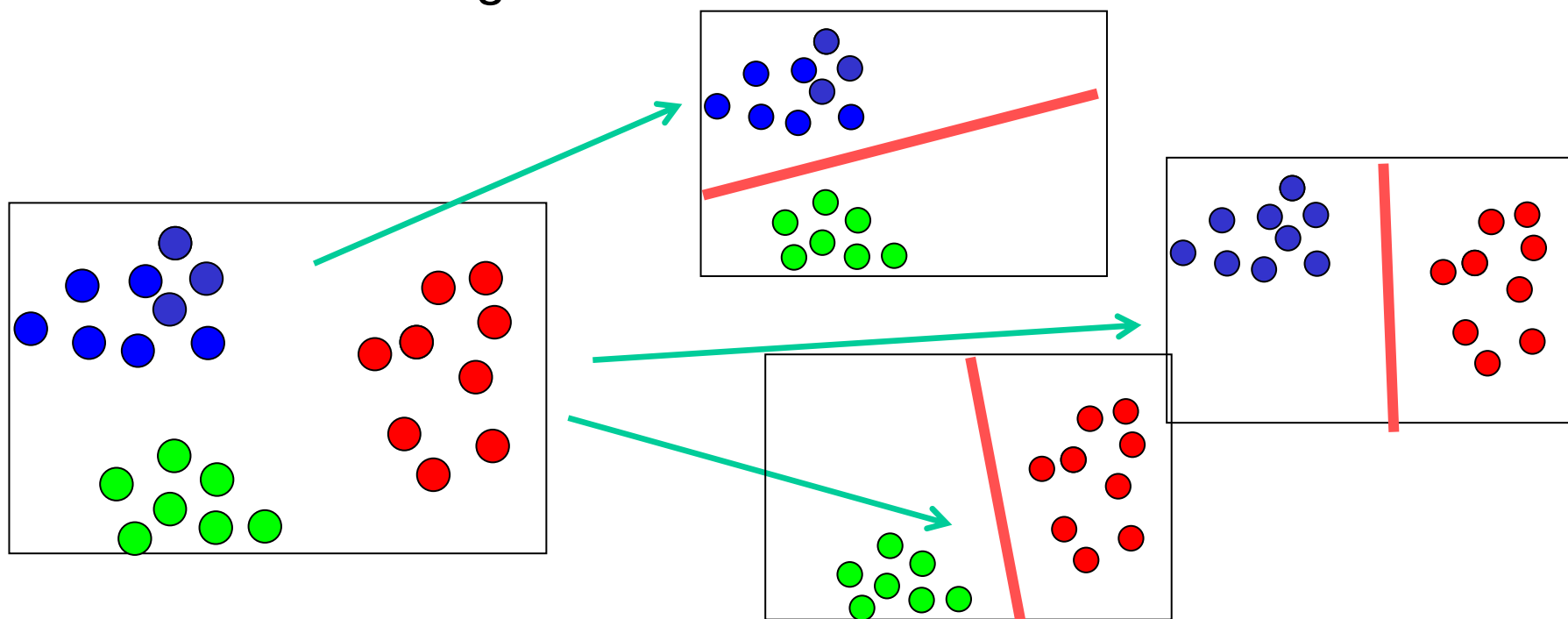
$$L_p = \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j x_i \cdot x_j$$

- Standard SVM algorithm just solves binary problems (i.e. two classes)
- Multi-class problem (three or more classes)
- Reduce (single) multi-class problem to multi binary problems
 - One vs. all → chose the classifier with the greatest margin
 - One vs. One → chose class selected most often

- One vs. All
 - Build binary classifiers that distinguish between class i and the rest ($i = 1, \dots, \text{\#numClasses}$)
 - Classifier with highest output $f(x)$ is the winner

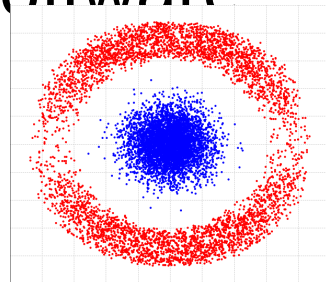


- One vs. One
 - Build binary classifier for each pair of classes
 - Class with highest number of votes wins



- *How many classifiers?*
 - Needs $|C|(|C|-1)/2$ classifiers (but smaller)

- High classification accuracy
 - Good generalisation, even though decision boundary in original space is complex
- Linear kernels: Good for sparse, high dimensional data, e.g. text mining
- Much research has been directed at SVM, and it's foundations → solid background
- Implementation available in open-source software (e.g. libSVM)



- Recurrent Neural Networks
- Support Vector Machines: Intro
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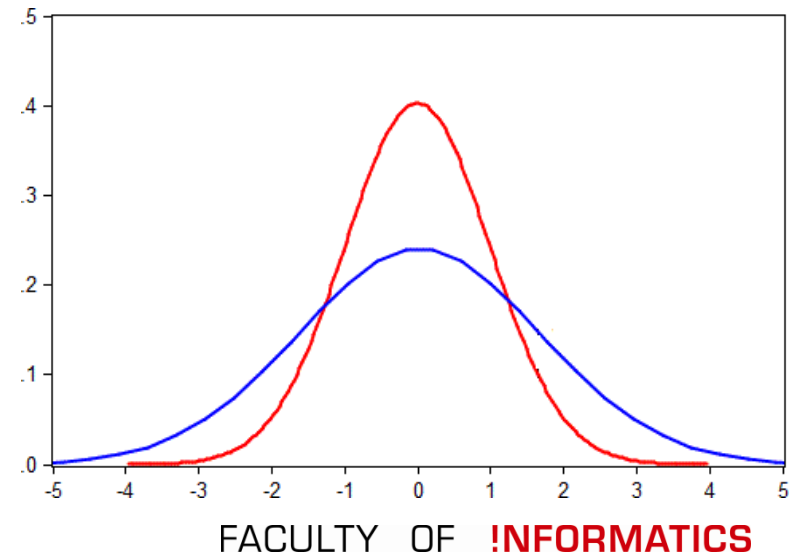
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- When is a classifier “better” than another?
 - If it achieves (even marginally) better accuracy?
(or performs better on any other evaluation measure used)
 - If it repeatedly performs better?
 - How much better does it need to be?
 - 1%, 2%, 3% ?
- Idea: test the **significance** of the difference
- If difference is statistically significant: we have a better justification to call one classifier better

- Performance of classifiers varies
 - Variations in selection of the test set
 - A classifier might outperform another on a specific test set
 - *even if they are identical on the whole population of test sets!*
 - Variations in the training set
 - Any classifier might outperform another on a specific training set (*even if equal otherwise*) (cf. variance/instability)
 - Randomness in the learning algorithm
 - Any classifier might outperform another with a specific random value (e.g. in the initialisation), *even if equal otherwise*

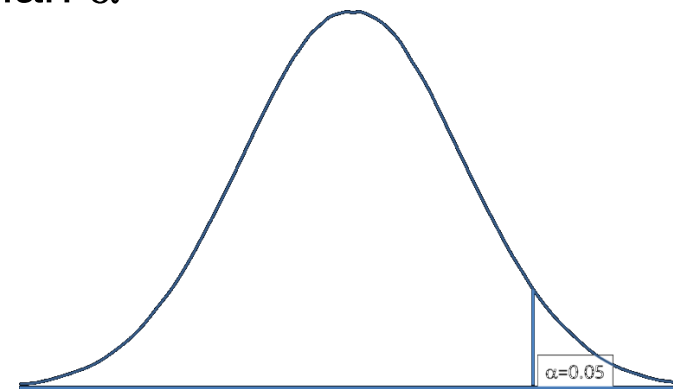
➔ Is the difference real, or due to chance?

- Testing the hypothesis whether two classifiers are statistically significantly different
 - *Basics of statistical tests in respective lectures*
- Null hypothesis
 - **Result** of two classifiers are samples **drawn from the same distribution**
 - If **not**: the performance of the classifiers is **different** with statistical significance



- Null hypothesis
 - *Result of two classifiers are samples drawn from the same distribution*
 - If **yes**: the performance of the classifiers is **not** different with statistical significance

- *Convention*
 - Reject null hypothesis (difference statistically significant)
 - If probability of null hypothesis less than α
 - It still might be a false rejection
 - But we have an error bound
 - α (level of significance) with common values of 0.05 or 0.01



- Simple test: McNemar's test
 - Based on χ^2 -test, compares classification output errors
 - E.g. two classifiers, simple train/test split
- Null hypothesis: both classifiers are equal
- Compute matrix:

# samples correct in A & B (N_{11})	# samples correct A, wrong B (N_{10})
# samples wrong A, correct B (N_{01})	# samples wrong in A & B (N_{00})

- Null hypothesis: $N_{01} = N_{10}$

Statistical Significance Testing

# samples correct in A & B (N_{11})	# samples correct A, wrong B (N_{10})
# samples wrong A, correct B (N_{01})	# samples wrong in A & B (N_{00})

Null hypothesis: $N_{01} = N_{10} = \frac{N_{01} + N_{10}}{2}$

$$\chi^2 = \frac{(N_{01} - \frac{N_{01} + N_{10}}{2})^2}{\frac{N_{01} + N_{10}}{2}} + \frac{(N_{10} - \frac{N_{01} + N_{10}}{2})^2}{\frac{N_{01} + N_{10}}{2}} = \frac{(N_{01} - N_{10})^2}{N_{01} + N_{10}}$$

Actually: $\chi^2 = \frac{(|N_{01} - N_{10}| - 1)^2}{N_{01} + N_{10}}$

Edward's correction

(as test statistic is discrete)

Statistical Significance Testing

- Compare with tabulated χ^2 values, e.g. for a level of significance of 5% (1 degree of freedom)

– If $\chi^2 > 3.841$:
reject null hypothesis, i.e. the classifiers are significantly different

Percentage Points of the Chi-Square Distribution

Degrees of Freedom	Probability of a larger value of χ^2								
	0.99	0.95	0.90	0.75	0.50	0.25	0.10	0.05	0.01
1	0.000	0.004	0.016	0.102	0.455	1.32	2.71	3.84	6.63
2	0.020	0.103	0.211	0.575	1.386	2.77	4.61	5.99	9.21
3	0.115	0.352	0.584	1.212	2.366	4.11	6.25	7.81	11.34
4	0.297	0.711	1.064	1.923	3.357	5.39	7.78	9.49	13.28
5	0.554	1.145	1.610	2.675	4.351	6.63	9.24	11.07	15.09
6	0.872	1.635	2.204	3.455	5.348	7.84	10.64	12.59	16.81
7	1.239	2.167	2.833	4.255	6.346	9.04	12.02	14.07	18.48
8	1.647	2.733	3.490	5.071	7.344	10.22	13.36	15.51	20.09
9	2.088	3.325	4.168	5.899	8.343	11.39	14.68	16.92	21.67
10	2.558	3.940	4.865	6.737	9.342	12.55	15.99	18.31	23.21
11	3.053	4.575	5.578	7.584	10.341	13.70	17.28	19.68	24.72
12	3.571	5.226	6.304	8.438	11.340	14.85	18.55	21.03	26.22
13	4.107	5.892	7.042	9.299	12.340	15.98	19.81	22.36	27.69
14	4.660	6.571	7.790	10.165	13.339	17.12	21.06	23.68	29.14
15	5.229	7.261	8.547	11.037	14.339	18.25	22.31	25.00	30.58
16	5.812	7.962	9.312	11.912	15.338	19.37	23.54	26.30	32.00
17	6.408	8.672	10.085	12.792	16.338	20.49	24.77	27.59	33.41
18	7.015	9.390	10.865	13.675	17.338	21.60	25.99	28.87	34.80
19	7.633	10.117	11.651	14.562	18.338	22.72	27.20	30.14	36.19
20	8.260	10.851	12.443	15.452	19.337	23.83	28.41	31.41	37.57
22	9.542	12.338	14.041	17.240	21.337	26.04	30.81	33.92	40.29
24	10.856	13.848	15.659	19.037	23.337	28.24	33.20	36.42	42.98
26	12.198	15.379	17.292	20.843	25.336	30.43	35.56	38.89	45.64
28	13.565	16.928	18.939	22.657	27.336	32.62	37.92	41.34	48.28
30	14.953	18.493	20.599	24.478	29.336	34.80	40.26	43.77	50.89
40	22.164	26.509	29.051	33.660	39.335	45.62	51.80	55.76	63.69
50	27.707	34.764	37.689	42.942	49.335	56.33	63.17	67.50	76.15
60	37.485	43.188	46.459	52.294	59.335	66.98	74.40	79.08	88.38

- Example: Dataset with 1,000 samples
- Run two classifiers
 1. Record N_{00} , N_{01} , N_{10} , N_{11}

- E.g.

500 (N_{11})	100 (N_{10})
100 (N_{01})	300 (N_{00})

2. Compute test statistic
3. Compare to tabulated value

500 (N_{11})	100 (N_{10})
100 (N_{01})	300 (N_{00})

$$\chi^2 = \frac{(|N_{01} - N_{10}| - 1)^2}{N_{01} + N_{10}} = \frac{(0 - 1)^2}{100 + 100} = \frac{1}{200} = 0,005$$

- $\chi^2_{1,0.05} = 3.841$

➔ don't reject, no significant difference

500 (N_{11})	20 (N_{10})
180 (N_{01})	300 (N_{00})

$$\chi^2 = \frac{(|N_{01} - N_{10}| - 1)^2}{N_{01} + N_{10}} = \frac{(160 - 1)^2}{180 + 20} = \frac{25281}{200} = 126.405$$

- $\chi^2_{1,0.05} = 3.841$

→ reject, significant difference

500 (N_{11})	116 (N_{10})
84 (N_{01})	300 (N_{00})

$$\chi^2 = \frac{(|N_{01} - N_{10}| - 1)^2}{N_{01} + N_{10}} = \frac{(32 - 1)^2}{85 + 115} = \frac{961}{200} = 4.805$$

- $\chi^2_{1,0.05} = 3.841$

→ reject, significant difference

500 (N_{11})	115 (N_{10})
85 (N_{01})	300 (N_{00})

$$\chi^2 = \frac{(|N_{01} - N_{10}| - 1)^2}{N_{01} + N_{10}} = \frac{(30 - 1)^2}{85 + 115} = \frac{841}{200} = 4.205$$

- $\chi^2_{1,0.05} = 3.841$

→ reject, significant difference

500 (N_{11})	114 (N_{10})
86 (N_{01})	300 (N_{00})

$$\chi^2 = \frac{(|N_{01} - N_{10}| - 1)^2}{N_{01} + N_{10}} = \frac{(28 - 1)^2}{86 + 114} = \frac{729}{200} = 3.645$$

- $\chi^2_{1,0.05} = 3.841$

➔ don't reject, no significant difference

500 (N_{11})	86 (N_{10})
114 (N_{01})	300 (N_{00})

$$\chi^2 = \frac{(|N_{01} - N_{10}| - 1)^2}{N_{01} + N_{10}} = \frac{(28 - 1)^2}{86 + 114} = \frac{729}{200} = 3.645$$

- $\chi^2_{1,0.05} = 3.841$

➔ don't reject, no significant difference

500 (N_{11})	80 (N_{10})
20 (N_{01})	350 (N_{00})

$$\chi^2 = \frac{(|N_{01} - N_{10}| - 1)^2}{N_{01} + N_{10}} = \frac{(60 - 1)^2}{80 + 20} = \frac{225}{100} = 3.481$$

- $\chi^2_{1,0.05} = 3.841$

➔ don't reject, no significant difference

500 (N_{11})	81 (N_{10})
19 (N_{01})	350 (N_{00})

$$\chi^2 = \frac{(|N_{01} - N_{10}| - 1)^2}{N_{01} + N_{10}} = \frac{(62 - 1)^2}{81 + 19} = \frac{3721}{100} = 3.721$$

- $\chi^2_{1,0.05} = 3.841$

➔ don't reject, no significant difference

Statistical Significance Testing

500 (N_{11})	82 (N_{10})
18 (N_{01})	350 (N_{00})

$$\chi^2 = \frac{(|N_{01} - N_{10}| - 1)^2}{N_{01} + N_{10}} = \frac{(64 - 1)^2}{82 + 18} = \frac{3969}{100} = 3.969$$

- $\chi^2_{1,0.05} = 3.841$

→ reject, significant difference

500 (N_{11})	86 (N_{10})
114 (N_{01})	300 (N_{00})

$$\chi^2 = \frac{(|N_{01} - N_{10}| - 1)^2}{N_{01} + N_{10}} = \frac{(28 - 1)^2}{86 + 114} = \frac{729}{200} = 3.645$$

- $\chi^2_{1,0.05} = 3.841$

➔ don't reject, no significant difference

Statistical Significance Testing

400 (N_{11})	136 (N_{10})
164 (N_{01})	300 (N_{00})

$$\chi^2 = \frac{(|N_{01} - N_{10}| - 1)^2}{N_{01} + N_{10}} = \frac{(28 - 1)^2}{136 + 164} = \frac{729}{300} = 2.43$$

- $\chi^2_{1,0.05} = 3.841$

➔ don't reject, no significant difference

- Paired Tests
 - When having several paired samples of training & test set
- *When?*
 - *E.g. cross validation, repeated holdout, ...*

- Paired t-Test:

$$t = \frac{\overline{X}_D - \mu_0}{\frac{S_D}{\sqrt{n}}}$$

- n: number of samples
- X_D : difference of performance in each run
 - e.g. difference in *accuracy* between classifier 1 and 2
- S_D : standard deviation of difference
- μ_0 : normally set to 0

- Paired t-Tests can be used e.g. with
 - Repeatedly splitting data into train & test set
 - Cross-validation
 - Number of (runs/folds – 1) → degrees of freedom
- Paired t-Test assumes independence of training and test sets
 - *Is this fulfilled above?*

Statistical Significance Testing

$$t = \frac{\bar{X}_D \sqrt{n}}{s_D}$$

Run	Classifier 1	Classifier 2
1	80.4	80.57
2	81.2	81.16
3	79.4	79.17
4	82.2	82.01
5	78.4	78
6	79.8	79.65
7	81.2	80.89
8	80.6	80.23
9	81.1	80.82
10	79.8	79.58

Statistical Significance Testing

$$t = \frac{\bar{X}_D \sqrt{n}}{s_D}$$

Run	Classifier 1	Classifier 2	Diff
1	80.4	80.57	-0.2
2	81.2	81.16	0.04
3	79.4	79.17	0.23
4	82.2	82.01	0.19
5	78.4	78	0.4
6	79.8	79.65	0.15
7	81.2	80.89	0.31
8	80.6	80.23	0.37
9	81.1	80.82	0.28
10	79.8	79.58	0.22

$$t = \frac{0.202 \sqrt{10}}{0.1679} = 3.8045$$

# samples	10	10	
Mean	80.41	80.208	0.2020
StdDev	1.0939	1.1442	0.1679

$$t = \frac{\bar{X}_D - \mu_0}{\frac{S_D}{\sqrt{n}}}$$

(Number of runs/folds – 1) →
Number of degrees of freedom

v	0.10	0.05	0.025	0.01	0.005	0.001
1.	3.078	6.314	12.706	31.821	63.657	318.313
2.	1.886	2.920	4.303	6.965	9.925	22.327
3.	1.638	2.353	3.182	4.541	5.841	10.215
4.	1.533	2.132	2.776	3.747	4.604	7.173
5.	1.476	2.015	2.571	3.365	4.032	5.893
6.	1.440	1.943	2.447	3.143	3.707	5.208
7.	1.415	1.895	2.365	2.998	3.499	4.782
8.	1.397	1.860	2.306	2.896	3.355	4.499
9.	1.383	1.833	2.262	2.821	3.250	4.296
10.	1.372	1.812	2.228	2.764	3.169	4.143
11.	1.363	1.796	2.201	2.718	3.106	4.024
12.	1.356	1.782	2.179	2.681	3.055	3.929
13.	1.350	1.771	2.160	2.650	3.012	3.852
14.	1.345	1.761	2.145	2.624	2.977	3.787
15.	1.341	1.753	2.131	2.602	2.947	3.733
16.	1.337	1.746	2.120	2.583	2.921	3.686
17.	1.333	1.740	2.110	2.567	2.898	3.646
18.	1.330	1.734	2.101	2.552	2.878	3.610
19.	1.328	1.729	2.093	2.539	2.861	3.579
20.	1.325	1.725	2.086	2.528	2.845	3.552
21.	1.323	1.721	2.080	2.518	2.831	3.527
22.	1.321	1.717	2.074	2.508	2.819	3.505
23.	1.319	1.714	2.069	2.500	2.807	3.485
24.	1.318	1.711	2.064	2.492	2.797	3.467
25.	1.316	1.708	2.060	2.485	2.787	3.450
26.	1.315	1.706	2.056	2.479	2.779	3.435
27.	1.314	1.703	2.052	2.473	2.771	3.421
28.	1.313	1.701	2.048	2.467	2.763	3.408
29.	1.311	1.699	2.045	2.462	2.756	3.396
30.	1.310	1.697	2.042	2.457	2.750	3.385
31.	1.309	1.696	2.040	2.453	2.744	3.375
32.	1.309	1.694	2.037	2.449	2.738	3.365
33.	1.308	1.692	2.035	2.445	2.733	3.356
34.	1.307	1.691	2.032	2.441	2.728	3.348
35.	1.306	1.690	2.030	2.438	2.724	3.340
36.	1.306	1.688	2.028	2.434	2.719	3.333
37.	1.305	1.687	2.026	2.431	2.715	3.326
38.	1.304	1.686	2.024	2.429	2.712	3.319
39.	1.304	1.685	2.023	2.426	2.708	3.313
40.	1.303	1.684	2.021	2.423	2.704	3.307
41.	1.303	1.683	2.020	2.421	2.701	3.301
42.	1.302	1.682	2.018	2.418	2.698	3.296
43.	1.302	1.681	2.017	2.416	2.695	3.291
44.	1.301	1.680	2.015	2.414	2.692	3.286
45.	1.301	1.679	2.014	2.412	2.690	3.281

- How does the size of the data set influence this?
 - *Respectively, does the size of the test set influence this?*
- Statistical significance testing “easier” with larger test set
 - I.e. we need less difference in observed measure to become statistically significantly different

- **Size of the data set: example**

red indicates stat. significant difference

Classifier	Accuracy
A (base)	59.17%
B	62.97% (+3.8%)
C	63.47% (+4.3%)

600 samples

Classifier	Accuracy
A (base)	66.32%
B	67.06% (+0.74%)
C	68.72% (+2.4%)

3,000 samples

- Summary:
 - Important to assure that “better” has a statistical basis
 - Adds more power to your claim ...
 - Not much extra effort when using ML toolkits (built-in)
- Statistically significant does not mean that we have achieved meaningful or significant contributions !
 - E.g an “improvement” from 15 to 18% accuracy might be statistically significant...
 - But both “performances” might be useless for a given task ...

- Janez Demsar, “On the appropriateness of Statistical Tests in Machine Learning”
- Null hypothesis can too easily be rejected with big test set size
 - Algorithm that intentionally misclassifies every 1000th sample
 - Statistically worse with a large enough sample size
- Sometimes: practically important differences, but not significant because test set was too small..

- Recurrent Neural Networks
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- Selecting a subset of the features / variables
- Motivation to do so?

- Selecting a subset of the features / variables
- Assumption: data set contains many *redundant* or *noisy* features
 - Redundant feature: no additional information
 - Noisy feature: no information at all, or even deceiving/wrong information
 - Eliminating noisy/irrelevant features → improving performance (effectiveness)

- Selecting a subset of the features / variables
- Problem: model **too complex**
 - Remove the features that contribute the least
 - **Simpler** models – higher **interpretability**, cf. bias-variance trade-off

- Selecting a subset of the features / variables
- Problem: **too many** variables/samples to (reasonably) compute
 - Remove the features that contribute the least
 - Reduce computational complexity – **improve runtime (efficiency)**
 - At the cost of performance, if features not redundant ...
 - Useful when dealing with very-high dimensional data
 - Important for models that have $> O(n)$ efficiency

- Often an issue with data represented as complex media types
- Text indexing: e.g. bag-of-words approach

- Every word → one feature!

- Several 100s of documents

- Number of features in the hundred thousands!

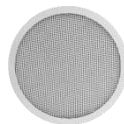
- Virtually impossible to compute....

	Word 1	Word 2	Word 3	Word 4	Word 5	...	Word n
Doc 1	1	0	0	2	3		0
Doc 2	2	0	0	0	2		2
Doc 3	1	3	0	0	1		5
Doc 4	2	0	0	2	0		2
Doc 5	5	4	0	0	1		0
...							0
Doc m	1	2	1	3	0		0

- Similar issues in several media types

General Process:

1. *Rank* features according to a measurement
 - **Utility** measurement
2. Select the number of features that is **suitable**
 - Other features are discarded / removed
 - Two approaches
 - Wrapper
 - Filter



- Ranking: categorisation of methods based on usage of **label**
 - Supervised selection
 - **Use class label** provided in training data
 - Unsupervised
 - Do **NOT** use class label provided in training data

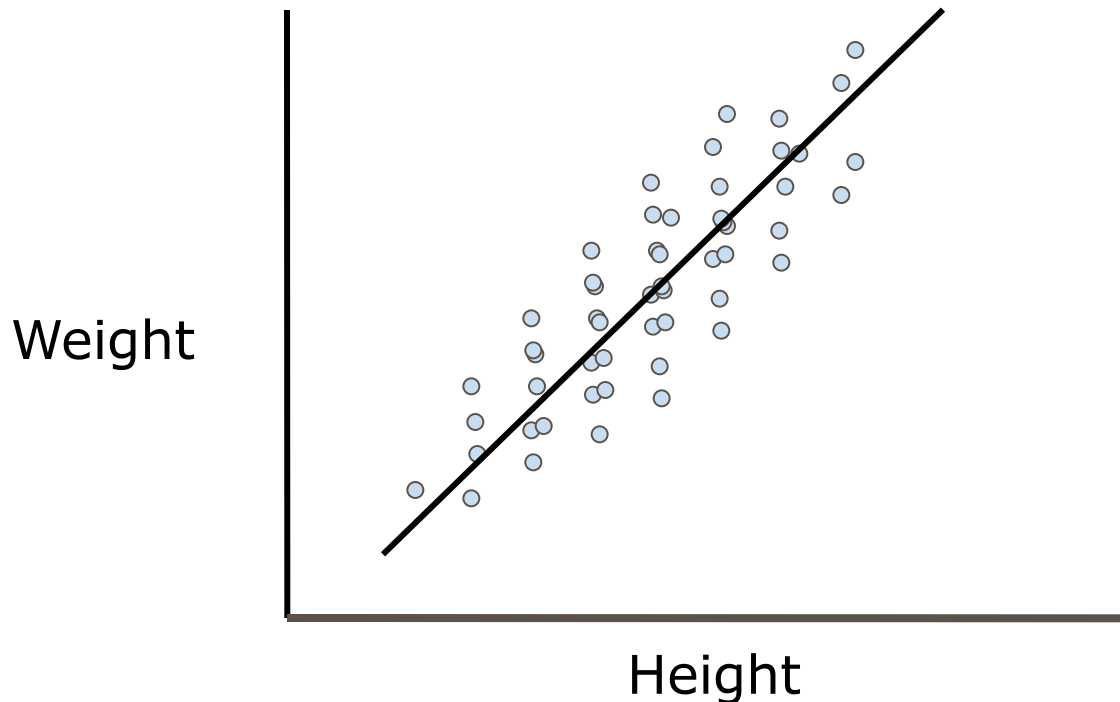
- Frequencies (text, image categorisation)
 - Select features on number of occurrences
 - Document frequency (# of docs)
 - Collection Frequency (total # occurrences)
 - Inverse Document Frequency (rareness of a term)
 - Inverse Collection Frequency
 - (Term Frequency x Document Frequency)

- Dimensionality Reduction methods, such as Principal Component Analysis (PCA)
 - Compute projection to lower dimensional space
 - Linear combination of features that explain the most variance
 - Via Singular Value Decomposition (SVD) / eigenvectors
 - Select highest ranked components (e.g. those that together explain 95% of the variance, top 5, ...)
 - (n.b: you can argue that PCA is not feature selection, as it projects into a different feature space first)
 - Select features from the original space that correspond to high absolute values in the component coefficients

- Data set might contain input variables that
 - Correlate poorly with output (**supervised** selection)
 - Directly depend on each other (**unsupervised** selection)
 - Might have unproportional weight on output prediction
 - Might be beneficial to treat such variables
- ➔ Finding dependencies with (pair wise) analysis of correlation as feature selection step
- E.g. Pearson correlation coefficient for linear dependence

Correlation Coefficient – examples

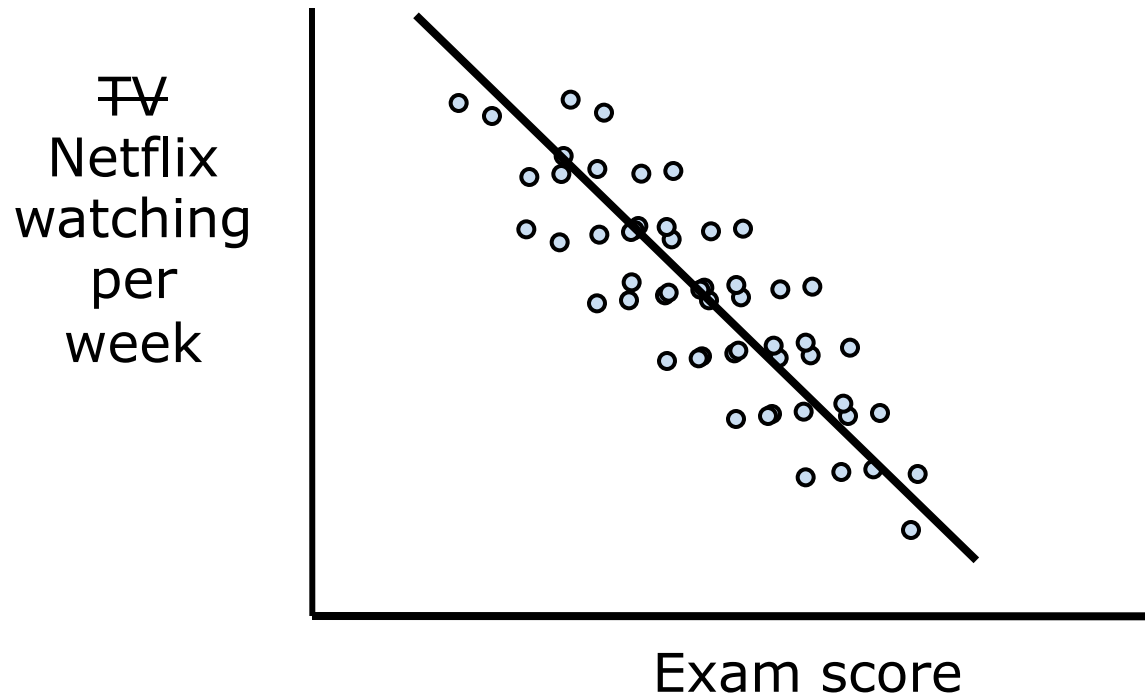
- High Degree of positive (direct) correlation



$$r = +.80$$

Correlation Coefficient – examples

- High degree of Negative Correlation



$$r = -.80$$

- Entropy / Information Gain
 - Based on information theory
 - c.f. Decision Trees

$$IG(X_A, X_B) = H(X) - p(x_A)H(X_A) - p(x_B)H(X_B)$$

- Information gain of each feature for ranking
- Highly related variations using e.g. average scores from Random Forests, etc.

- Mutual information (MI)
 - MI of a class and a feature
 - How much information the presence/absence of a feature contributes to correct classification
 - Also based on information theory

$$\sum_{e_f \in \{0,1\}} \sum_{e_c \in \{0,1\}} P(U = e_f, C = e_c) \log_2 \frac{P(U = e_f, C = e_c)}{P(U = e_f)P(C = e_c)}$$

- U: vector contains feature f (yes/no)
- C: document is in class c (yes/no)

- Using maximum likelihood estimates

feature f present & class c (N_{11})	feature f not present & class c (N_{10})
feature f present & not class c (N_{01})	feature f not present & not class c (N_{00})

$$\frac{N_{11}}{N} \log_2 \frac{N N_{11}}{N_{1.} N_{.1}} + \frac{N_{01}}{N} \log_2 \frac{N N_{01}}{N_{0.} N_{.1}} + \frac{N_{10}}{N} \log_2 \frac{N N_{10}}{N_{1.} N_{.0}} + \frac{N_{00}}{N} \log_2 \frac{N N_{00}}{N_{0.} N_{.0}}$$

- **MI = 0** if ...
 - Feature distributed same in class & whole set
- **MI is max** if ...
 - Feature only present if the item is in the class

- Odds ratio
 - Association (non-independence) between two variables
 - Based on probabilistic
- χ^2 feature selection
 - Independence of occurrence of feature and class
- At least 10 more frequently used measures !
 - “A Comparative Study on Feature Selection in Text Categorisation”. Yiming Yang & Jan O. Pedersen, 1997
 - “An Extensive Empirical Study of Feature Selection Metrics for Text Classification”, George Foreman, 2003
 - “Combination of Feature Selection Methods for Text Categorisation”. Robert Neumayer, Rudolf Mayer, and Kjetil Nørnvåg. 33rd European Conference on Information Retrieval 2011

- Algorithm

1. *Rank* features according to a measurement

- Utility measurement

2. Select the number of features that is **suitable**

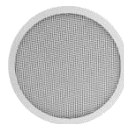
- Other features are discarded / removed

- Two approaches

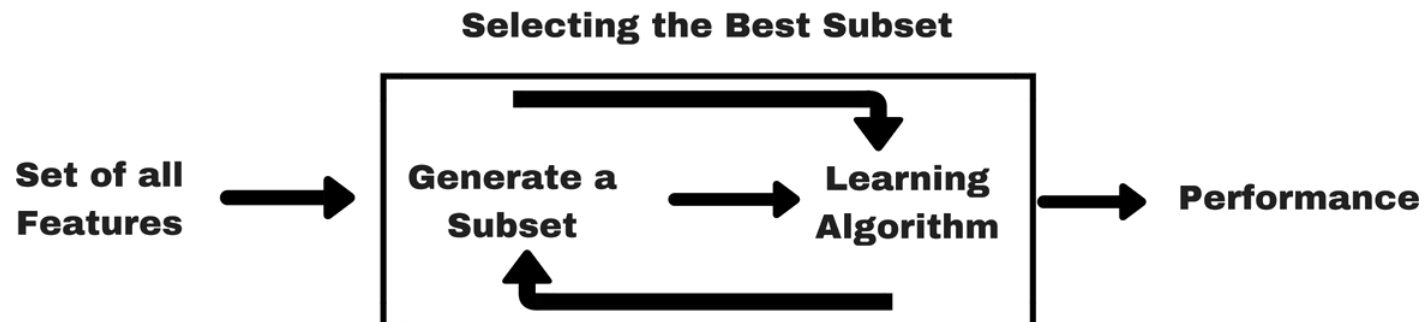
- Wrapper



- Filter



- Wrapper strategy
 - Use search strategies to select subsets
 - Evaluate subset by building models and computing e.g. Accuracy
 - Select feature subset that got the best performance evaluation
 - Similar to model selection



- Many different search strategies
 - Exhaustive search
 - Genetic algorithms
 - Hillclimbing
 - Greedy forward selection / backward elimination
 - Random
 - ...
- *(Dis)advantages of wrapper method?*

- Filter strategy
 - Don't evaluate on a model
 - Use proxy measure that is fast to compute instead
 - E.g. Information gain
- *Implications?*
 - Feature set not tuned to the model finally applied
 - *Is that good?*

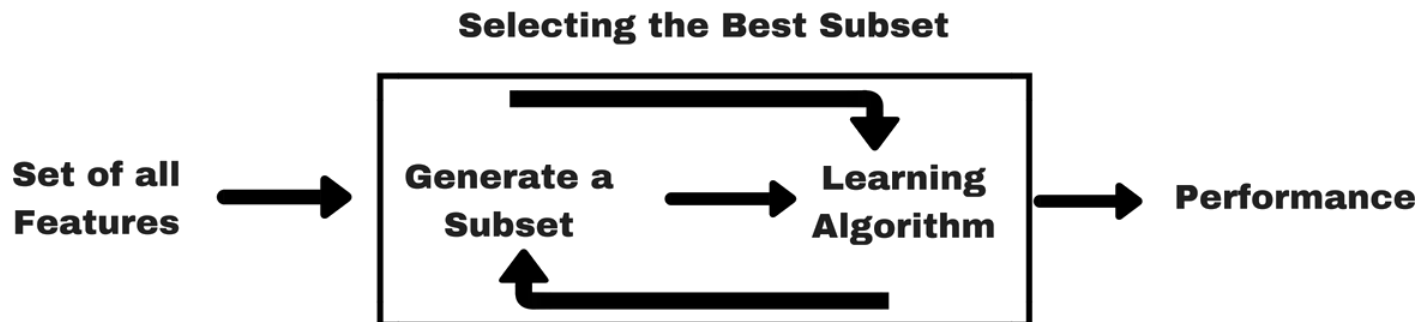


Wrapper vs. Filter

- Filter



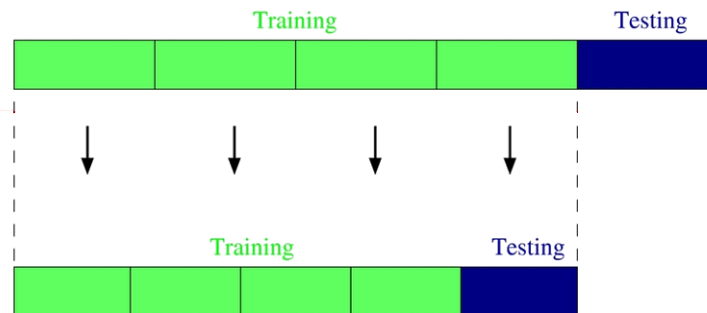
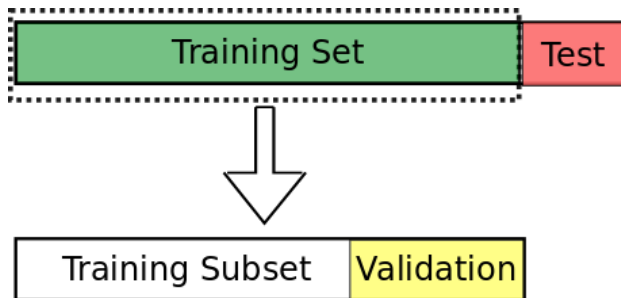
- Wrapper



- Filter measures relevance by correlation with dependent variable
 - Wrapper measure the direct utility of subset when training a model
- Filter methods much faster compared to wrapper
- Filter methods not tailored to a specific task
 - *Wrapper methods prone to overfitting?*

Feature selection methods

- On which set to do feature selection?
 - *Complete set?*
 - *Training set only?*



- Reduce computational complexity
 - Can bring enormous gains
 - Especially for classifiers that have polynomial complexity on the number of features
- Improve classification results
 - Sometimes, really depends on the data
- Trade-off less complexity vs. accuracy
 - Sometimes worth to sacrifice a bit of accuracy

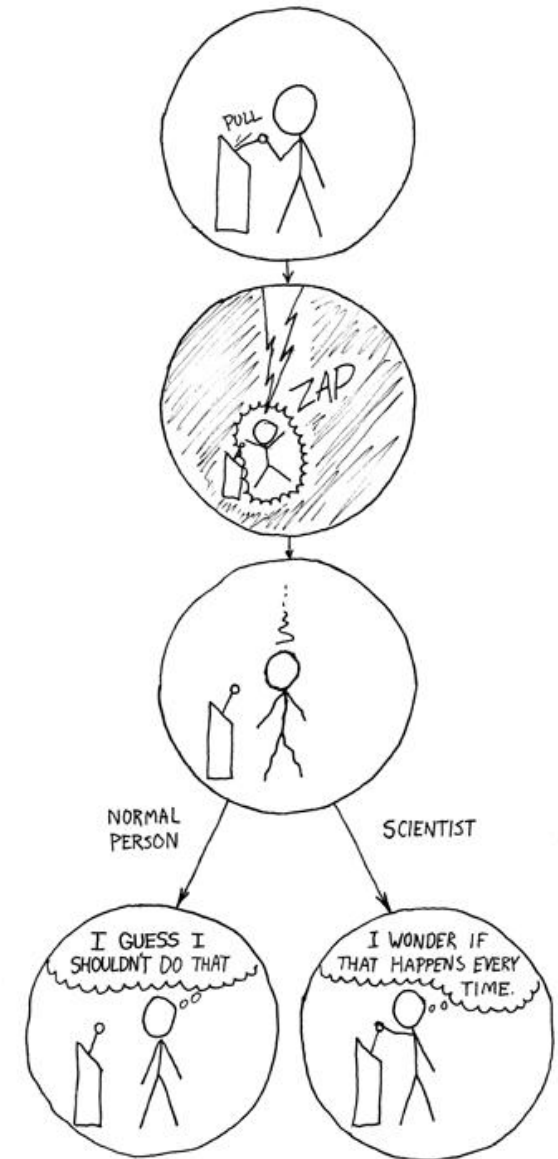
- Recurrent Neural Networks
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...

- Core to the scientific method
- Focus
 - NOT on detecting misconduct
 - But on the will to produce good work

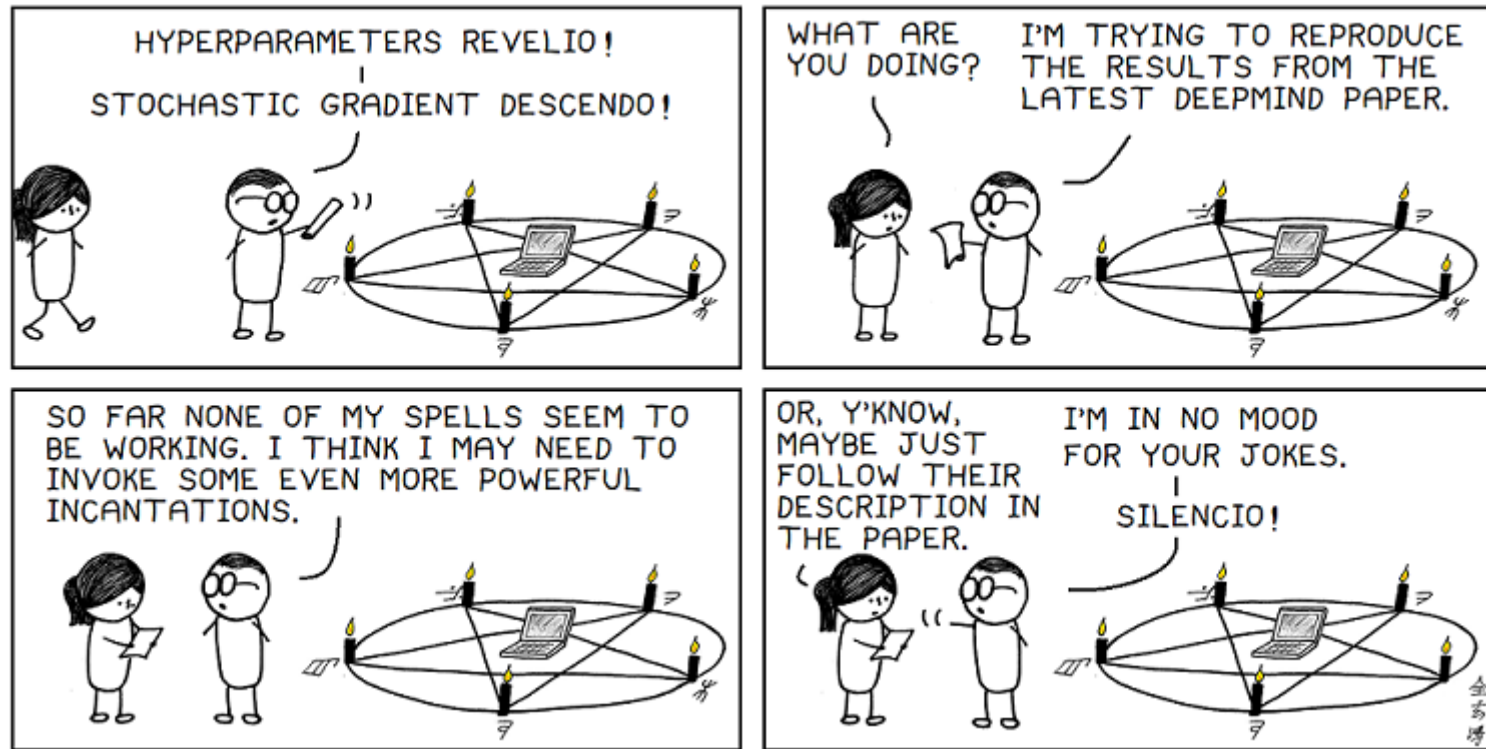
*More details especially in
the Data Science curriculum*

(Experiment Design, Data Stewardship)



<https://xkcd.com/242/>

Reproducibility



- Related terms often wrongly used interchangeably

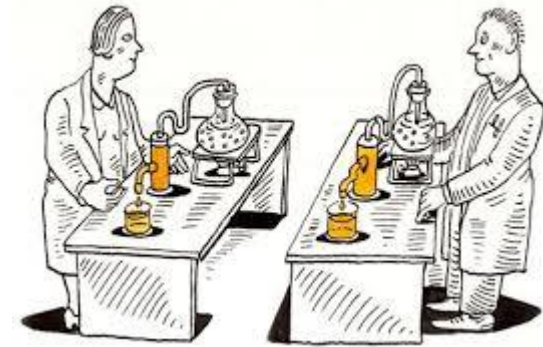
Term	Environment	Researcher	Complete workflow	V&V of workflow re-execution
replicate	same	different	yes	yes
repeat	same	same	yes	yes
reproduce	different	same/different	yes	NA
re-execute	same/different	same/different	yes	no
rerun	same/different	same/different	yes	no
reuse	same/different	same/different	no	(yes)

- Check 'Terminology' section:

<https://www.acm.org/publications/policies/artifact-review-badging>



- We want to be able to achieve the same results as in a previous experiment (run)
- This is difficult, for several reasons
 - *Examples?*
 - Initialisation of algorithms (random number generation)
 - Order of data samples (e.g. for Perceptron)
 - Version of software used
 - APIs: changes in Python 2 vs 3;
 - Bug fixes, improvements, change in default (unset) values, etc..
 - Version of operating system (e.g. different C++ library..)
 - Preprocessing, feature extraction, other tools
 - Any many more ...



- 30 research papers analysed
- Papers lack information on
 - datasets used
 - parameters (randomisation)
 - software environment
- Code is usually not provided
 - only generic method explanation

Reproducibility in ML Workshop at the 34th International Conference on Machine Learning (ICML 2017), Sydney, Australia.

<https://openreview.net/pdf?id=By4l2PbQ->

Reproducibility in Machine Learning-Based Studies: An Example of Text Mining

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Abstract

Reproducibility is an essential requirement for computational studies including those based on machine learning techniques. However, many machine learning studies are either not reproducible or are difficult to reproduce. In this paper, we consider what information about text mining studies is crucial to successful reproduction of such studies. We identify a set of factors that affect reproducibility based on our experience of attempting to reproduce six studies proposing text mining techniques for the automation of the citation screening stage in the systematic review process. Subsequently, the reproducibility of 30 studies was evaluated based on the presence or otherwise of information relating to the factors. While the studies provide useful reports of their results, they lack information on access to the dataset in the form and order as used in the original study (as against raw data), the software environment used, randomization control and the implementation of proposed techniques. In order to increase the chances of being reproduced, researchers should ensure that details about and/or access to information about these factors are provided in their reports.

1 Introduction

Independent verification of published claims for the purpose of credibility confirmation, extension and building a 'body of knowledge' is a standard scientific practice [13]. Machine learning methods based research are not excluded from this strict scientific research requirement. However, it may sometimes be hard or even impossible to replicate computational studies of this nature [12]. This is why the minimum standard expected of any computational study is for it to be reproducible [11].

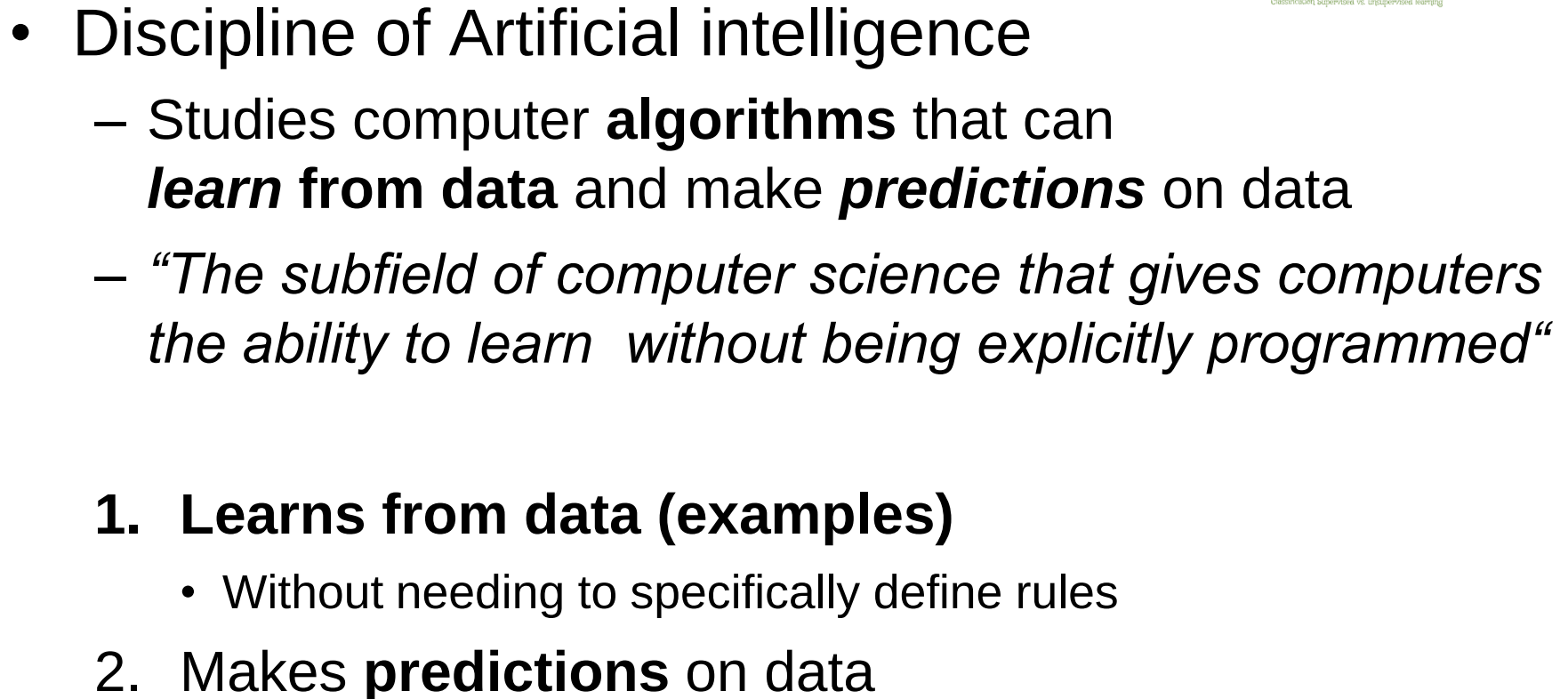
In order for a study to be reproduced, an independent researcher will need at least full information and artefacts of the experiment - datasets, experiment parameters, similar software and hardware environment etc., as used in the original study. However, the experience in studies today shows a lack of sufficient information that can enable an independent researcher reproduce majority of the studies successfully.

- Students reproduce papers submitted to ML conference
 - Academia, Google & DeepMind, Facebook, Microsoft, and Amazon
- Papers are anonymously published in advance
- Comments are public
- Goal: put pressure on authors and improve reproducibility
- The software a computer runs when reproducing an algorithmic experiment, as well as the configuration of that software and the data used, are comparable to the impact of gravity and temperature in the physical world. These elements provide context for an experiment, and need to be replicated to understand how and why the experiment works”



- Recurrent Neural Networks
- Support Vector Machines: Intro
- Statistical Significance Testing
- Feature Selection
- Challenge: Reproducibility
- Summary
- Advertisements: ML lectures, internships, thesis,

...

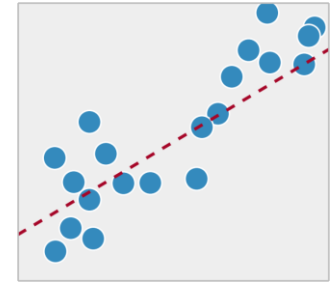


Machine Learning – scope

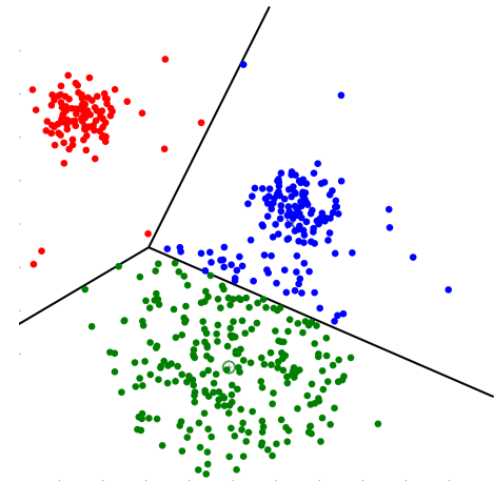
- Can be roughly grouped into

- Supervised learning

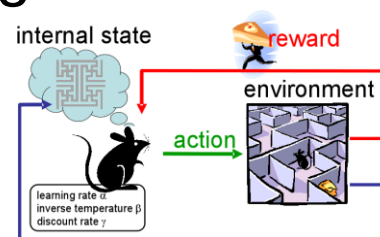
- Regression & classification



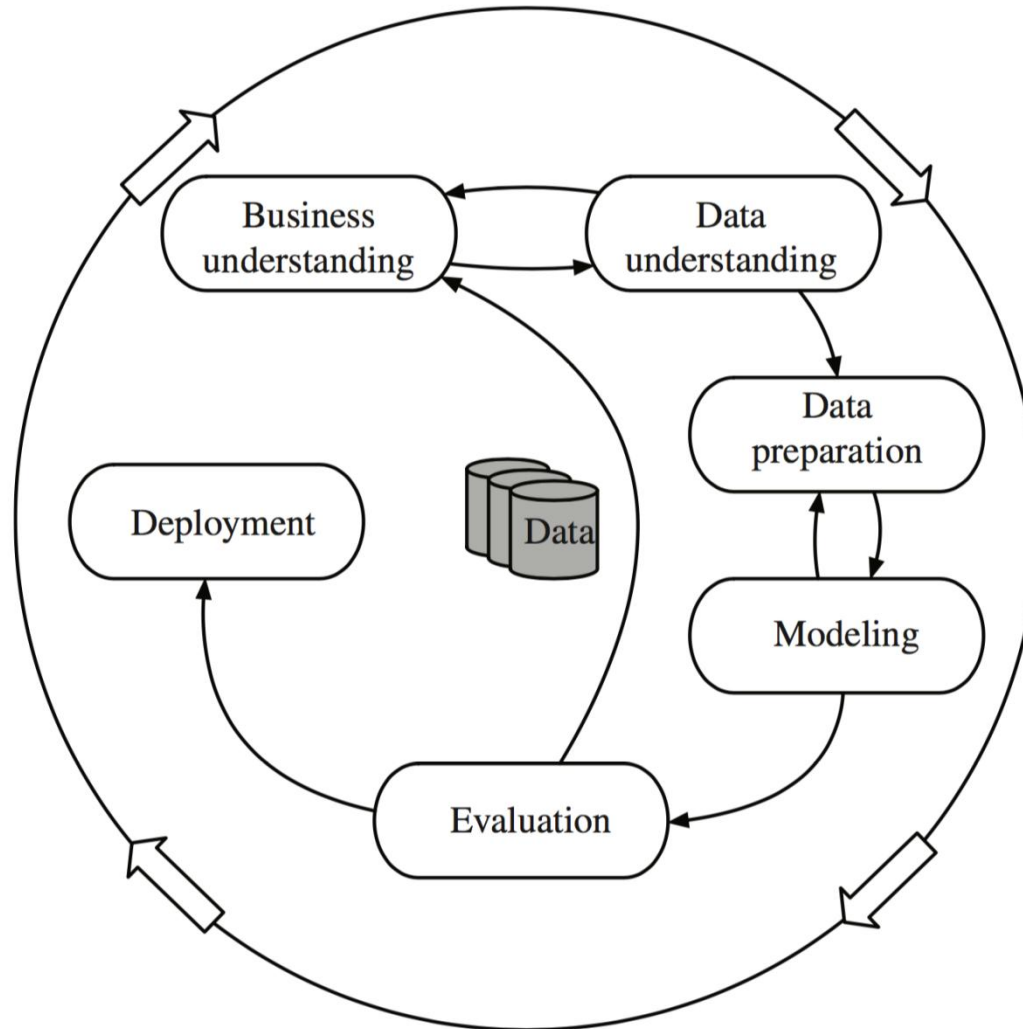
- Unsupervised learning



- Reinforcement learning



General analytical process



Important take-away lessons

- Machine learning (especially Deep Learning) is a very hyped term at the moment
 - Many techniques are decades old – but combined in a novel way
 - Some are rebranded – sometimes, called deep learning – but actually “just” Neural Networks from the 80s ...

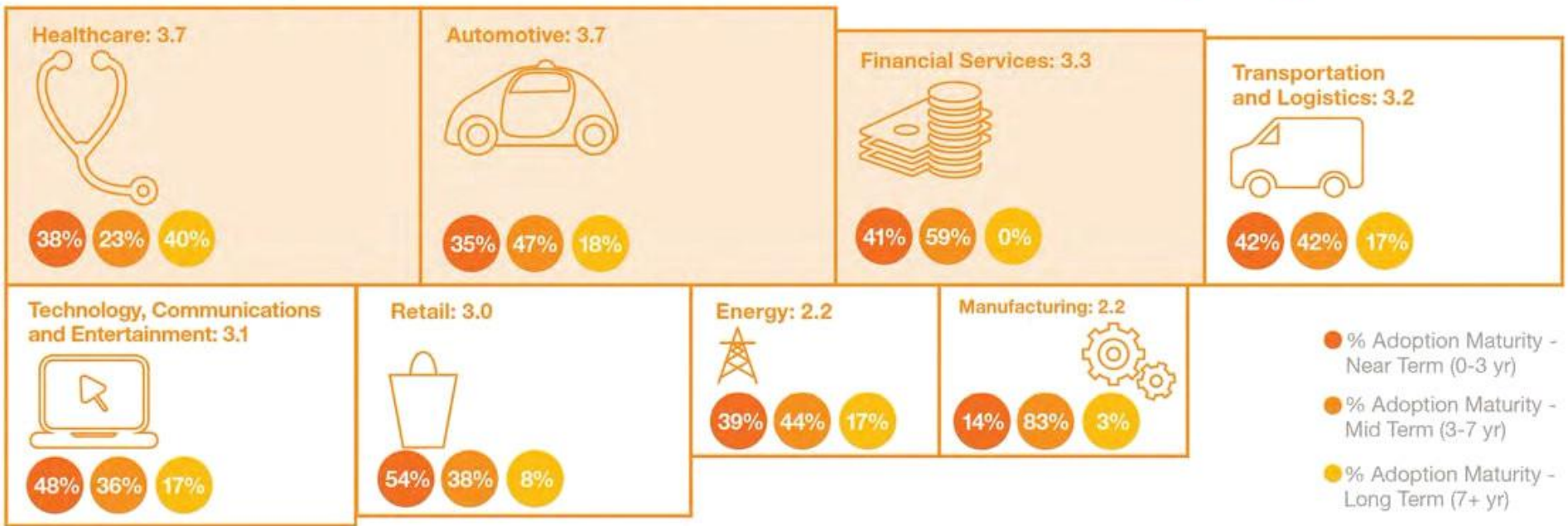
- But: **significant improvements** are achieved
 - Lots of computational power
 - Lots of (labelled) data!



Important take-away lessons

- A lot of market expectation

Sizing the prize – What's the potential for AI to impact your sector?



Scores based on PwC's AI Impact Index evaluation. Potential scores range from 1-5, with 5 indicating the highest potential impact due to AI, and 1 being the lowest.

www.pwc.com/ai
 #AIrevolution

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Important take-away lessons

.....

- Not every task with data is Machine Learning!
 - Often it is much simpler statistical tasks, such as
 - Collection, interpretation and presentation of data
 - Descriptive: summarise data (mean, standard deviation, probability density, correlation analysis)
 - Predictive modelling: e.g. simple (linear) regression
- Often the first step is explorative (Data Mining)
 - Understanding what is in the data
 - Discovering patterns in large datasets
 - Unsupervised
 - Cluster analysis, outlier detection, association rule mining

Important take-away lessons

- Are GPT & co general artificial intelligence?



HIT THE PAUSE BUTTON?

A collective of prominent tech industry leaders, including Elon Musk, Steve Wozniak, and others, have recently advocated for a pause in developing artificial intelligence (AI) technologies such as ChatGPT.



Important take-away lessons

- Machine Learning doesn't solve everything, everywhere
 - ML is not Artificial Intelligence
 - It is primarily very efficient at *pattern recognition*
- Often, the difficult challenge is to
 - Formulate a problem as a ML task
 - E.g. to define that we want to automatically assign text documents to categories
 - And to gather the data!
- The algorithm still doesn't "think on its own"
 - It can't by itself figure out what the task is, what to learn...
 - Just "throwing data at Machine Learning" doesn't work



Important take-away lessons

.....

- A lot of analytical effort goes into the steps “surrounding” Machine Learning
- Analysing the problem / research question
- Identifying, collecting & cleaning the data
- Presentation of results, and incorporation of results into the business
 - How to convince decision makers to
 - Trust your results?
 - Actually use them?
 - ...

Important take-away lessons



What society thinks I do



What my friends think I do



What other computer scientists think I do



What mathematicians think I do

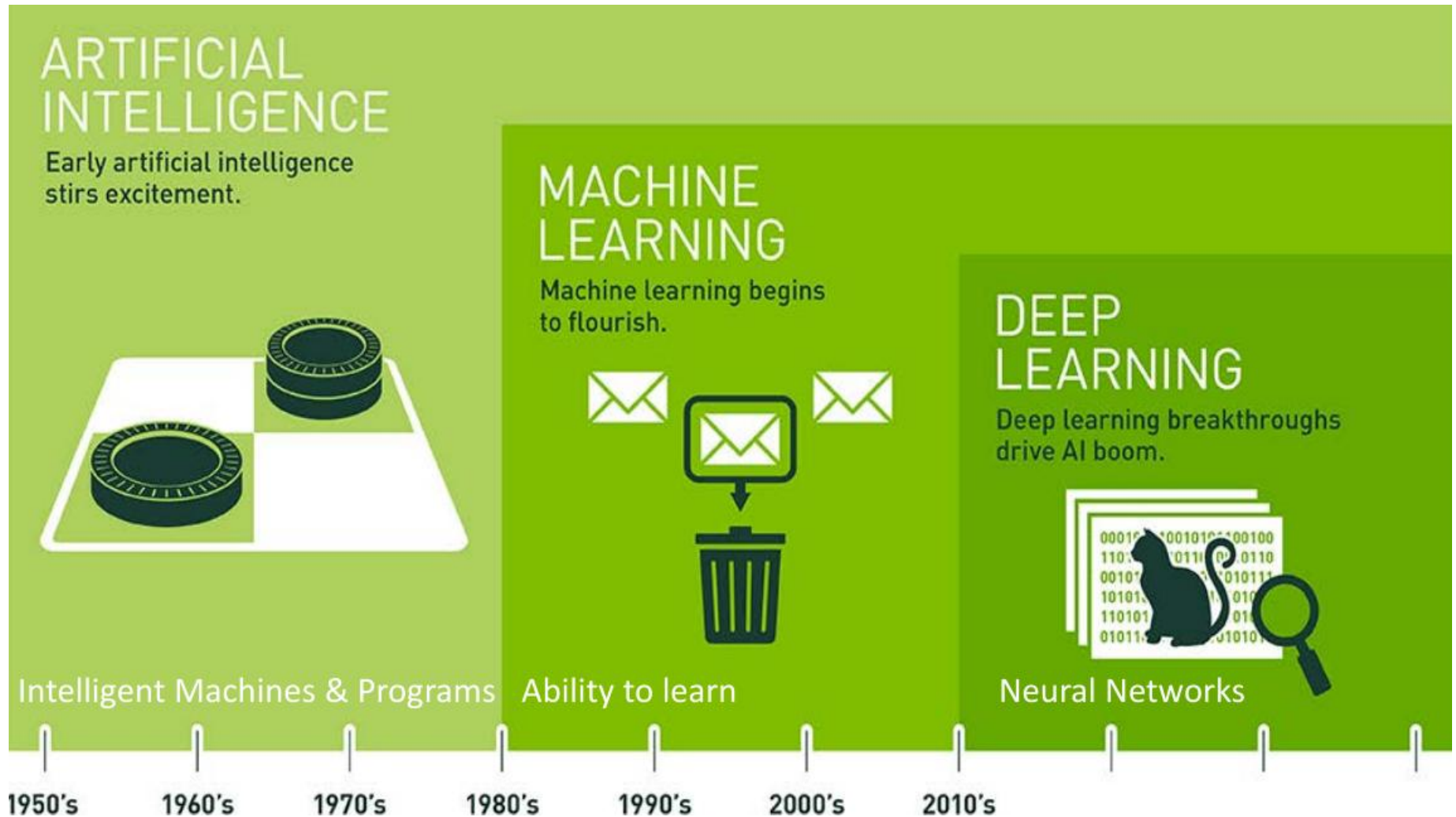


What I think I do

```
from theano import *
```

What I actually do

Summary: ML through the years





Mat Velloso
@matvelloso



Difference between machine learning and AI:

If it is written in Python, it's probably machine learning

If it is written in PowerPoint, it's probably AI

♡ 20.8K 4:25 AM - Nov 23, 2018



💬 7,460 people are talking about this



Important take-away lessons

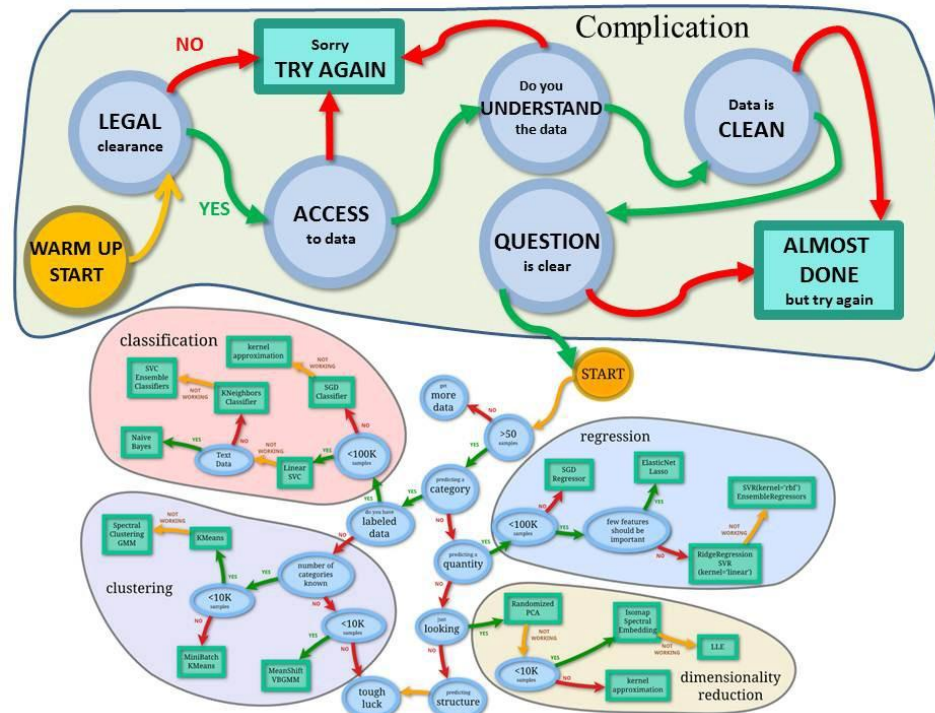
.....

BIG DATA IS LIKE **TEENAGE
SEX: EVERYONE** TALKS
ABOUT IT, **NOBODY** REALLY
KNOWS HOW TO DO IT,
EVERYONE THINKS
EVERYONE ELSE IS DOING IT,
SO **EVERYONE** CLAIMS THEY
ARE DOING IT...

DAN ARIELY

- True as well for Machine Learning, Deep Learning, ..

- We covered in this lecture
 - Data preparation & transformation (scaling, coding, ...)
 - Evaluation – why, how, caveats
 - Various learning models (classification & regression)
 - k-NN
 - Bayesian Learning, Naive Bayes, Markov Models
 - Decision Trees, Random Forests
 - Ensembles
 - SVMs
 - Perceptron, Neural Networks, Deep Learning
- *Which one is the best?*



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- For master or PhD thesis, various topics possible
 - Feature selection, ensembles
 - Music analysis
 - Bio-informatics topics
 - Data privacy & utility
 - Distributed / Federated Machine learning
 - Adversarial Machine Learning
 - Reproducibility of machine learning, ...
 - Or your topic of choice!
- Interdisciplinary project – co-supervision

➔ Contact me if you are interested!

(rudolf.mayer@tuwien.ac.at)

- Self-Organising Systems (188.413)
 - Unsupervised: Self-Organising Maps, Genetic Algorithms, Agents, ...
 - **Winter semester**

- Advanced Information Retrieval (188.980)
 - Feature extraction from text & music, (genre) classification
 - **Summer semester**

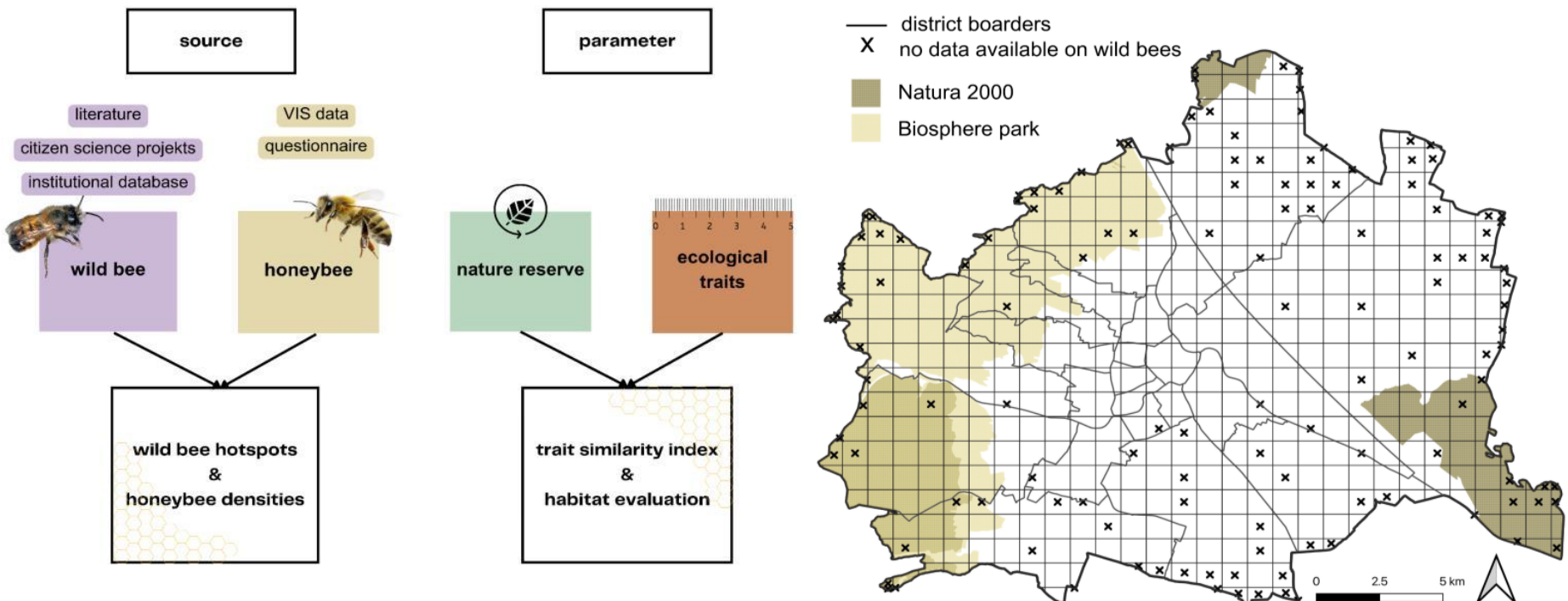
- Problem Solving and Search in Artificial Intelligence (181.190)
 - AI search techniques, learning in search
 - **Summer semester**

- Security, Privacy and Explainability in Machine Learning (194.055)
 - Attacks on ML; explaining ML models; data privacy for machine learning
 - **Summer semester**

- <https://www.ml.tuwien.ac.at/teaching/>
 - E.g. theoretical Foundations and Research Topics in Machine Learning (194.100)

Advertisement: **Interdisciplinary Project**

- Topic on modelling & visualising habitat quality for wild and honey bees
- Main supervisor: Sophie Kratschmer, BOKU



Advertisement: **Internship opportunity**

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- FEMTech Internship opportunity
 - Target group: **female students in technology**
 - Duration: 4-6 months, **paid** (start possible up to Feb 2027)
 - Can be combined with work on a thesis or project in your curriculum
 - Topic should fit to the Research at [SBA Research](#) (most topics in ML will somehow fit)
- More info: <https://www.ffg.at/femtech-praktika>
- Interested?
 - ➔ **Contact me** (rudolf.mayer@tuwien.ac.at)
 - ➔ Next application will be in early March; we will send out information on the exact process beginning of February

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Questions ?